

Chapter 8 Combined and Special-Purpose FACTS Controllers: UPFC, IPFC, Phase-Angle Regulators, IPC, Current Limiters and Voltage Limiters

Learning Objectives

After working through this chapter, the reader should be able to:

- derive the active and reactive power delivered by a line equipped with a unified power flow controller (UPFC), construct its circular control region, and compare it with the regions attainable with series compensation or a phase-angle regulator of the same voltage rating;
- explain the real-power exchange between the shunt and series converters of the UPFC, size both converters, model the DC-link energy, and describe the control modes, control structure and the Inez installation;
- analyse the interline power flow controller (IPFC), including the prime-converter constraint, real-power transfer between lines, voltage-margin sharing and the generalized multi-converter controllers;
- derive the behaviour of in-phase voltage regulators, quadrature boosters and ideal phase shifters, and analyse thyristor tap-changers, including their waveforms, harmonics, commutation constraints and ternary winding arrangements;
- derive the power characteristics of the interphase power controller (IPC), its design equations and operating region, and explain the assisted phase-shifting transformer and the decoupling interconnector;
- explain series-resonant, bridge-type and transformer-coupled solid-state fault current limiters, solid-state breakers and transfer switches, and estimate symmetrical and peak fault currents with and without them;
- describe the metal-oxide varistor law and the thyristor-controlled voltage limiter, and calculate clamping levels, inrush transients and absorbed energy;
- formulate research problems on modular converter architectures, coordinated dispatch, impedance-based stability, current-limited operation and protection interaction, and choose a validation method that matches the claim being made.

Conventions. Underlined symbols denote RMS fundamental phasors; a few early equations, such as (8.1) and (8.4), use plain symbols where the phasor meaning is clear from the context. Physical circuit equations are per phase, so $\underline{V}\underline{I}^*$ is multiplied by 3 for balanced three-phase power, while per-unit equations already use a common three-phase base. With balanced line-to-line RMS voltages, $V_{LL,1}V_{LL,2}/X$ gives three-phase power directly, but a current still needs the factor $\sqrt{3}$. Series injection is written as a voltage rise, $\underline{V}_s' = \underline{V}_s + \underline{V}_{pq}$, the opposite of a series-drop convention. Converter powers P_{se} and P_{sh} are positive from converter to AC network; the IPC powers Q_s and Q_R are positive into the controller at their own terminals. Angles inside derivatives and integrals are in radians. Sections 8.1 to 8.7 form the core of the chapter. Section 8.8 is intended for advanced and research study and assumes familiarity with small-signal control, impedance methods and protection.

8.1 The Unified Power Flow Controller (UPFC)

8.1.1 Concept and circuit

Gyugyi proposed the UPFC in 1991 as a controller that can, within its ratings, act simultaneously or selectively on all the parameters that govern power flow in a line: voltage, impedance and phase angle

[1], [2], [3], [4], [5]. Figure 8.1 shows its circuit. Two voltage-source converters (VSCs) share a DC capacitor. Converter 2 is connected in series with the line through an insertion transformer and injects a voltage V_{pq} of controllable magnitude $0 \leq V_{pq} \leq V_{pq,max}$ and controllable angle $0 \leq \rho < 2\pi$. Converter 1 is connected in shunt through a coupling transformer. Its first task is to supply or absorb, through the DC link, the real power that converter 2 exchanges with the line. It can also generate or absorb reactive power independently, exactly as a STATCOM does, to regulate the bus voltage.

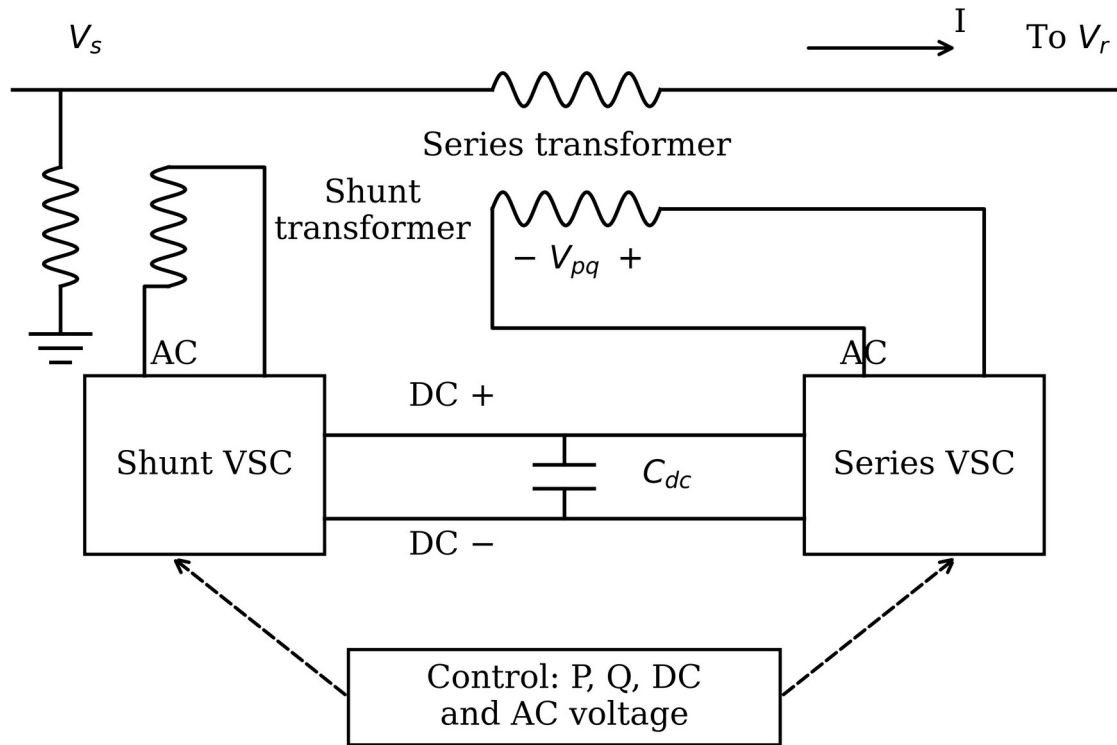


Figure 8.1. UPFC circuit: shunt and series voltage-source converters sharing a DC link. Series polarity – on the sending side and + on the line side denotes the voltage rise used in this chapter.

The UPFC is an energy-coupled combination of the shunt VSC of Chapter 6 and the series VSC of Chapter 7. An independent series converter with only a DC capacitor can sustain no net average real-power delivery apart from drawing its losses. Storage or another powered DC port changes that restriction. In the UPFC, the shunt converter exchanges real power with its AC bus and the common DC link supports the series converter's in-phase voltage component. This allows two-dimensional receiving-end P-Q control within the combined ratings.

The injection has components in phase and in quadrature with line current. The quadrature component exchanges reactive power locally at the series AC port; it does not transfer average reactive power through the DC link. The in-phase component exchanges real power. DC current transfers average real power between converters, while capacitor energy absorbs temporary imbalance and conversion losses must be supplied. Changes in this stored energy are accounted for by Equation (8.7). The ideal series source can emulate voltage, impedance and angle regulation; actual response is set by controls, transformer dynamics and protection, and cannot be summarized by a fixed number of cycles.

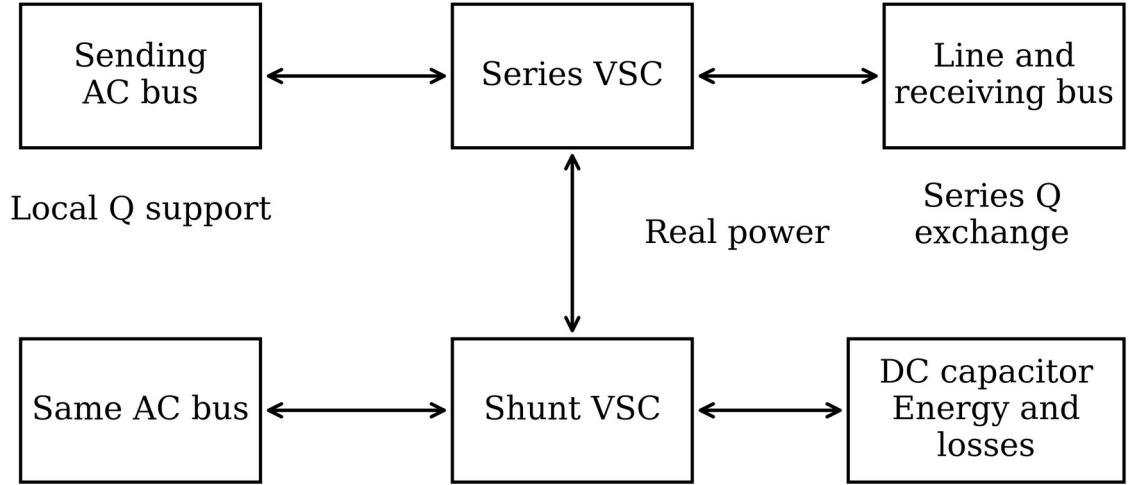


Figure 8.2. Energy and control paths in a UPFC. Only the real-power imbalance is exchanged through the common DC link; reactive power is handled locally at each converter.

The shunt converter therefore has two simultaneous duties. The first is energetic: it regulates the DC-link voltage by supplying the real power demanded by the series converter plus the losses. The second is local AC support: within its current rating it injects or absorbs reactive current to regulate the sending-bus voltage. Both duties share one current limit, so they cannot be treated as independent, unlimited services.

8.1.2 Power-flow equations

Consider the two-machine system with the UPFC at the sending end of a lossless line of reactance X . Take the receiving voltage as the reference, $V_r = V_r \angle 0$, the sending voltage as $V_s = V_s \angle \delta$, and write the series injection as $V_{pq} = V_{pq} \angle (\delta + \rho)$, so that ρ is measured from the sending-end voltage. Kirchhoff's voltage law around the line gives

$$I = \frac{V_s + V_{pq} - V_r}{jX} \quad (8.1)$$

where I is the line current phasor. The complex power delivered to the receiving end, $S_r = V_r I^*$, follows from the expanded current:

From phasors to the power circle. Set $\vartheta = \delta + \rho$. Expanding Equation (8.1) gives $I = [V_s \sin \delta + V_{pq} \sin \vartheta] / X - j[V_s \cos \delta + V_{pq} \cos \vartheta - V_r] / X$. Therefore $S_r = V_r I^* = (V_r / X) \{ V_s \sin \delta + V_{pq} \sin \vartheta + j[V_s \cos \delta + V_{pq} \cos \vartheta - V_r] \}$. Its real and imaginary parts give Equation (8.2). In particular $(P_r - P_0, Q_r - Q_0) = (V_r V_{pq} / X)(\sin \vartheta, \cos \vartheta)$; squaring and adding uses $\sin^2 \vartheta + \cos^2 \vartheta = 1$, rather than a new power-flow assumption.

$$P_r = \frac{V_s V_r}{X} \sin \delta + \frac{V_r V_{pq}}{X} \sin(\delta + \rho), Q_r = \frac{V_s V_r \cos \delta - V_r^2}{X} + \frac{V_r V_{pq}}{X} \cos(\delta + \rho) \quad (8.2)$$

where P_r and Q_r are the active and reactive power delivered to the receiving bus. For $V_s = V_r = V$ the uncompensated values are $P_0 = (V^2 / X) \sin \delta$ and $Q_0 = -(V^2 / X)(1 - \cos \delta)$, and Equation (8.2) can be written as

$$(P_r - P_0)^2 + (Q_r - Q_0)^2 = \left(\frac{V V_{pq}}{X} \right)^2 \quad (8.3)$$

As ρ sweeps a full turn at fixed V_{pq} , the operating point traces a circle centred on the uncompensated point (P_0, Q_0) with radius $V V_{pq}/X$. Allowing any magnitude up to $V_{pq,max}$ fills the disk. The centre slides along the uncompensated locus as δ changes, but the radius does not depend on δ (Figure 8.3). Even at $\delta=0$, where an uncompensated line carries nothing, the UPFC can drive up to $V V_{pq,max}/X$ of active power in either direction.

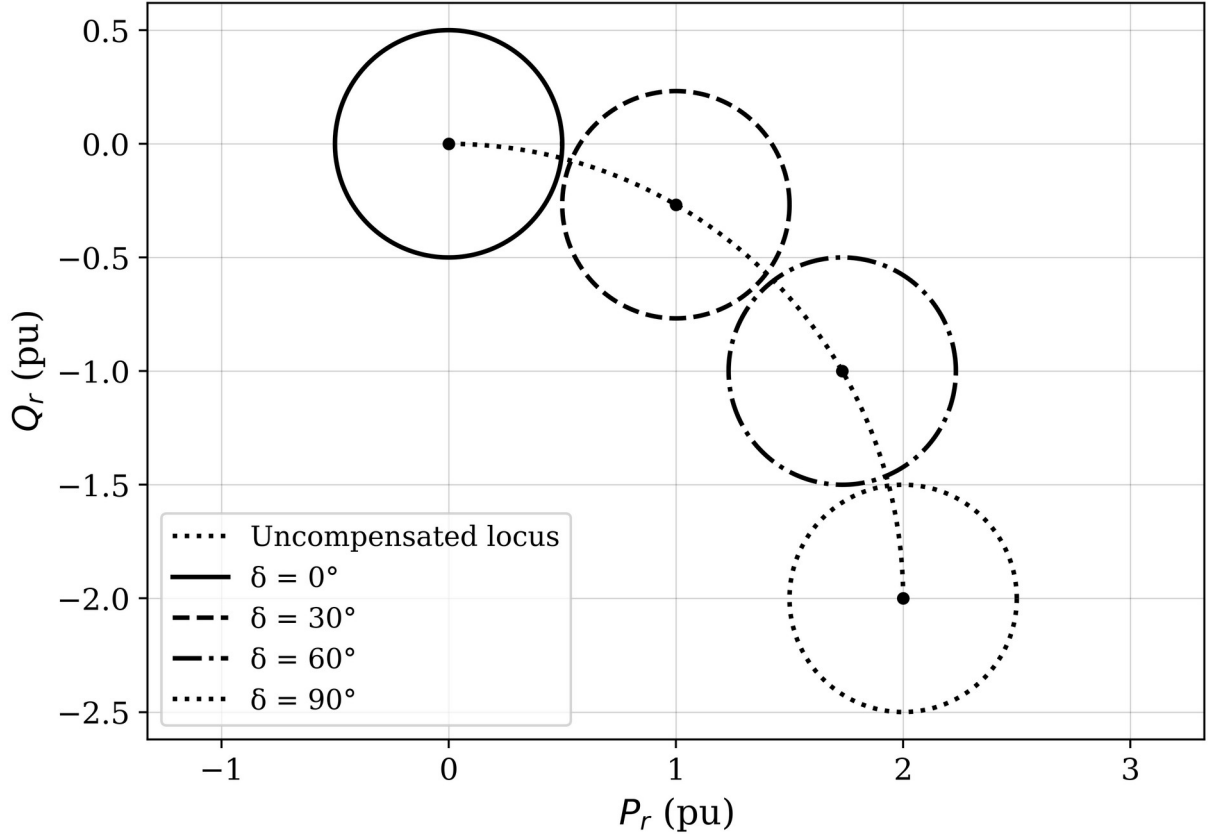


Figure 8.3. UPFC control regions in the P_r - Q_r plane at $\delta=0^\circ, 30^\circ, 60^\circ$ and 90° ($V=1$ pu, $X=0.5$ pu, $V_{pq,max}=0.25$ pu).

The attainable active power at any angle therefore lies in the band $P_0 \pm V V_{pq,max}/X$ (Figure 8.4). Maximum power at a given angle is obtained with $\sin(\delta+\rho)=1$, that is, with V_{pq} leading the receiving voltage by 90° ; at that point the reactive power is unchanged from Q_0 . Figure 8.5 shows how both receiving-end powers vary as the injection angle rotates. The active and reactive responses are in quadrature, which is the reason a single-angle device cannot set both of them independently.

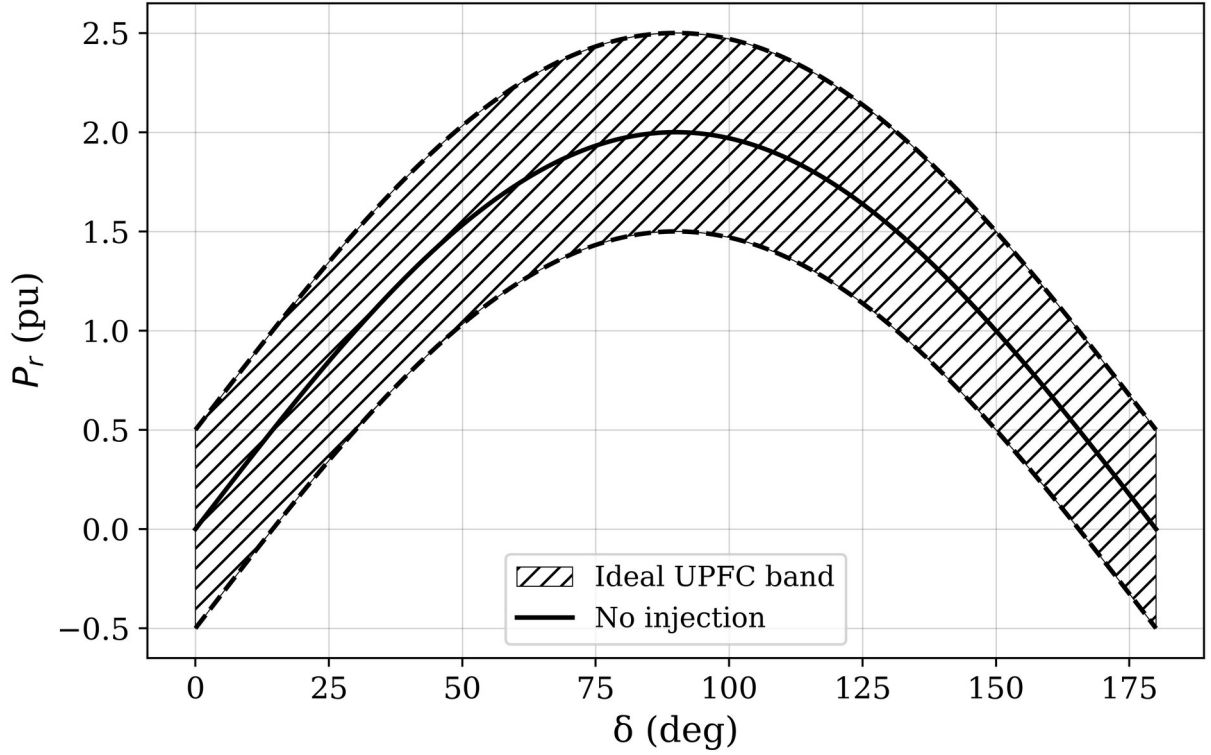


Figure 8.4. Range of transmittable active power with a UPFC versus transmission angle ($V = 1$ pu, $V_{pq,max} = 0.25$ pu, $X = 0.5$ pu).

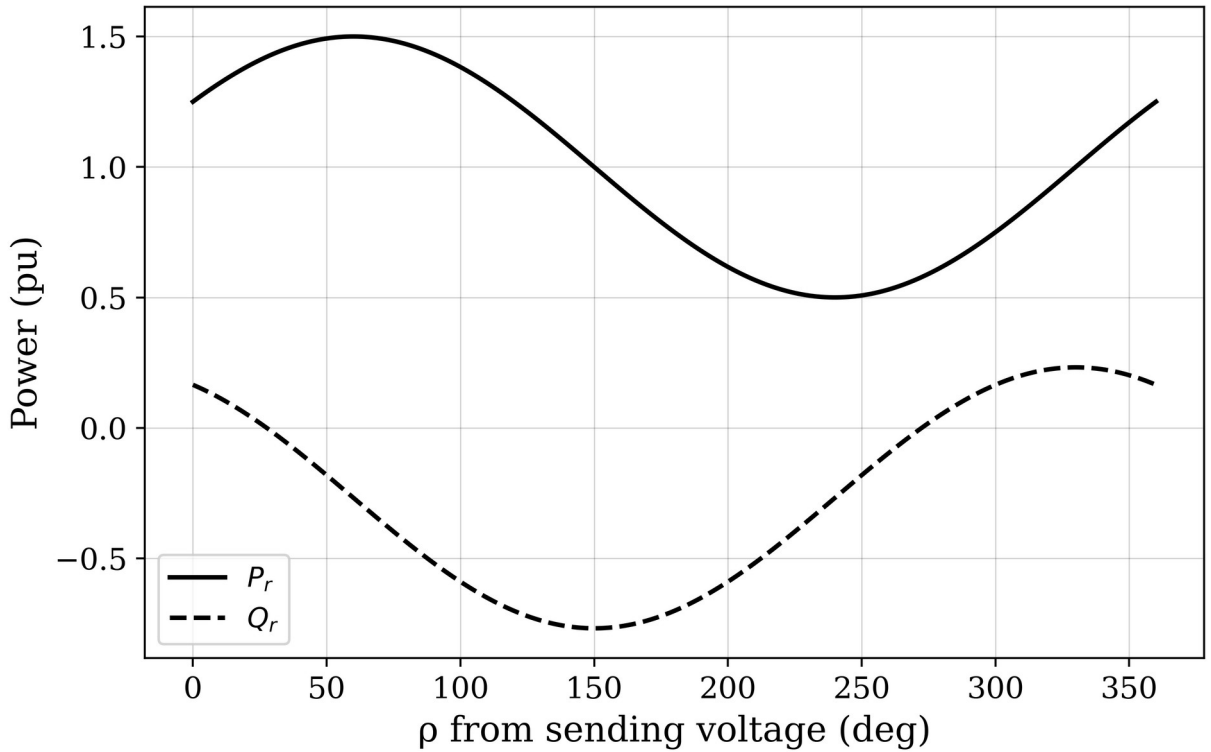


Figure 8.5. Receiving-end active and reactive power versus injection angle for $V = 1$ pu, $X = 0.5$ pu, $\delta = 30^\circ$ and $V_{pq} = 0.25$ pu. The two curves are the projections of the control circle of Equation (8.3).

The disk is a kinematic result for an ideal, lossless network. A real UPFC occupies only the part of it that also satisfies the series-voltage limit, the line-current limit, the series- and shunt-converter MVA

limits, the DC-link voltage band and the semiconductor thermal limits. These constraints are introduced in Sections 8.1.4 and 8.1.6.

Example 8.1. UPFC control circle

For $V=1.0$ pu, $X=0.5$ pu, $\delta=30^\circ$ and $V_{pq,max}=0.25$ pu, find the uncompensated operating point, the radius of the ideal control circle and the attainable range of P_r .

Solution. $P_0=(V^2/X)\sin\delta=2\sin 30^\circ=1.00$ pu and $Q_0=-2(1-\cos 30^\circ)=-0.268$ pu. The radius is $R=V V_{pq,max}/X=0.25/0.5=0.50$ pu. The extreme active powers lie at the horizontal ends of the disk, so $P_{r,max}=1.50$ pu and $P_{r,min}=0.50$ pu. Every point of the disk is geometrically reachable, but converter current, series-transformer voltage and shunt-converter real-power capability must still be checked before a point is accepted as feasible.

8.1.3 Comparison with single-parameter controllers

Figure 8.6 compares devices under the same series-injection magnitude limit at $\delta=30^\circ$. With zero average series real power, an SSSC follows the line $Q_r=-P_r\tan(\delta/2)$ for equal end voltages on the forward-current branch. An ideal constant-magnitude phase shifter traces an arc as $\delta+\sigma$ changes, with $2V|\sin(\sigma/2)|\leq V_{pq,max}$. A quadrature booster is a different regulator and also changes voltage magnitude. The UPFC can vary both phasor components and fills the ideal disk. Current, DC balance and converter MVA can reduce every one of these geometrical sets [6], [7].

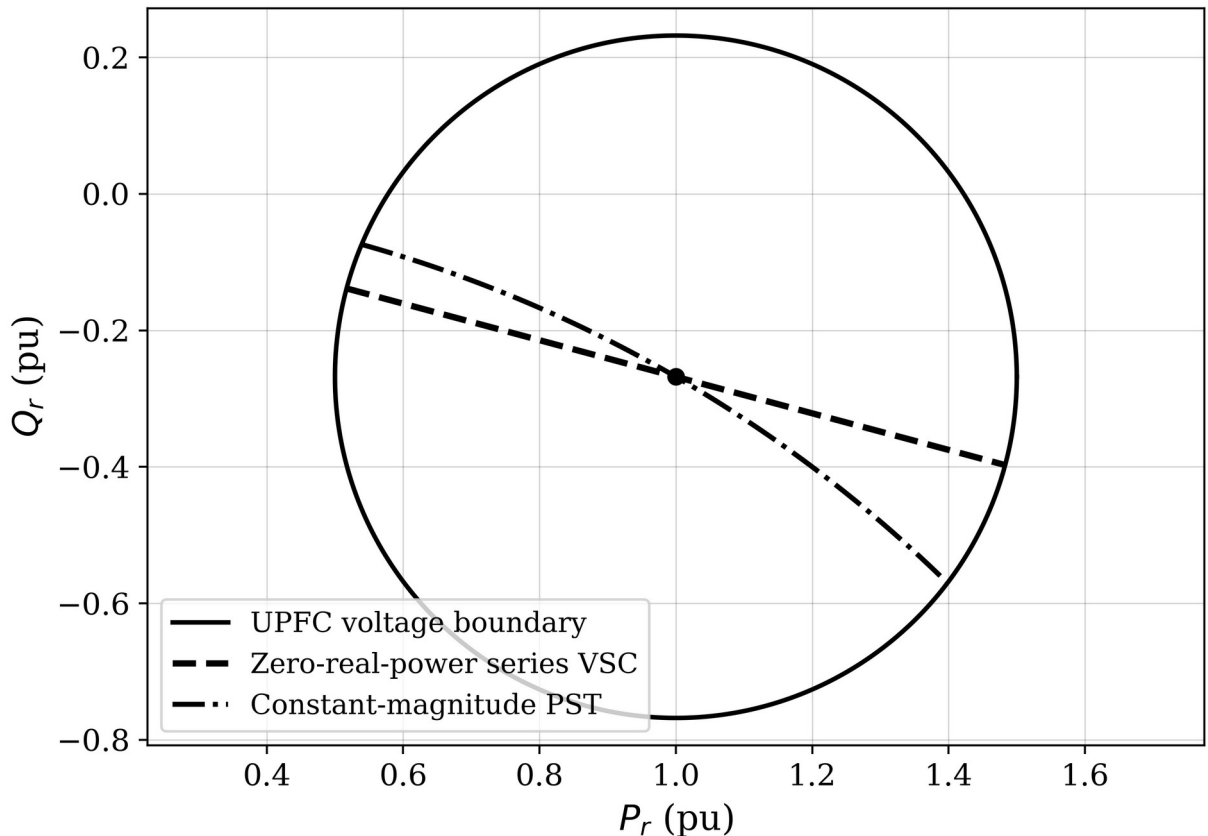
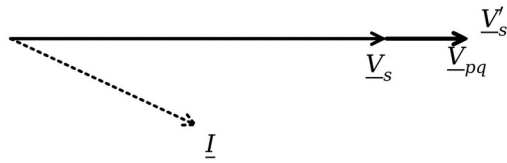


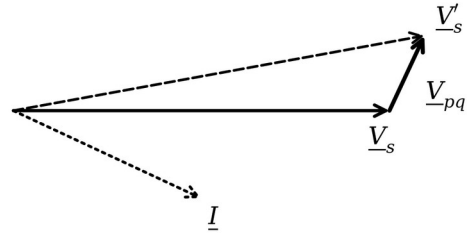
Figure 8.6. Equal-voltage comparison at $\delta = 30^\circ$, $X = 0.5$ pu and $|V_{pq}| \leq 0.25$ pu: UPFC disk boundary, zero-real-power SSSC segment, and ideal constant-magnitude phase-shifter arc. Device ratings can further restrict these loci.

The same series source reproduces single-purpose functions by constraining $\underline{V}_{pq}$ (Figure 8.7). Voltage regulation aligns it with $\underline{V}_s$. Reactance emulation sets $\underline{V}_{pq} = -jX_{inj}\underline{I}$: positive X_{inj} is inductive and negative X_{inj} is capacitive. Thus capacitive **voltage rise** leads the current by 90° . Ideal phase regulation uses $\underline{V}_{pq} = \underline{V}_s(e^{j\sigma} - 1)$ and preserves $|\underline{V}_s|$. Automatic P-Q control computes both components subject to the feasible region. These voltage-rise signs must be reversed if a voltage-drop convention from Chapter 7 is used.

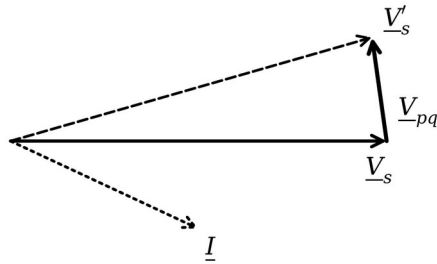
(a) Bus-aligned rise



(b) Capacitive rise (leads $\underline{I}$ by 90°)



(c) Constant-magnitude phase shift



(d) General injection

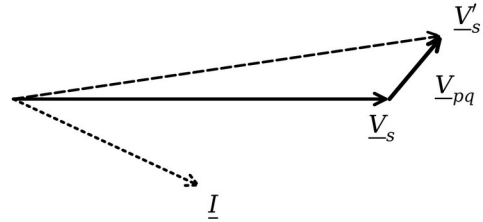


Figure 8.7. Voltage-rise phasors for four control modes. The dotted current arrows indicate direction on a separate scale; they are not voltage vectors.

8.1.4 Internal power exchange and ratings

The complex power supplied by the series converter to the line is

$$S_{se} = P_{se} + jQ_{se} = \underline{V}_{pq} \underline{I}^* \quad (8.4)$$

where P_{se} and Q_{se} are the real and reactive power delivered by the series converter. In a lossless UPFC in steady state the DC link must balance:

$$P_{sh} + P_{se} = 0 \quad (8.5)$$

where P_{sh} is the real power delivered by the shunt converter to the AC system. The shunt converter therefore draws P_{se} from the bus (or returns it) in addition to its own reactive output Q_{sh} , and its rating must satisfy

$$S_{sh} = \sqrt{P_{se}^2 + Q_{sh}^2} \leq V_{bus} I_{sh,max}, S_{se} = V_{pq} I \leq V_{pq,max} I_{max} \quad (8.6)$$

Figure 8.8 plots receiving-end power and series-port power. Example 8.25 shows that a 0.067 pu link transfer accompanies a 0.5 pu increase in line power; the link supplies converter exchange, not the full transmitted power. For a selected point, $|S_{se}| = |V_{pq}|I$ on a common base. The voltage and current maxima define an upper rating envelope, and simultaneous maxima need not be the actual worst feasible operating point. Shunt rating includes the active balancing duty and reactive support. Equal converter ratings are a project choice rather than a physical requirement.

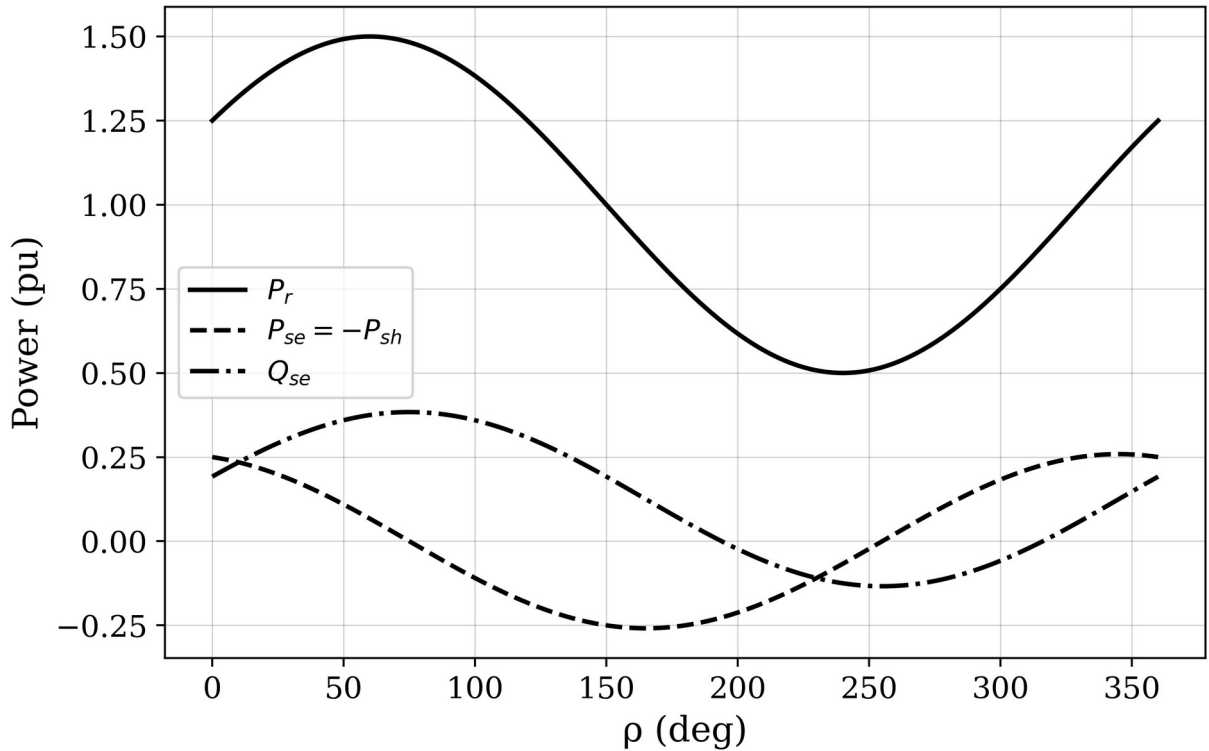


Figure 8.8. Receiving-end power and series-converter real and reactive power versus injection angle ($\delta = 30^\circ$, $V_{pq} = 0.25$ pu, $X = 0.5$ pu).

Example 8.2. Series-converter real-power requirement

A UPFC series converter injects 0.12 pu at an angle of 30° with respect to a 0.90 pu line current. Find the converter real and reactive power on the same three-phase base.

Solution. With the current as reference, $S_{se} = V_{pq} I^{\angle} = 0.12 \times 0.90 \angle 30^\circ = 0.108 \angle 30^\circ$ pu. Hence $P_{se} = 0.108 \cos 30^\circ = 0.0935$ pu and $Q_{se} = 0.108 \sin 30^\circ = 0.0540$ pu. The shunt converter must supply about 0.0935 pu of real power through the DC link, plus losses. The 0.054 pu of reactive power is produced inside the series converter and does not cross the link.

Example 8.3. Shunt-converter current allocation

A shunt VSC connected to a 1.0 pu bus must draw 0.18 pu of real power to support the UPFC DC link and inject 0.24 pu of reactive power for voltage support. Determine its apparent-power duty and current.

Solution. $S_{sh} = \sqrt{0.18^2 + 0.24^2} = 0.30$ pu. At $V = 1.0$ pu the current magnitude is also 0.30 pu. Once the converter reaches its current limit, real-power balancing and reactive support compete for the same current and cannot be allocated independently.

8.1.5 Operating and control modes

The shunt converter operates in one of two modes. In VAR control it follows a reactive power (or reactive current) reference. In automatic voltage control, the usual mode, it adjusts reactive current to regulate the bus voltage at the point of connection along a droop characteristic. In either mode a DC-voltage regulator adds the active-current component that balances the series-converter real power and the losses.

The series converter offers five functional modes. They are control objectives, not different power circuits:

- *Direct voltage injection.* The converter injects the commanded V_{pq} and ρ directly. Special cases are an injection in quadrature with the current (SSSC-like reactive compensation) and an injection in phase with the bus voltage (voltage regulation).
- *Bus voltage regulation.* The injection is kept in phase with the sending-end voltage and its magnitude regulates the voltage at the line-side terminal.
- *Line impedance compensation.* The injection is made proportional to the line current with a chosen phase, emulating a series reactance and, if desired, a series resistance.
- *Phase-angle regulation.* The injection is chosen so that the line-side voltage is shifted by a commanded angle without change of magnitude.
- *Automatic power-flow control.* The most general mode, in which closed outer loops around the measured line P and Q compute the series voltage, so that the commanded flows are maintained in spite of network changes.

8.1.6 Dynamic model, control structure and limits

For dynamic studies the DC capacitor cannot be treated as an algebraic power-transfer path. Its stored energy is $W_{dc} = 1/2 C_{dc} V_{dc}^2$. With positive P_{sh} and P_{se} denoting real power delivered from the converters to the AC system, the energy balance is

$$\frac{dW_{dc}}{dt} = \frac{d}{dt} \left(1/2 C_{dc} V_{dc}^2 \right) = - (P_{sh} + P_{se} + P_{loss}) \quad (8.7)$$

and for small deviations about the operating voltage V_{dc0} ,

$$C_{dc} V_{dc0} \frac{d\Delta V_{dc}}{dt} \approx - (\Delta P_{sh} + \Delta P_{se}) \quad (8.8)$$

Equation (8.5) is therefore valid only after the energy transient has settled. A temporary mismatch moves V_{dc} , and the shunt-converter DC-voltage loop supplies the active current that restores it.

Example 8.4. DC-link energy deviation

A 20 mF UPFC DC capacitor operates at 20 kV. For 15 ms the series converter draws 6 MW more real power from the link than the shunt converter supplies. Neglect losses and find the DC voltage after the event.

Solution. Initial energy $W_1 = 1/2 C V_1^2 = 0.5 \times 0.020 \times (20\,000)^2 = 4.00$ MJ. The deficit is $6 \times 10^6 \times 0.015 = 0.09$ MJ, so $W_2 = 3.91$ MJ and $V_2 = \sqrt{2W_2/C} = 19.77$ kV, a 1.1 % dip. The linearized Equation (8.8) gives $\Delta V_{dc} \approx -0.09 \times 10^6 / (0.020 \times 20\,000) = -225$ V, which agrees closely.

Figure 8.9 shows a typical vector-control structure. A phase-locked loop provides the reference angle for both converters. For the series converter, P_{ref} and Q_{ref} are converted into d - and q -axis line-current references, and a current controller with decoupling computes the series voltage, limited to $V_{pq,max}$. For the shunt converter, a DC-voltage regulator produces the active-current reference that keeps the link charged, and an AC-voltage (or VAR) regulator produces the reactive-current reference. Supplementary loops add power-oscillation damping and transient-stability functions to the series reference.

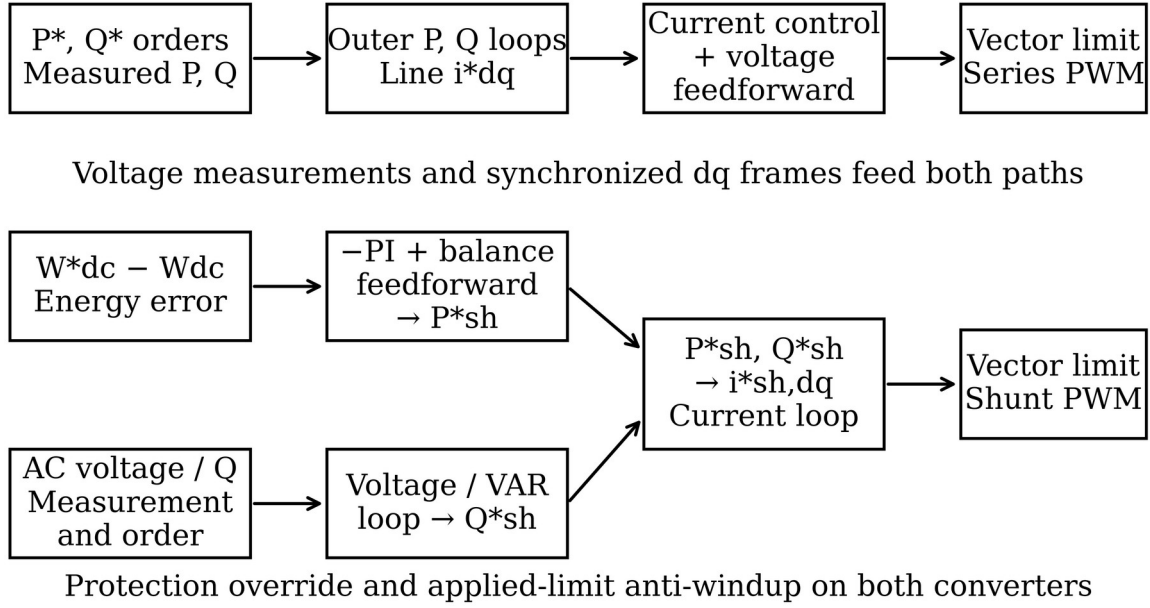


Figure 8.9. UPFC control structure: series converter in automatic power-flow control, shunt converter regulating DC and AC voltage.

The inner current or voltage loops must be appreciably faster than the outer power, AC-voltage and DC-voltage loops. Bandwidth separation reduces interaction, but it does not remove the coupling that exists through the network and the DC link. Anti-windup is required wherever a reference saturates.

The ideal control disk is not a converter capability curve. At every requested operating point the controller must check series-voltage magnitude, line current, series-converter MVA, shunt-converter current, transformer MVA, DC-link voltage and semiconductor temperature. In normal operation the outer P-Q loop may have full authority. During a severe voltage depression or fault, current limiting overrides it, because semiconductor survival has priority over tracking a power-flow reference. A common priority order is converter protection and DC-link energy first, then the remaining current margin for reactive support. The exact order is manufacturer and application dependent and should be represented explicitly in EMT studies.

An explicit capability check. For unequal as well as equal terminal voltages define

$P_0 = V_s V_r \sin \delta / X$ and $Q_0 = (V_s V_r \cos \delta - V_r^2) / X$. On the receiving-voltage reference,

$I = \sqrt{P_r^2 + Q_r^2} / V_r$ and $V_{pq} = (X / V_r) \sqrt{(P_r - P_0)^2 + (Q_r - Q_0)^2}$. The series limit requires

$(P_r^2 + Q_r^2) [(P_r - P_0)^2 + (Q_r - Q_0)^2] \leq (V_r^2 S_{se,max} / X)^2$. Since $\underline{V}_{pq} = V_r + j X \underline{I} - \underline{V}_s$, direct multiplication by $\underline{I}^c$ gives

$P_{se} = [(V_r - V_s \cos \delta) P_r + V_s \sin \delta Q_r] / V_r = [(P_r - P_0)(V_r - V_s \cos \delta) + V_s \sin \delta (Q_r - Q_0)] / V_r$. The

plus sign on the reactive-power term follows this chapter's voltage-rise and receiving-power

conventions. The shunt must meet $P_{sh} = -P_{se} - P_{loss}$ in steady state and $P_{sh}^2 + Q_{sh}^2 \leq (V_{bus} I_{sh,max})^2$. These relations clip the ideal disk.

A typical start-up precharges the DC link, checks transformers and buses, synchronizes the controls, establishes DC voltage with the shunt converter, and then releases series injection from bypass. During a severe fault, semiconductor blocking alone does not remove transformer-secondary current. A designed fast bypass carries that current; mechanical bypass/isolation and arrester paths provide subsequent protection. Natural thyristor recovery, stored magnetic energy and post-fault DC balance determine the restart sequence. The actual manufacturer's protection scheme must be represented in switching studies.

Figure 8.10 is an analytical sketch rather than a measured or simulated trace. It illustrates separate P-Q transients and a temporary DC-energy disturbance. A sustained DC-voltage drift implies average power imbalance; a complete dynamic study must include the shunt balancing loop, current limits, measurement delay and transformer dynamics.

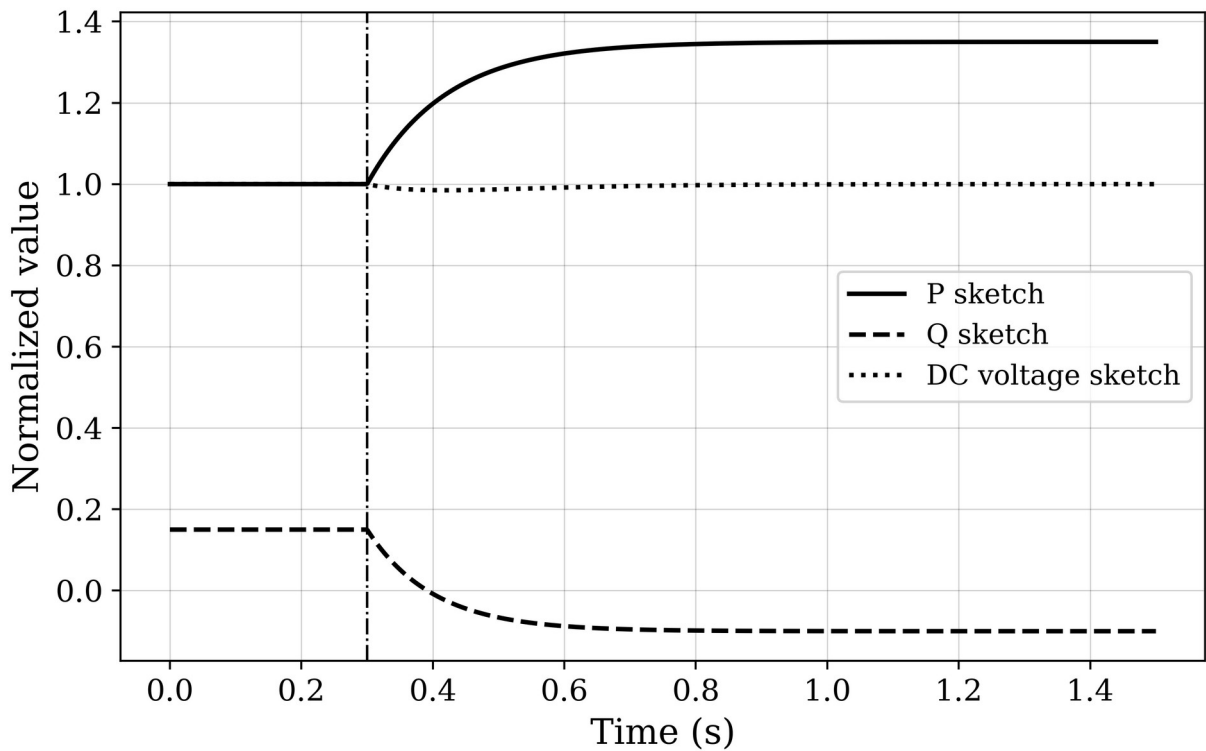


Figure 8.10. Analytical response sketch. At $t = 0.3$ s let $u = \max(t - 0.3, 0)$: $P = 1 + 0.35(1 - \exp(-u/0.12))$, $Q = 0.15 - 0.25(1 - \exp(-u/0.10))$, $W_{dc} = 1 - 0.03(u/0.12)\exp(1 - u/0.12)$, and $V_{dc} = \sqrt{W_{dc}}$, in normalized units.

For damping, use the swing equation $M d \Delta \omega / dt = \Delta P_m - \Delta P_e - D \Delta \omega$. A positive component of ΔP_e **in phase with positive speed deviation** adds damping. An angle-proportional component primarily changes synchronizing stiffness. An angle measurement can feed a POD controller, but its filtering and phase compensation must produce the required speed-phase component after network and actuator dynamics. During faults protection overrides normal flow control; restoration must respect current, voltage and post-fault stability limits.

8.1.7 UPFC with a fixed phase shifter

A UPFC can be combined with a fixed, mechanically tapped phase-shifting transformer. The fixed shifter moves the centre of the control circle to the point that corresponds to the shifted angle, and the UPFC provides fast control around that new centre. The motivation is rating reduction. If the steady

requirement is V_{fix} and the dynamic correction is ΔV , the converter need only be rated for the envelope of ΔV rather than for $V_{fix} + \Delta V$. The price is flexibility, because the fixed component cannot be removed electronically during an emergency.

Example 8.5. Fixed phase shifter plus UPFC

A corridor requires a steady 12° phase shift but only $\pm 4^\circ$ of fast modulation for damping and dispatch. If an ideal fixed phase shifter supplies 12° , what range must the UPFC series converter provide, and what is the approximate rating saving on a 1 pu voltage base?

Solution. The fast converter covers -4° to $+4^\circ$ about the fixed 12° shift, so the total is 8° to 16° . Without the fixed shifter, $2 \sin 8^\circ = 0.27835$ pu is needed at the upper angle. With it, the maximum residual chord is $2 \sin 2^\circ = 0.06980$ pu, about 25.1% of that voltage. This is a voltage-envelope reduction of about 74.9%; an MVA saving of the same fraction additionally assumes unchanged current and comparable transformer ratios. Recheck current and shunt real-power duty at the actual operating points.

8.1.8 Field experience: the Inez UPFC

The first UPFC was installed at the Inez substation of American Electric Power in eastern Kentucky and was commissioned in mid-1998, after its shunt STATCOM stage had entered service in 1997 [8], [9]. The AEP/EPRI/Westinghouse project used two GTO voltage-source converters, each with a **160 MVA apparent-power rating**, and harmonic-neutralizing multi-pulse transformer arrangements. Their reconfigurable modes include STATCOM, SSSC and UPFC. The project supported a heavily loaded 138 kV area and controlled flow on a high-capacity line. Later converter building blocks were used in NYPA's Marcy convertible static compensator (Section 8.2.4). Nanjing and Suzhou later used modular multilevel conversion (Section 8.8.1). Its rating, energy-balance and transformer-protection lessons remain relevant whichever semiconductor and waveform-synthesis method a newer design adopts.

8.2 The Interline Power Flow Controller (IPFC)

8.2.1 Concept

Gyugyi, Sen and Schauder proposed the IPFC to compensate several lines leaving one substation [10]. Each line is given its own series converter, and all converters share a common DC link (Figure 8.11). Each converter can provide reactive series compensation to its own line, as an SSSC would. In addition, because the converters share the DC link, any converter can deliver real power to its line provided another converter draws the same real power from its own line. The IPFC can therefore transfer real power between lines: it can relieve an overloaded line by shifting active power to an underloaded one, equalize both active and reactive flows, and compensate resistive as well as reactive voltage drops.

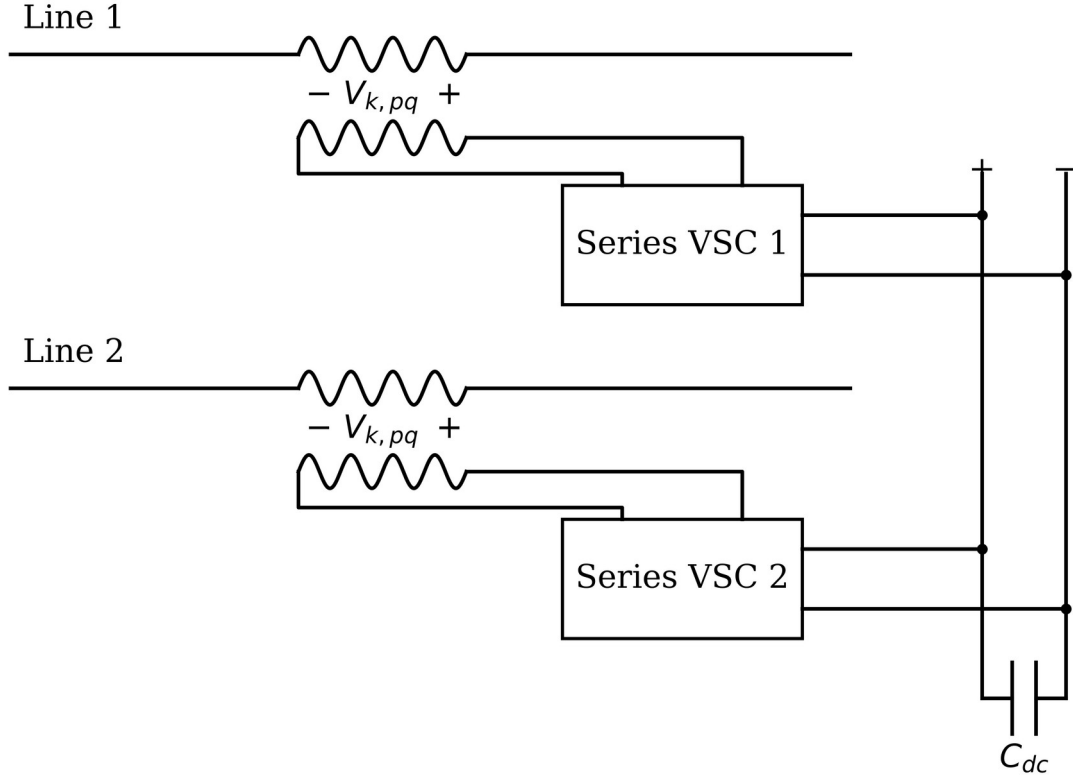


Figure 8.11. Two-line IPFC equivalent: separate series transformer AC ports and a common two-rail DC link. Black dots mark connected rail junctions; an unmarked crossing is not a connection. Series polarity – on the sending side and + on the line side denotes the voltage rise used in this chapter.

The common DC link creates no energy. In an IPFC without a shunt converter or storage, the algebraic sum of the average real powers exchanged by all series converters must be zero apart from losses and transient changes of capacitor energy. For N series converters (Figure 8.12)

$$\sum_{k=1}^N P_{k,pq} + \frac{dW_{dc}}{dt} + P_{loss} = 0 \quad (8.9)$$

All series powers are positive from converter to line. Their sum is not an external energy source: it must cover losses and DC-energy change according to Equation (8.9). A common DC link couples the line controls even when each AC series injection is separately commanded.

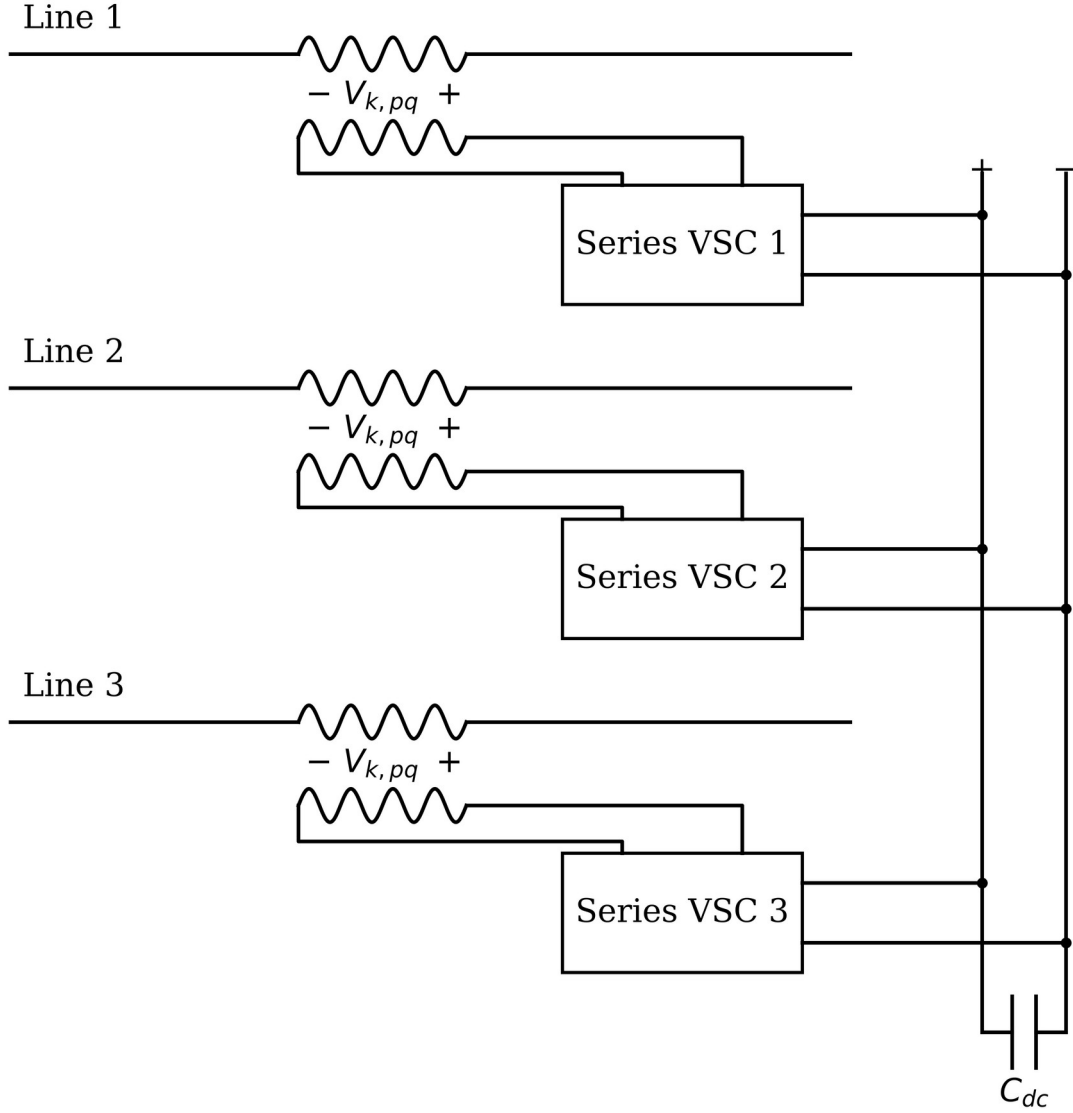


Figure 8.12. Three-line IPFC equivalent with separate series AC ports and a common two-rail DC link. Black dots mark rail junctions; unmarked crossings are not connections. Series polarity – on the sending side and + on the line side denotes the voltage rise used in this chapter.

8.2.2 Constrained operation of the prime converter

In a two-converter IPFC, converter 1 (the prime converter) is chosen to control line 1 fully, with its injected voltage V_{1pq} free in magnitude and angle. Its line then follows Equation (8.2) and Equation (8.3) exactly as with a UPFC. The difference lies in where the real power comes from. The series power supplied by converter 1 to line 1 must be absorbed by converter 2:

$$P_{1pq} = \text{Re}(V_{1pq} I_1^*), P_{1pq} + P_{2pq} = 0 \quad (8.10)$$

where P_{1pq} and P_{2pq} are the real powers delivered to lines 1 and 2 by their converters and I_1 is the current of line 1. Converter 2 must therefore inject into line 2 a voltage whose component in phase with I_2 absorbs exactly P_{1pq} . That fixes one degree of freedom of converter 2. The other, its quadrature component, remains free for reactive compensation of line 2, but only within the voltage margin left inside its circular limit (Figure 8.13):

$$V_{2p} = -\frac{P_{1pq}}{I_2}, |V_{2q}| \leq \sqrt{V_{2,max}^2 - V_{2p}^2} \quad (8.11)$$

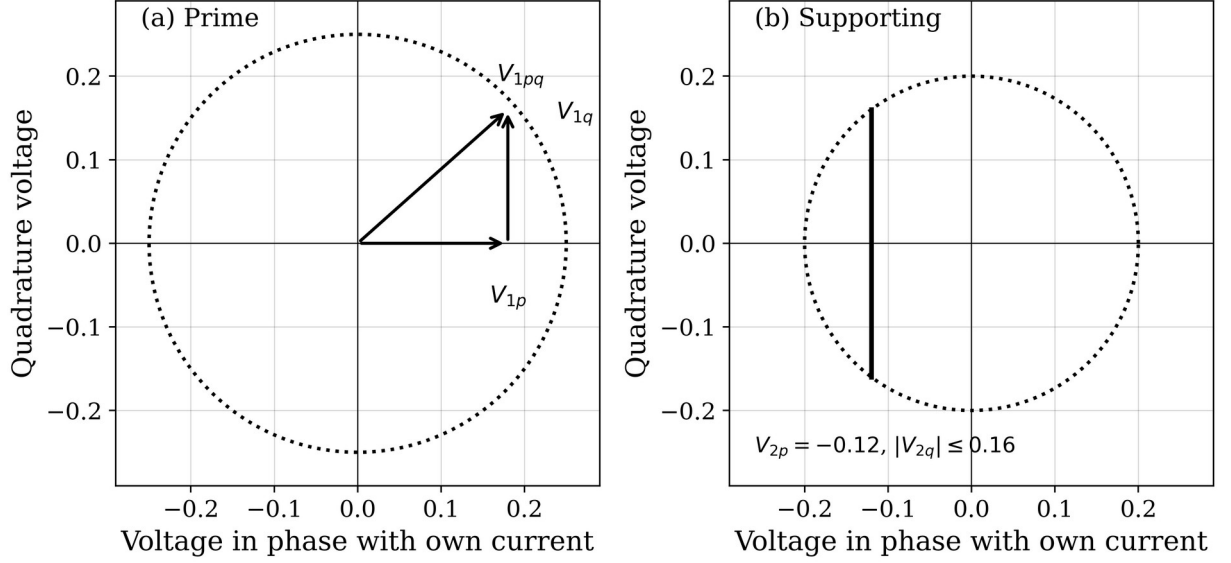


Figure 8.13. IPFC injection vectors on each line's own current reference, with $I_1 = 1$ pu and $I_2 = 1.5$ pu. The supporting converter absorbs 0.18 pu real power: $V_{2p} = -0.12$ pu and $|V_{2q}| \leq 0.16$ pu for $|V_{2pq}| \leq 0.20$ pu.

The prime-line disk is feasible only where the supporting converter can absorb or deliver the required real power: $|P_{1pq}| \leq V_{2,max} I_2$ in the lossless steady model. This is a straight strip in the prime-line P-Q plane when the end-voltage phasors and the supporting power bound are fixed, intersected with the voltage disk and all current/MVA constraints (Figure 8.14). Example 8.32 requires 0.19053 pu support while holding $Q_{1r} = 0$ and reaching $P_{1r} = 1.42214$ pu. Its two supporting-line phasor solutions have different receiving powers, currents and reactive exchanges.

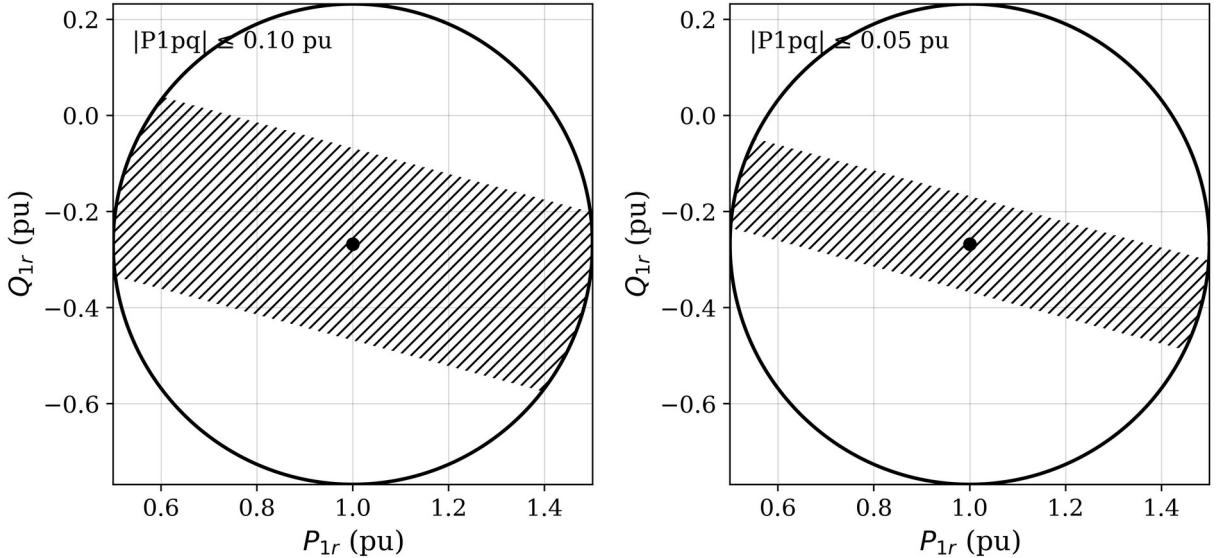


Figure 8.14. Prime-line UPFC disk clipped by explicit supporting real-power limits $|P_{1pq}| \leq 0.10$ and 0.05 pu (hatched feasible strips). $V_s = V_r = 1$ pu, $\delta = 30^\circ$, $X = 0.5$ pu, $|V_{1pq}| \leq 0.25$ pu. These are power-exchange constraints; current and MVA constraints are additional.

In a multi-line installation the prime role need not be permanently assigned. A supervisory controller can redistribute real-power duty according to line loading, converter temperature and remaining voltage margin (Figure 8.15).

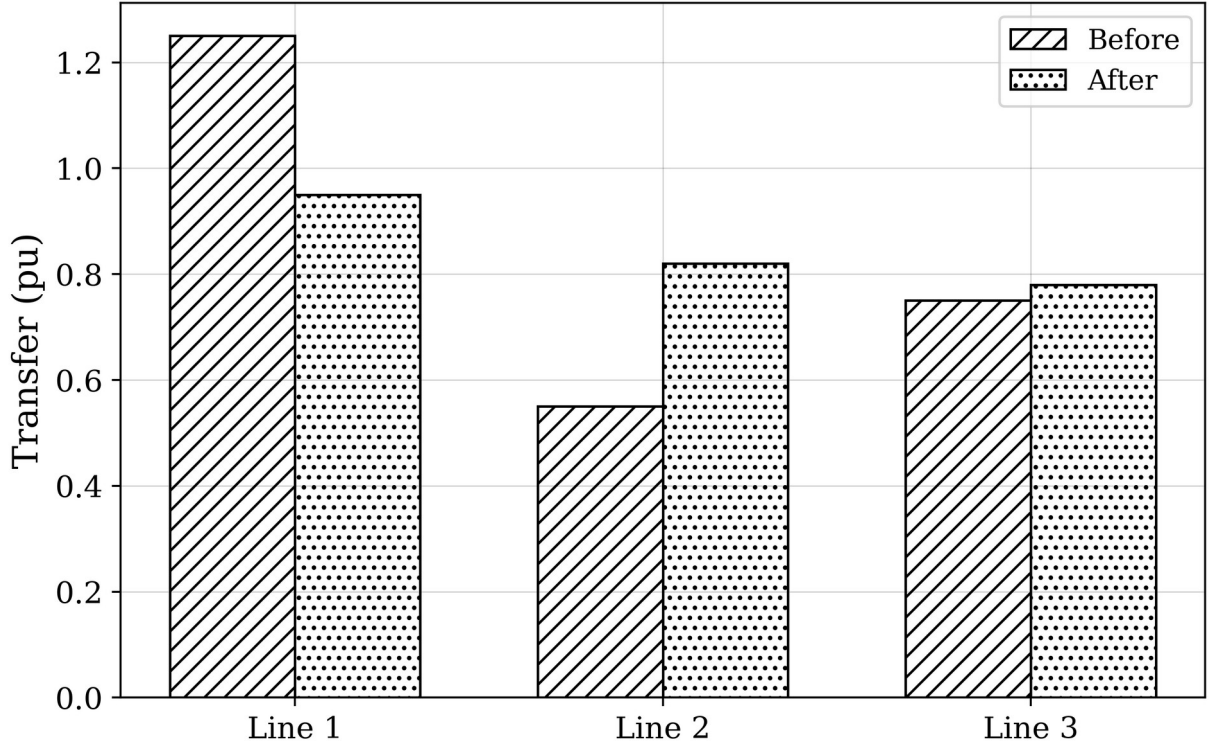


Figure 8.15. Illustrative three-line redistribution with unchanged total transfer 2.55 pu. These values demonstrate the objective, not a computed network optimization.

Example 8.6. Real-power redistribution in a three-converter IPFC

Three series converters share a common DC link. Converter 1 delivers 0.18 pu of real power to line 1 and converter 2 absorbs 0.07 pu from line 2. Neglecting losses and DC-energy change, find the real power of converter 3.

Solution. From Equation (8.9), $P_1 + P_2 + P_3 = 0$. Taking delivery to a line as positive, $P_3 = -(0.18 - 0.07) = -0.11$ pu: converter 3 must absorb 0.11 pu from line 3. An IPFC can redistribute real power, but it cannot supply net real power to all lines at once unless a shunt converter, storage or another real-power port is added.

Example 8.7. Remaining quadrature margin of a supporting converter

A supporting converter is limited to $V_{2,max} = 0.20$ pu and carries $I_2 = 1.0$ pu. It must absorb 0.12 pu of real power for the prime line. Find the largest quadrature component it can still inject, and repeat for a line current of 1.5 pu.

Solution. At $I_2 = 1$ pu, $V_{2p} = -0.12$ pu and $|V_{2q}| \leq \sqrt{0.20^2 - 0.12^2} = 0.16$ pu. At $I_2 = 1.5$ pu, $V_{2p} = -0.08$ pu and the remaining magnitude is $\sqrt{0.04 - 0.0064} = 0.1833$ pu. Higher supporting current reduces the in-phase voltage needed for the same real-power exchange, **provided** its current, MVA and thermal limits remain admissible. It does not automatically make the higher-loaded line the best donor.

8.2.3 Generalized and multi-converter controllers

The UPFC and the IPFC are special cases of a family built from voltage-source converters on a common DC bus. A generalized UPFC (GUPFC) adds a shunt converter to an IPFC: the shunt converter regulates the bus voltage and supplies the net real-power imbalance of the series converters,

so that each line can be controlled fully. A generalized IPFC with n series converters satisfies the single steady-state constraint

$$P_{sh} + \sum_{k=1}^n P_{k,pq} + P_{loss} - P_{storage,in} = 0 \quad (8.12)$$

Here $P_{k,pq}$ and P_{sh} are positive from their converters to the AC network, $P_{storage,in}$ is positive from storage into the DC bus, and $P_{loss} \geq 0$. Thus Equation (8.12) states $P_{sh} + \sum P_{k,pq} + P_{loss} - P_{storage,in} = 0$ in steady state. If drawn shunt power is useful, define $P_{sh,in} = -P_{sh}$ explicitly. An n -series pure IPFC has at most $2n - 1$ independently assignable series-voltage variables after its single average DC balance; current, energy and modulation limits further reduce capability. A GUPFC adds a controllable real-power port, not unlimited energy.

Example 8.8. IPFC DC-link voltage drift

A 10 mF DC capacitor operates at 15 kV. For 20 ms the algebraic converter-power imbalance removes 3 MW from the capacitor. Neglect losses and find the final DC voltage.

Solution. Initial energy $0.5 \times 0.010 \times 15\,000^2 = 1.125$ MJ. The energy removed is $3 \times 10^6 \times 0.020 = 0.060$ MJ, leaving 1.065 MJ, so $V_{dc2} = \sqrt{2 \times 1.065 \times 10^6 / 0.010} = 14.59$ kV. Even a short mismatch produces a visible DC excursion, which is why dynamic IPFC models must not replace the capacitor by a constant-voltage source.

8.2.4 The convertible static compensator

NYPA's Marcy 345 kV convertible static compensator uses two 100 MVA converters, a 200 MVA shunt transformer and two 100 MVA series transformers. Switching permits STATCOM, SSSC, UPFC and IPFC configurations [11], [12]. The STATCOM stage was operating in 2001; the complete CSC was finished in July 2004 [13]. Reconfiguration requires a specified switching and protection sequence; functional convertibility does not mean arbitrary instantaneous changes between energized topologies.

8.3 Voltage and Phase-Angle Regulators (TCVR and TCPAR)

8.3.1 Principle

A voltage regulator adds an in-phase voltage and changes magnitude. A phase-angle regulator changes phase. Figure 8.16 distinguishes an in-phase regulator, a quadrature booster and an ideal constant-magnitude phase shifter. A quadrature injection $\underline{V}_\sigma = jV_\sigma$ gives $\sigma = \tan^{-1}(V_\sigma/V_s)$ and $|\underline{V}'_s| = \sqrt{V_s^2 + V_\sigma^2}$. For an ideal phase shifter, $\underline{V}_\sigma = V_s(e^{j\sigma} - 1)$; for $\sigma > 0$ its angle is $(\pi + \sigma)/2$, whereas a negative shift uses the reflected chord. The injection **magnitude** is

$$V_\sigma = 2V_s |\sin(\sigma/2)| \quad (8.13)$$

Geometrically, the injection is the chord between two equal-radius phasors separated by σ .

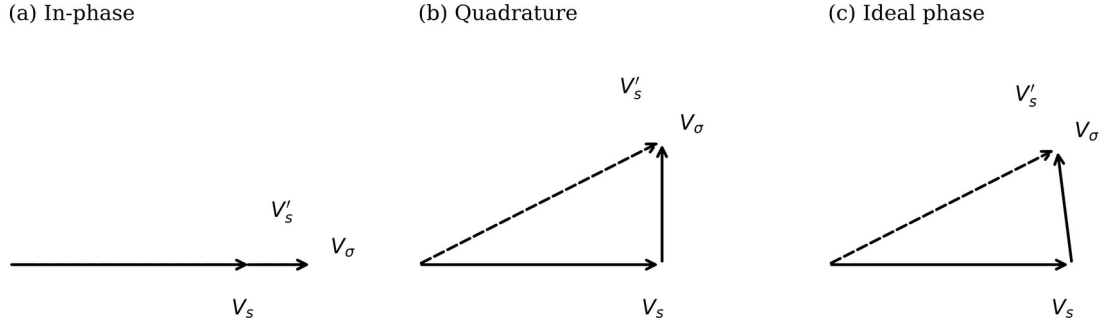


Figure 8.16. Voltage-regulator phasors: bus-aligned injection, quadrature booster with changed magnitude, and ideal phase shift with unchanged magnitude. Dashed arrows are resultant voltages.

With an ideal phase shifter at the sending end of a lossless line, the transmitted power becomes

$$P = \frac{V^2}{X} \sin(\delta \pm \sigma) \quad (8.14)$$

Here σ is the signed phase shift, positive when it increases the sending-to-receiving line angle; equivalently use the boost/buck sign with a nonnegative shift magnitude. An ideal constant-magnitude phase shifter shifts the power-angle curve without changing its peak at fixed voltages and reactance. It sets loop flow in a meshed network. During acceleration, additional electrical power can reduce the accelerating area; after fault clearance it can increase decelerating power when the operating point and shift permit. Benefit is constrained by converter or tap limits and the resulting trajectory. Damping modulation must have the required speed-phase component rather than follow angle deviation indiscriminately.

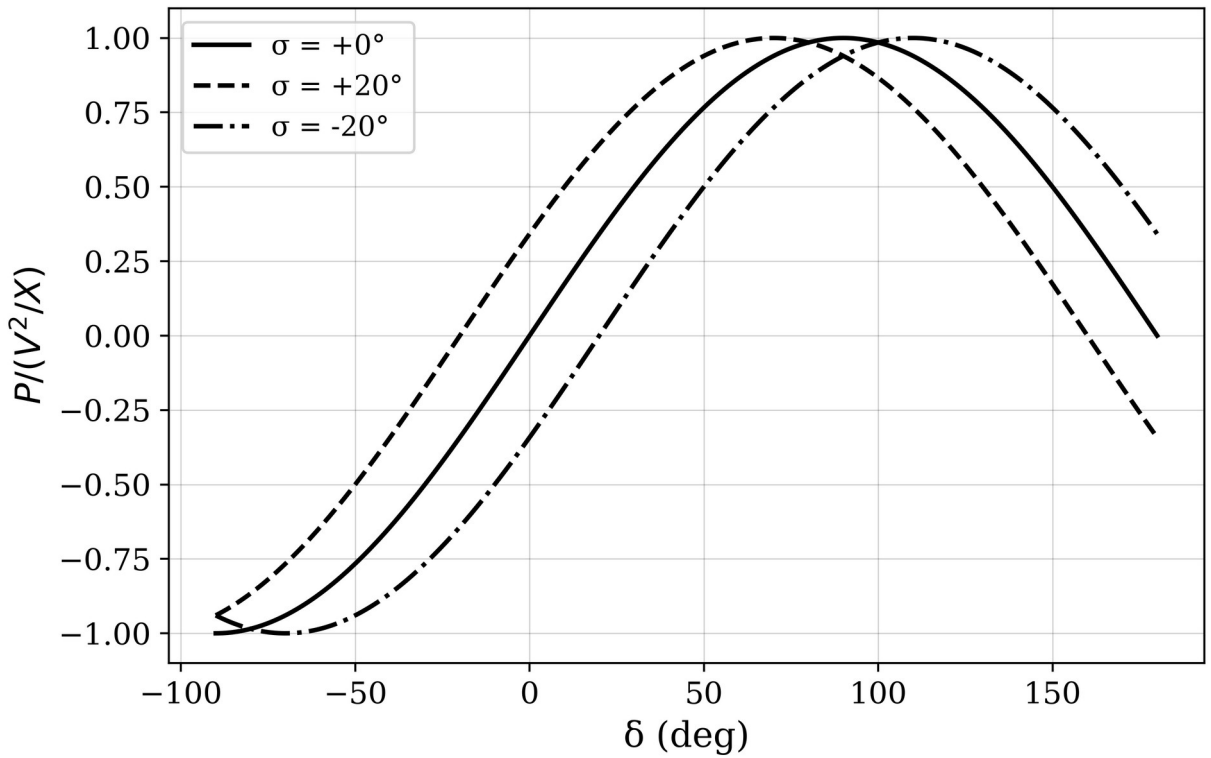


Figure 8.17. Power-angle curves with a phase-angle regulator: the curve shifts sideways by σ without change of peak.

A quadrature booster on a lossless line delivers $P = (V_{s,eff} V_r / X) \sin(\delta + \sigma)$ with $V_{s,eff} = \sqrt{V_s^2 + V_\sigma^2}$, so it slightly raises the reactive power flow as well. For small V_σ/V_s the magnitude error is second order, which is why the approximation is acceptable at modest angles, but the difference matters for rating and reactive-power calculations. In meshed networks with $X \gg R$, an in-phase regulator mainly controls reactive loop flow and a quadrature regulator mainly controls active loop flow.

Example 8.9. Quadrature-booster magnitude error

A 1.0 pu bus receives a quadrature injection of 0.20 pu. Find the resulting voltage magnitude and phase shift, and the injection an ideal phase shifter needs for the same angle.

Solution. $|V'| = \sqrt{1 + 0.2^2} = 1.0198$ pu and $\sigma = \tan^{-1} 0.2 = 11.31^\circ$. The shift is accompanied by a 1.98 % magnitude rise. An ideal phase shifter needs $2 \sin(5.655^\circ) = 0.197$ pu at an angle of 95.7° to V_s and leaves the magnitude at 1.0 pu.

Transformer arrangements and rating. A single-core PST combines excitation and series functions magnetically; a two-core PST has a separate exciting transformer and series transformer. A symmetric arrangement can approximate equal input/output voltage magnitudes, while an asymmetric or quadrature arrangement generally changes magnitude as well as angle. A tapped exciting winding and reversing switch select boost/buck injection derived from other phases. The line-side series winding carries full line current. Its injection duty is approximately $3 V_{inj,ph} I$, but line insulation, leakage reactance, winding loading and fault withstand must also be specified. A mechanical OLTC regulates dispatch slowly; thyristor or hybrid transfer adds valve losses, commutation logic and protection. Constant-magnitude phasor models do not reproduce all these hardware details [3], [6], [14].

8.3.2 Thyristor tap-changers

A mechanical OLTC selects transformer taps slowly and must maintain load-current continuity while controlling the tap-to-tap circulating current. Thyristor-assisted, fully static and hybrid arrangements replace different parts of that operation. Two ordinary antiparallel SCR tap valves alone do **not** implement arbitrary interior-angle transfer with a resistive load: withdrawing a gate does not turn off a conducting SCR. Figure 8.18 and Example 8.40 instead analyze an **ideal selectable-tap waveform**, $v_o = V_1 \sin \theta$ for $0 \leq \theta < \alpha$, $v_o = V_2 \sin \theta$ for $\alpha \leq \theta < \pi$, with half-wave antisymmetry. It is a harmonic calculation; a realizable circuit needs voltage-driven current transfer with limited overlap, forced commutation or turn-off devices and a verified transfer sequence [14].

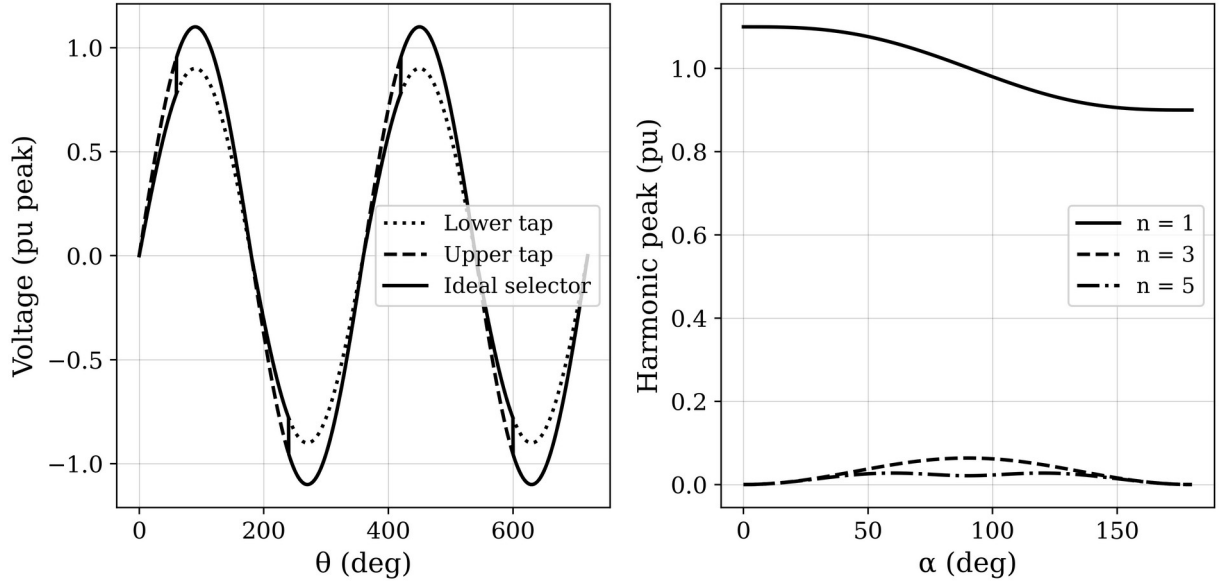


Figure 8.18. Ideal selectable-tap waveform (0.9 and 1.1 pu peak, $\alpha = 60^\circ$) and exact Fourier harmonic magnitudes versus α . This is a mathematical selector model, not a thyristor-circuit simulation.

In a transition-reactor/resistor arrangement the incoming path is selected through a limiting impedance while the outgoing path still carries current. The tap difference drives a circulating component whose permitted polarity must reduce the outgoing valve current to zero. The controller then allows its specified recovery time before applying blocking voltage and completing the transfer. The temporary overlap is **controlled**, not an uncontrolled tap short circuit. The sequence must preserve load-current continuity, restrict di/dt and dv/dt , and include fallback on failed commutation. Tap-up and tap-down sequences depend on current direction, tap-voltage polarity, leakage/transition impedance and SCR recovery. Blocking the outgoing valve before the incoming path conducts would simply interrupt the load current. Natural-current-zero whole-cycle tap selection is simpler but still requires explicit load and transition conditions.

Fourier bridge. With half-wave antisymmetry, the full-cycle coefficient is

$$a_n = \left[1 - (-1)^n \right] \pi^{-1} \int_0^\pi v_o(\theta) \cos(n\theta) d\theta \text{ and } b_n \text{ follows by replacing cosine with sine. Hence all even harmonics and the DC term vanish. For odd } n \text{ only,}$$

harmonics and the DC term vanish. **For odd n only,**

$$a_n = (2/\pi) \left[V_1 \int_0^\alpha \sin \theta \cos(n\theta) d\theta + V_2 \int_\alpha^\pi \sin \theta \cos(n\theta) d\theta \right], \text{ with the corresponding sine integral for } b_n.$$

The peak harmonic is $A_n = \sqrt{a_n^2 + b_n^2}$; write the fundamental as $A_1 \sin(\theta + \phi_1)$, where

$\phi_1 = \tan^{-1}(b_1/a_1)$. Partial THD omits unlisted orders. For this ideal zero-mean waveform, Parseval

$$\text{gives } THD = \sqrt{2 V_{rms}^2 / A_1^2 - 1} \text{ with } V_{rms}^2 = \pi^{-1} \int_0^\pi v_o^2 d\theta. \text{ Actual commutation and magnetic}$$

distortion need their own waveform model.

Whole-cycle steady tap selection avoids repetitive phase chopping, so ideal sinusoidal taps generate no steady harmonics; tap changes still create transients, and magnetic nonlinearity can generate distortion. Independently selectable sections in the ratio 1:3:9:..., each able to add, subtract or bypass its winding voltage, provide balanced-ternary levels. With L such sections the ideal level count is 3^L (Figure 8.19). This is a signed winding-selection law; an ordinary thyristor bridge still needs a safe AC commutation topology and transition sequence, rather than PWM-style gate turn-off.

$$N_{levels}=3^L, \Delta V_{max}=\frac{3^L-1}{2} V_{step} \quad (8.15)$$

where L is the number of winding sections, V_{step} the voltage of the smallest section and ΔV_{max} the largest injected voltage in each direction. Applied to an in-phase winding, this gives a thyristor-controlled voltage regulator (TCVR); applied to a quadrature winding excited from the other two phases, a thyristor-controlled phase-angle regulator (TCPAR).

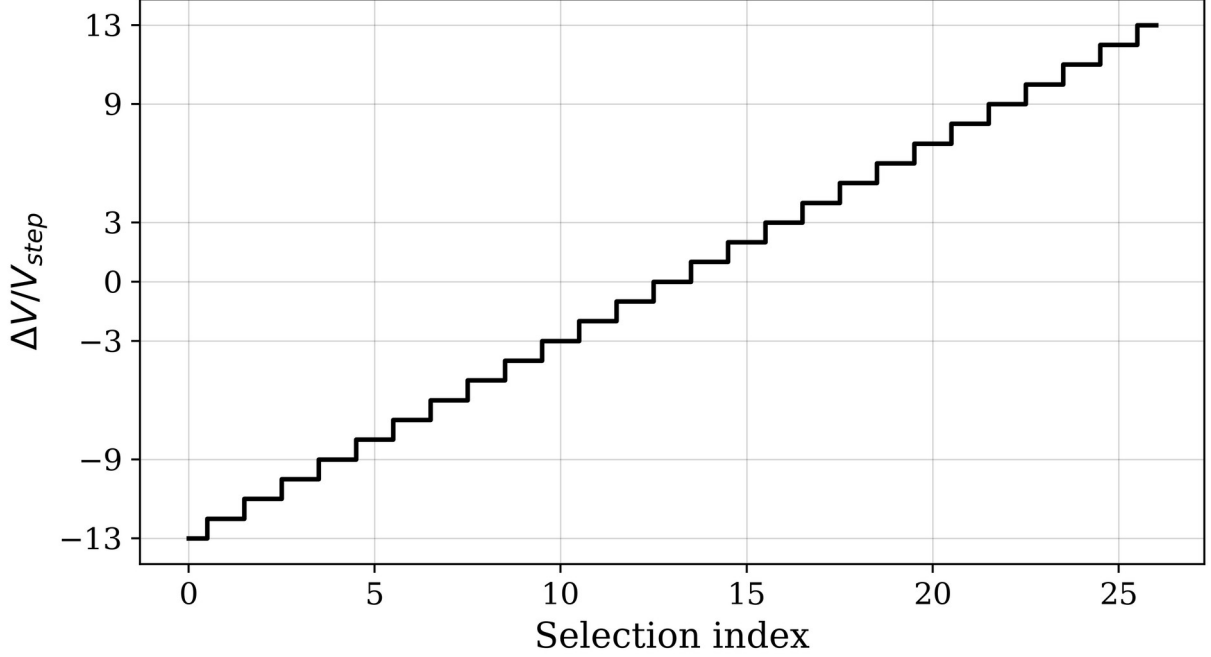


Figure 8.19. All 27 balanced-ternary voltage levels for sections in the ratio 1:3:9, each selected as positive, zero or negative. Selection index orders the levels; it is not time.

Example 8.10. Ternary regulator

A three-section ternary regulator has $V_{step}=4$ kV. Find the section voltages, the number of levels and the maximum positive injection.

Solution. The sections are 4, 12 and 36 kV. The number of levels is $3^3=27$, and the maximum positive injection is $4 \times (27-1)/2=52$ kV. Ternary proportioning obtains many levels with few sections, but insulation, transformer complexity and valve coordination still govern the practical design.

8.3.3 Converter-based regulators

A series VSC used for bus-aligned voltage or constant-magnitude phase regulation generally exchanges real power because its voltage is not constrained to quadrature with current. A shunt converter, storage or another powered port supports it. Mechanically tapped PSTs with small converter trimming can reduce electronic rating while retaining fast modulation. A Sen transformer synthesizes selectable series voltage from tapped windings energized by a shunt transformer [15]. It is an electromagnetic multi-parameter controller; its achievable points are discrete, and dynamic response depends on its tap-transfer implementation. Transformer cost or speed advantages must be compared for a specified project rather than assumed universally.

8.4 The Interphase Power Controller (IPC)

8.4.1 Motivation and concept

The IPC combines passive inductive and capacitive branches fed by deliberately phase-shifted voltages [16], [17]. It can shape a tie's power-angle sensitivity and, in decoupling variants, restrict incremental fault transfer. Its ideal linear fundamental-frequency circuit generates no switching harmonics, but reactors, capacitors and transformers still have losses, insulation and maintenance requirements. Transformer saturation, detuning, capacitor protection and switching transients belong in the equipment model. Figure 8.20 is its one-phase equivalent, not a full three-phase wiring diagram.

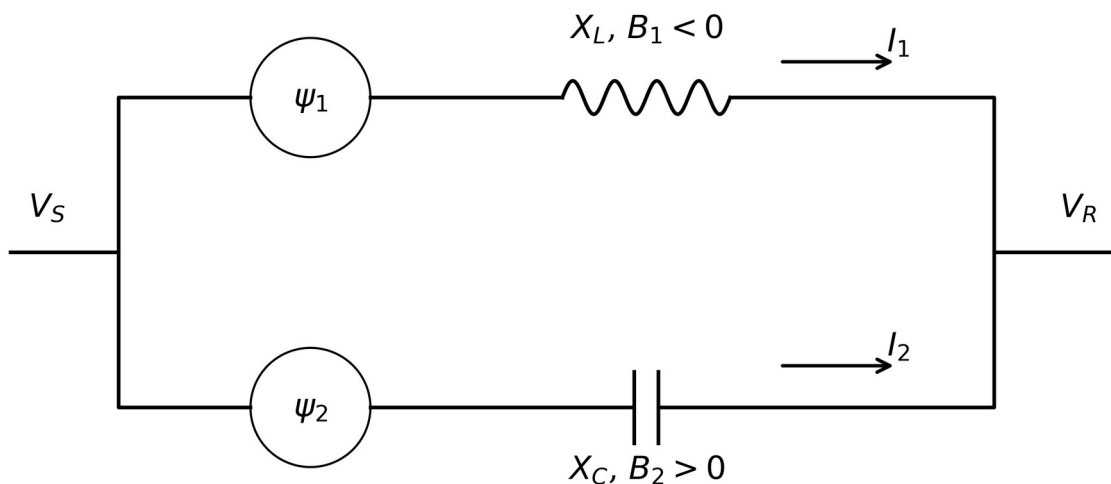


Figure 8.20. Single-phase IPC equivalent with phase-shifted inductive and capacitive parallel branches. Both branch currents are defined toward the receiving terminal.

8.4.2 Power characteristics

Let the branch susceptances be B_1 (inductive, $B_1 = -1/X_L < 0$) and B_2 (capacitive, $B_2 = +1/X_C > 0$). Branch k sees an angle $\delta_{Bk} = \delta_{SR} - \psi_k$, where $\delta_{SR} = \delta_S - \delta_R$. Summing the two branch flows gives

$$P = -V_S V_R (B_1 \sin \delta_{B1} + B_2 \sin \delta_{B2}) \quad (8.16)$$

$$-Q_S = \sum_{k=1,2} (V_S^2 - V_S V_R \cos \delta_{Bk}) B_k, Q_R = \sum_{k=1,2} (V_R^2 - V_S V_R \cos \delta_{Bk}) B_k \quad (8.17)$$

where P is the transmitted active power and Q_S and Q_R are the reactive powers at the sending and receiving ends. Subtracting the two reactive equations gives $-Q_S = (B_1 + B_2)(V_S^2 - V_R^2) + Q_R$, so that with $B_1 = -B_2$ the reactive powers at the two ends are equal and opposite whatever the voltages.

In the IPC120 the branches of each phase are connected, through a Y-y6 transformer, to the inverted voltages of the two other phases, which lie at $\psi_1 = -60^\circ$ and $\psi_2 = +60^\circ$ from the phase voltage; the two connection points are thus 120° apart. With $|B_1| = |B_2| = 1/X$ and $V'_S = V_S/n$ for transformer ratio n , Equation (8.16) and Equation (8.17) combine into

$$\begin{aligned} P &= P_{max} \cos(\delta_{SR} - (\psi_1 + \psi_2)/2) \sin((\psi_2 - \psi_1)/2) \\ Q_R &= -P_{max} \sin(\delta_{SR} - (\psi_1 + \psi_2)/2) \sin((\psi_2 - \psi_1)/2) \end{aligned} \quad (8.18)$$

with $P_{max}=2V_S'V_R/X$. For the symmetric IPC120 this is $P=0.866P_{max}\cos\delta_{SR}$. The power is a cosine of the angle and is flat near $\delta_{SR}=0$: over $\pm 25^\circ$ it changes by less than 10 % (Figure 8.21). An ordinary tie line, whose power is a sine of the angle, would swing from strong forward to strong reverse flow over the same range. The IPC also delivers non-zero power at zero angle, because its branches see the phase-shifted voltages.

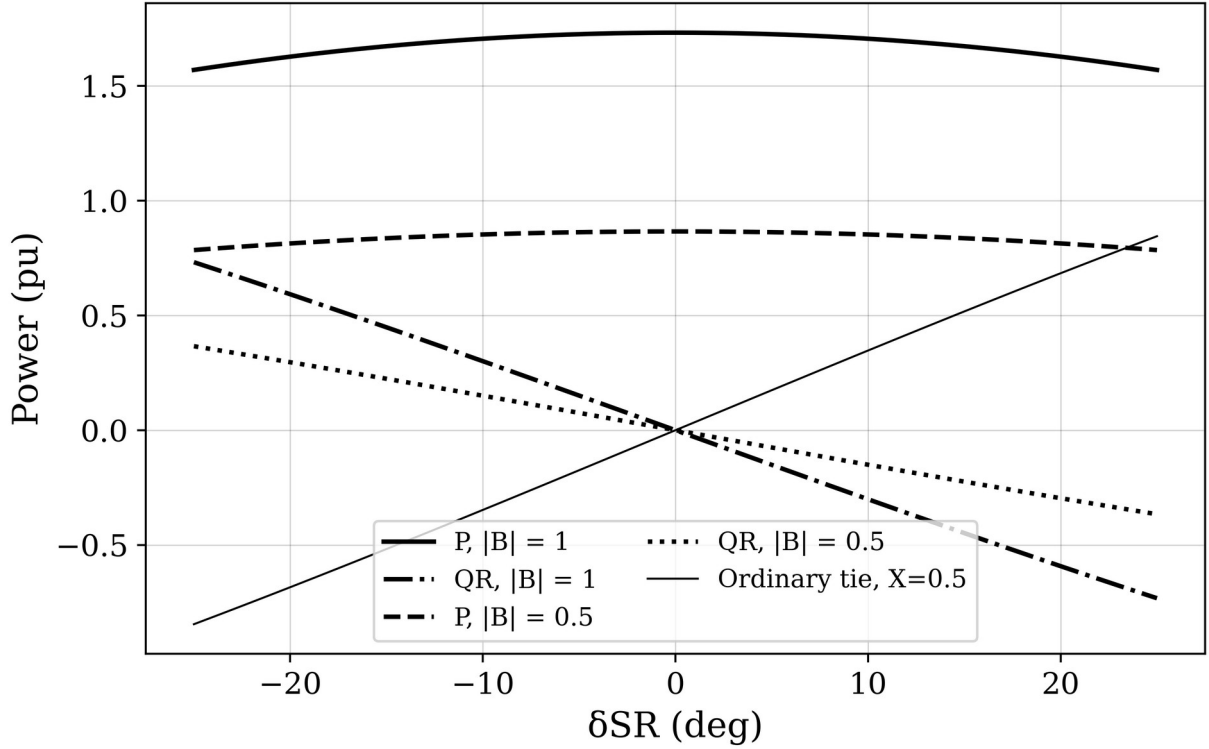


Figure 8.21. IPC120 terminal powers for unit voltages and opposite branch susceptances, compared with an ordinary $X = 0.5$ pu tie. $P = \sqrt{3}|B|\cos\delta_{SR}$ and $QR = -\sqrt{3}|B|\sin\delta_{SR}$, with QR positive into the controller.

The net characteristic is the sum of branch powers: with $B_1=-1$, $B_2=+1$ and shifts $-60^\circ, +60^\circ$, $P_1=\sin(\delta_{SR}+60^\circ)$ and $P_2=\sin(60^\circ-\delta_{SR})$. Their sum is $\sqrt{3}\cos\delta_{SR}$ (Figure 8.22). The slopes are opposite near zero, but each sine is not monotone over an arbitrary interval. Internal reactive circulation and branch currents can be large even when net MW is nearly flat; terminal net power does not determine either branch's rating.

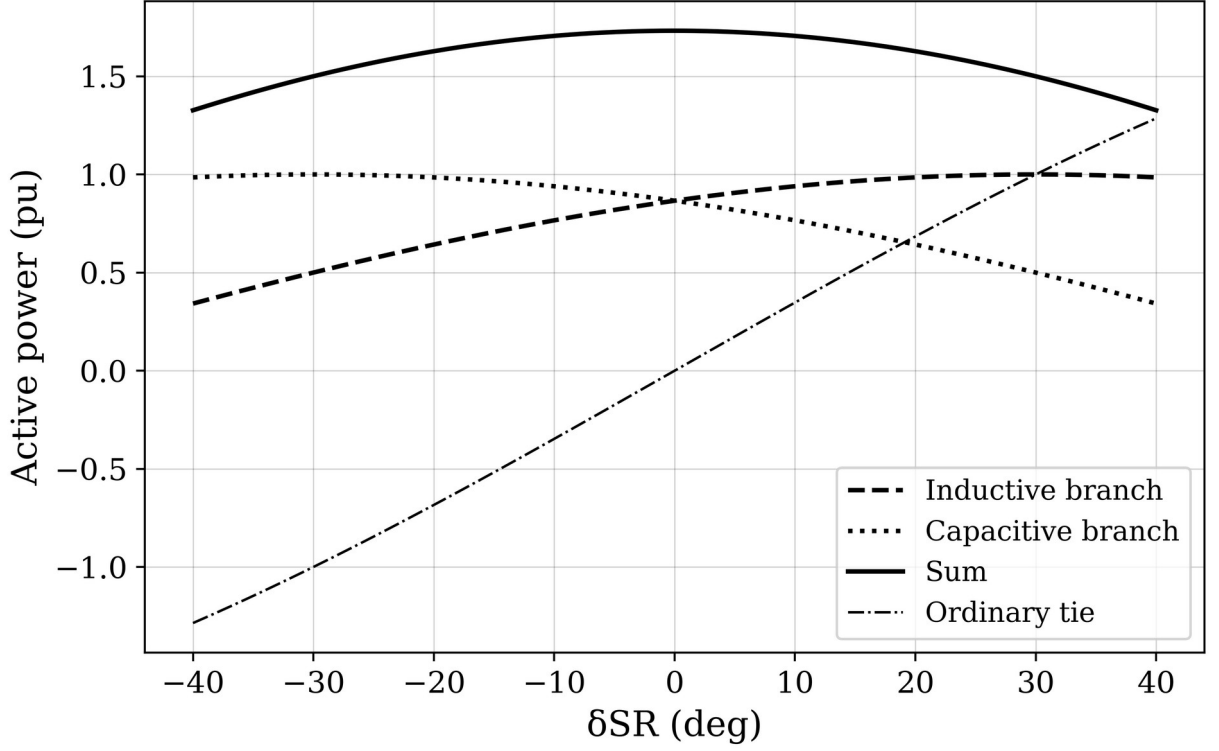


Figure 8.22. IPC120 inductive and capacitive branch active powers and their sum for unit voltages and $|B| = 1$ pu. The ordinary tie curve uses $X = 0.5$ pu.

Example 8.11. IPC branch-current rating

An IPC transfers 300 MW at unity net power factor on a 230 kV three-phase link, but internal branch currents are 1.35 times the line current because of reactive circulation. Estimate the line current and the branch-current rating.

Solution. $I = 300 \times 10^6 / (\sqrt{3} \times 230 \times 10^3) = 753$ A. Each branch should be rated for at least $1.35 \times 753 = 1017$ A before overload margin is added. The net terminal power factor reveals nothing about this internal circulation; branch equipment must be rated from the branch phasors.

8.4.3 Design equations, fault behaviour and variants

For specified P, Q_R , Equations (8.16)–(8.17) are two linear equations in B_1, B_2 . With $d_k = \delta_{SR} - \psi_k$, write $\begin{bmatrix} -V_S V_R \sin d_1 & -V_S V_R \sin d_2 \\ V_R^2 - V_S V_R \cos d_1 & V_R^2 - V_S V_R \cos d_2 \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \end{bmatrix} = \begin{bmatrix} P \\ Q_R \end{bmatrix}$. For the IPC120 shifts, its determinant is proportional to $V_S V_R (V_S - 2V_R \cos \delta_{SR})$. Equations (8.19)–(8.20) require nonzero voltages and $V_S - 2V_R \cos \delta_{SR} \neq 0$. If it vanishes, use the original matrix: inconsistent targets have no solution, while consistent targets need an additional design constraint for a unique solution. The inverse formula is

$$B_1 = \frac{P(2V_R - V_S \cos \delta_{SR} - \sqrt{3}V_S \sin \delta_{SR}) - Q_R V_S (\sqrt{3} \cos \delta_{SR} - \sin \delta_{SR})}{\sqrt{3}V_S V_R (V_S - 2V_R \cos \delta_{SR})} \quad (8.19)$$

$$B_2 = \frac{-P(2V_R - V_S \cos \delta_{SR} + \sqrt{3}V_S \sin \delta_{SR}) - Q_R V_S (\sqrt{3} \cos \delta_{SR} + \sin \delta_{SR})}{\sqrt{3}V_S V_R (V_S - 2V_R \cos \delta_{SR})} \quad (8.20)$$

Substituting Example 8.43, $P = 1.67303$ pu and $Q_R = -0.44829$ pu at 15° , returns $B_1 = -1$, $B_2 = +1$. Fault decoupling must be derived from **both** branch currents. Define $I_k = j B_k (\underline{V}_{Sk} - \underline{V}_R)$ toward the receiving bus. Then $I_R = j B_1 \underline{V}_{S1} + j B_2 \underline{V}_{S2} - j (B_1 + B_2) \underline{V}_R$. When $B_1 + B_2 = 0$, receiving-voltage dependence cancels at the fundamental; neither branch current is thereby zero. A bolted receiving fault leaves $I_R = j B_1 \underline{V}_{S1} + j B_2 \underline{V}_{S2}$, which is generally nonzero for distinct shifted sending sources. Thus decoupling limits incremental fault transfer; it does not in general eliminate the fault contribution. Component stress, unbalanced faults and source-voltage changes require a three-phase transient study.

IPC variants have different objectives. IPC120 uses $\pm 60^\circ$ internal shifts and matched susceptance magnitudes; an injection IPC uses smaller transformer-generated shifts. A decoupling interconnector (DI) is tuned to parallel resonance at the fundamental, and an optimized fault-current-limiting transformer (FCLT) adapts that principle for parallel transformer capacity. Their special fault behavior depends on topology, excitation and protection, and is not a generic zero-current consequence of Equation (8.18). During disturbances resonant component voltages and stored energy require MOV and switching protection. Transmission APST arrangements retain sufficient network coupling for stability. The simplest APST adds a parallel reactive branch to a PST. Plattsburgh's 115 kV APST entered commercial service in June 1998: parallel inductors increased summer transfer from 105 to 140 MW in the reported operating condition [18]. A capacitive parallel branch serves a different boost configuration; the project model decides which branch is useful.

8.4.4 Operating region and retrofitting of PSTs

For a PST leakage reactance X_T , the small-angle approximation is $P \approx V_S V_R (\delta_{SR} - \psi) / X_T$, with angles in radians. A parallel branch contributes $V_S V_R \delta_{SR} / X_p$. The total is $V_S V_R [\delta_{SR} (1/X_T + 1/X_p) - \psi / X_T]$. At unit voltages and $X_p = -X_T$, the ideal slope cancels and $P = -\psi / X_T$ is independent of external angle in this linearized model. An inductor steepens the slope; a capacitor can reduce or cancel it. Current, tap and capacitor/reactor limits still bound the region. Figure 8.23 uses explicit parameters so the matched case is actually flat; partial cancellation has a residual slope.

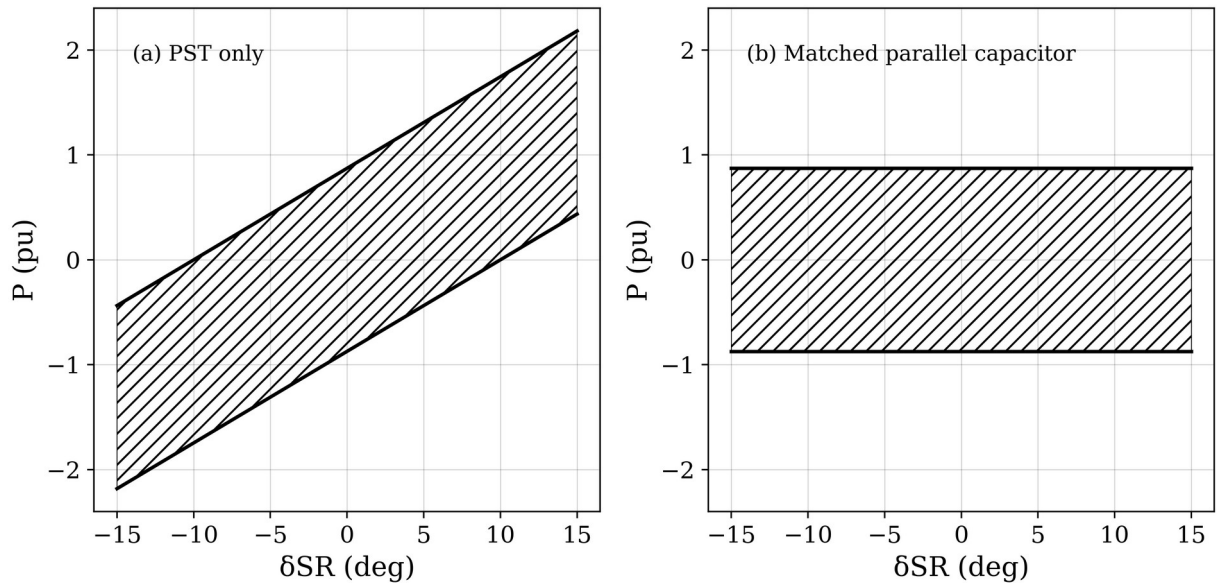


Figure 8.23. Small-angle PST/APST transfer bands: $V_s = V_r = 1$ pu, $X_T = 0.2$ pu, $|\psi| \leq 10^\circ$. Adding matched $X_P = -0.2$ pu cancels the δ_{SR} slope in this ideal model; the hatched bands show the tap range.

8.5 Solid-State Current Limiters, Breakers and Transfer Switches

8.5.1 Fault current limitation

A fault current limiter changes the prospective current trajectory before clearance. Single-shot fuses and pyrotechnic limiters must be replaced; repeatable approaches include switched-impedance, solid-state/hybrid, saturable-core and superconducting devices [19]. A solid-state device's total time to effective limitation includes sensing, decision, commutation and circuit response. It reduces the first peak only if that action is early enough for the actual fault inception and X/R ratio. A quoted operating time of a few milliseconds therefore does not by itself guarantee a lower first peak.

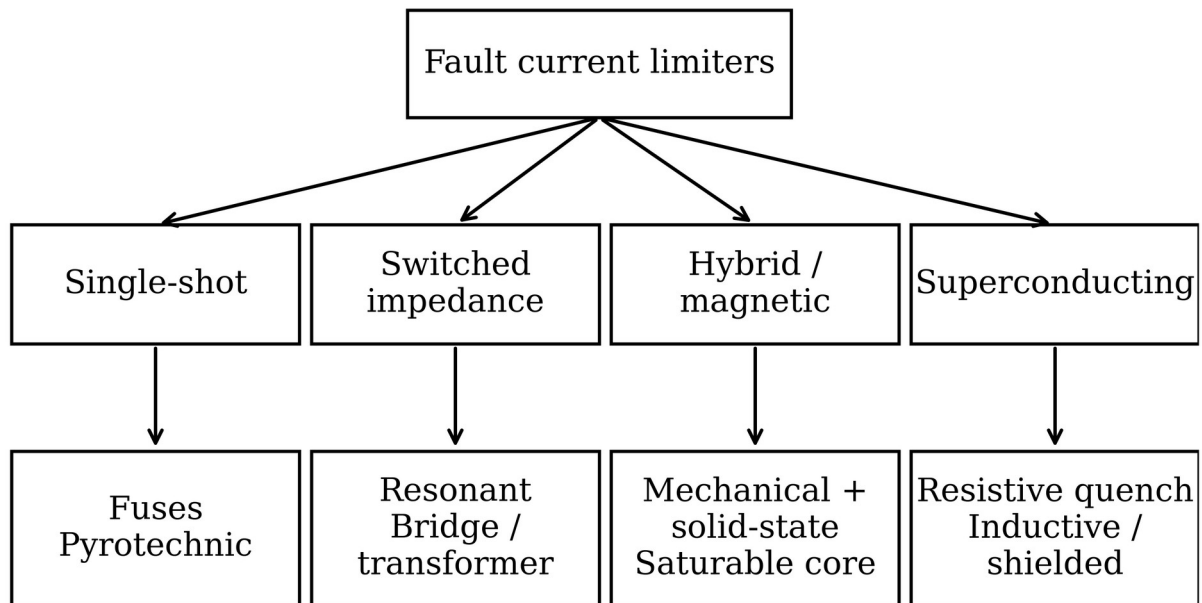


Figure 8.24. Classification of fault-current-limiting technologies. This chapter concentrates on solid-state and hybrid implementations and retains superconducting limiters for comparison.

A good limiter presents very small impedance and loss in normal operation, responds before the prospective current reaches its first peak, inserts enough impedance during the fault, coordinates with relays and breakers, withstands the associated energy and recovery voltage, and returns to service quickly after clearance. These requirements conflict. A large limiting impedance improves current reduction but raises recovery voltage and stored energy; a semiconductor path improves speed but adds conduction loss and thermal stress.

For preliminary design the symmetrical fault current follows from the Thevenin equivalent, $I_f = V_{ph} / |Z_{th} + Z_{lim}|$. For a predominantly inductive source the required limiting reactance is

$$X_{lim} = \frac{V_{ph}}{I_{f,lim}} - X_s \quad (8.21)$$

Equation (8.21) assumes a balanced, predominantly inductive source, nominal voltage and already inserted limiting impedance. With source resistance retained, $X_{lim} = \sqrt{(V_{ph}/I_{f,lim})^2 - R_s^2} - X_s$, provided a nonnegative real value is required. This is a symmetrical RMS screen. The familiar radial-loop peak approximation below uses the final loop R/X , not its pre-insertion value. IEC 60909-0:2016 is the historical basis of that approximation [20]; IEC 60909-0:2026 has superseded it [21],

and a project calculation must apply the current edition, voltage factors and network method. A switched limiter's changing topology additionally requires time-domain calculation.

$$i_p = \kappa \sqrt{2} I_k'', \kappa \approx 1.02 + 0.98 e^{-3R/X} \quad (8.22)$$

so that for $X/R=15$ the peak is $1.82\sqrt{2}=2.58$ times the symmetrical RMS value [20]. A limiter intended to relieve breaker making duty or semiconductor stress must therefore be checked in the time domain.

Example 8.12. Required limiter impedance and peak current

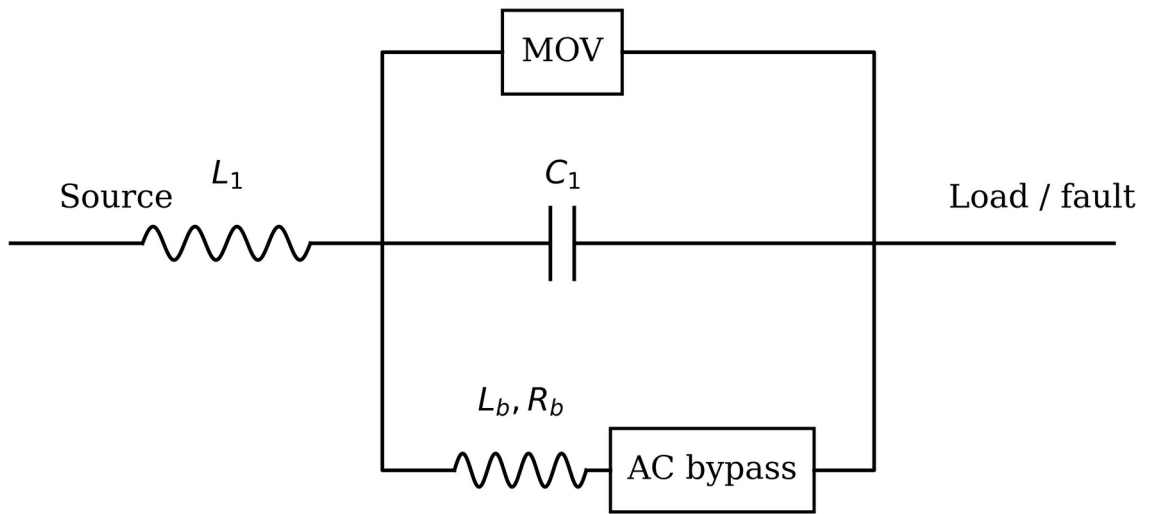
A 33 kV system has a per-phase Thevenin impedance of $0.127 + j1.90 \Omega$ ($X/R=15$). Find the additional reactance that reduces the symmetrical three-phase fault current to 5 kA, and compare the first peaks with and without the limiter.

Solution. $V_{ph} = 33/\sqrt{3} = 19.0526$ kV. Retaining $R_s = 0.127 \Omega$, $X_{lim} = \sqrt{(19.0526/5)^2 - 0.127^2} - 1.90 = 1.9084 \Omega$; neglecting R gives 1.9105Ω . Initially $I_k'' = 19.0526/|0.127 + j1.90| = 10.005$ kA. The radial-loop approximation gives $\kappa = 1.822$ and $i_p = 25.78$ kA. With the final reactance 3.8084Ω , $\kappa = 1.907$ and the fully inserted-loop peak estimate is 13.48 kA. This does not give the first peak of a limiter inserted after a detection delay. Use switching equations for that trajectory, capacitor discharge, MOV energy and breaker recovery voltage.

8.5.2 Series-resonant SSFCL

A series-resonant limiter has $\omega^2 L_1 C_1 = 1$, so its ideal fundamental reactances cancel in normal service. Its inductor and capacitor individually carry substantial voltage and energy; harmonics, detuning and component losses do not cancel. Closing the capacitor bypass on a fault leaves predominantly X_{L1} in series. A discharge-limiting reactor and MOV protect the bypassed capacitor (Figure 8.25). Ordinary SCRs turn on rapidly but release only at current zero; gate withdrawal is not interruption. The explicit switched-circuit illustration in Figure 8.26 shows why source impedance, inception angle, detection delay and capacitor discharge must all be included in a first-peak study.

Reproducible limiter illustration. Figure 8.26 uses an 11 kV line-to-line RMS, 50 Hz balanced source, $v_s = V_m \cos \omega t$, with per-phase peak $V_m = \sqrt{2}(11000)/\sqrt{3}$ V. Parameters are $R_s = 0.12 \Omega$, $L_s = 2$ mH, a 20Ω prefault load, $L_1 = 10$ mH and $C_1 = 1013.21$ μ F. A bolted fault replaces the load at 40 ms; the bypass closes at 43 ms through $L_b = 0.1$ mH and $R_b = 0.1 \Omega$. Integrate $(L_s + L_1)\dot{i} = v_s - (R_s + R_{load})i - v_C$, $C_1 \dot{v}_C = i - i_b$, and, after closure, $L_b \dot{i}_b = v_C - R_b i_b$, with $i_b = 0$ before closure. For initialization define the **complex peak amplitude** $\hat{I} = V_m / (R_s + R_{load} + j\omega L_s)$ and $\hat{V}_C = \hat{I} / (j\omega C_1)$; at $t=0$ use $i = \Re \hat{I}$ and $v_C = \Re \hat{V}_C$. These hats are peak amplitudes, unlike the RMS phasors used in the power-flow sections. The prospective fault reference omits the limiter but retains the same prefault line-current initial condition. The model omits active MOV conduction and breaker clearing; peaks over 40–120 ms are 14.964 kA prospective and 2.440 kA limited.



Bypass closes after detection; capacitor discharge is limited

Figure 8.25. Series-resonant fault-current limiter equivalent. The controlled AC bypass and discharge impedance L_b, R_b are across C_1 , in parallel with its MOV. The MOV is a protection path, not included as active in the illustrative model of Figure 8.26.

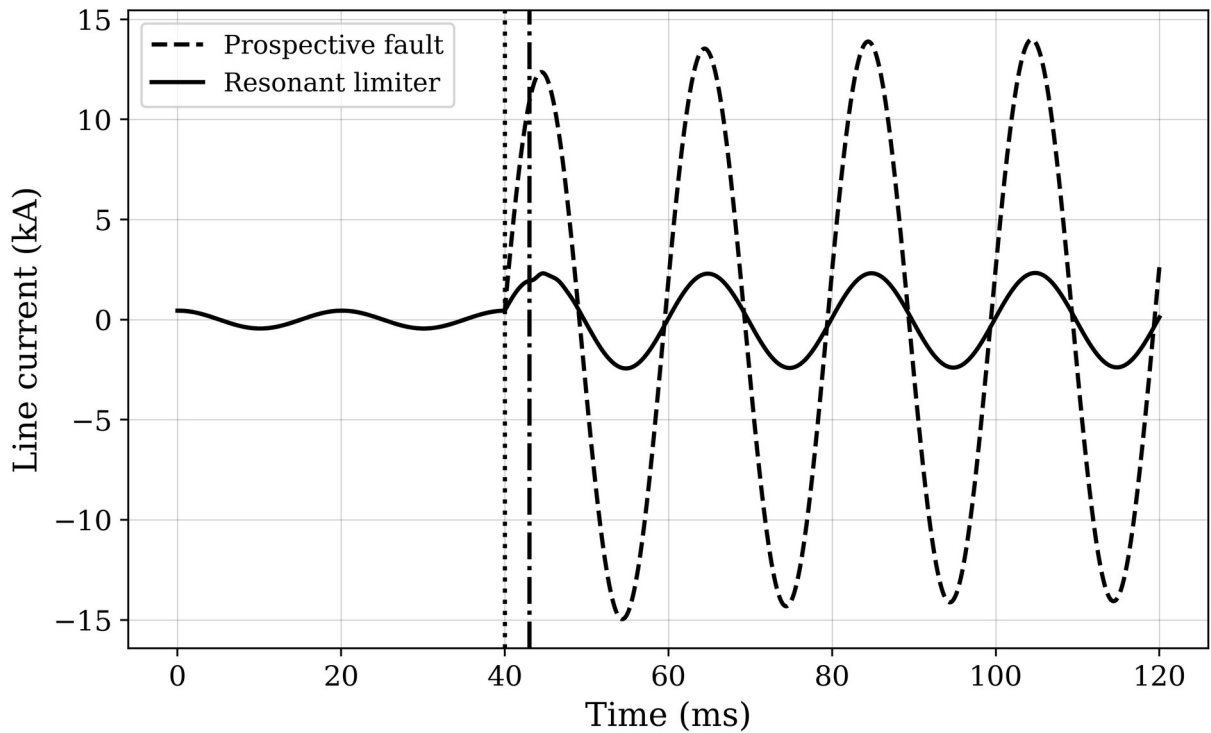
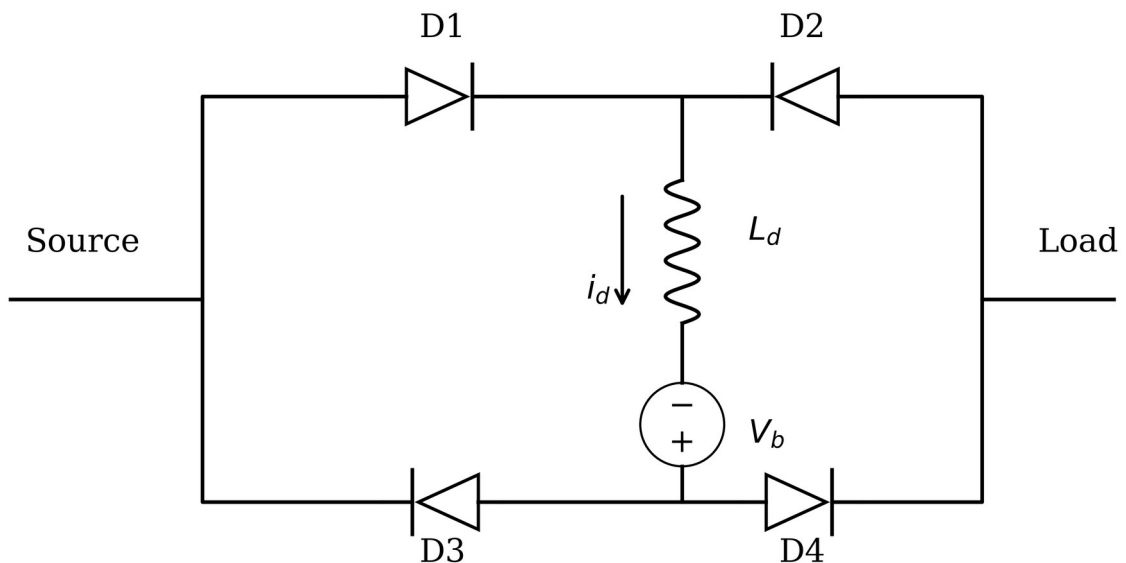


Figure 8.26. Reproducible switched-circuit illustration with the equations and parameters stated in Section 8.5.2: fault at 40 ms and capacitor bypass at 43 ms. Limited and prospective peaks over 40–120 ms are approximately 2.440 and 14.964 kA. These are model outputs, not equipment test results.

8.5.3 Bridge-type and transformer-coupled limiters

Figure 8.27 shows a complete one-phase biased rectifier bridge in series with the AC circuit. The DC reactor current i_d is positive. When $|i| < i_d$, simultaneous conduction in four ideal arms can divide their currents as $(i_d + i)/2$ and $(i_d - i)/2$; both are nonnegative and the ideal AC-port voltage is zero. At $|i| = i_d$, one pair extinguishes and a diagonal pair carries the fault current through the DC reactor, limiting its rise. For an unbiased unidirectional bridge the reactor is not automatically precharged above load peak; start-up and bias supply must be provided. Figure 8.28 is an explicit ideal complementarity illustration. Reactor current can increase from rectified fault energy, so limiting action still needs coordinated clearance. Ordinary thyristors cannot interrupt a nonzero reactor current merely by gate blocking; a current-zero, forced-commutation or turn-off-device path and energy discharge are needed [3], [22].

Bridge waveform model. Figure 8.28 uses an ideal initial reactor bias $i_d = 1.5$ kA. A balanced 11 kV line-to-line RMS, 50 Hz source has $v_s = V_m \cos \omega t$, $V_m = \sqrt{2}(11\,000)/\sqrt{3}$ V, $R_s = 0.2 \Omega$, $L_s = 2$ mH and $L_d = 20$ mH. The prefault resistive load is 8.75946Ω , giving a 1.0 kA peak line current lagging the source by 4.01153° . A bolted fault removes that load at 40 ms. While $|i| < i_d$, $L_s \dot{i} = v_s - R_s i$ and $\dot{i}_d = 0$. At an outward current bound, $i = \text{sgn}(i) i_d$ and $(L_s + L_d) \dot{i}_d = \text{sgn}(i) v_s - R_s i_d$, until the current can return inside the bound. Reactor freewheeling is ideal; no bias-source power is added during the fault. The first-order complementarity integration uses $2 \mu\text{s}$ steps; halving the step changes the computed maximum current by less than 0.01%. Arm losses, bias-supply dynamics, clearance and energy discharge must be added for design.



AC terminals: left and right; DC terminals: top and bottom

Bias maintains initial reactor current; fault energy raises it

Figure 8.27. DC-biased four-diode bridge reactor limiter, one-phase equivalent. The bias polarity supports downward reactor current; the AC terminals are left/right and DC terminals top/bottom. SCR and switched-reactor variants require their own gating and protection.

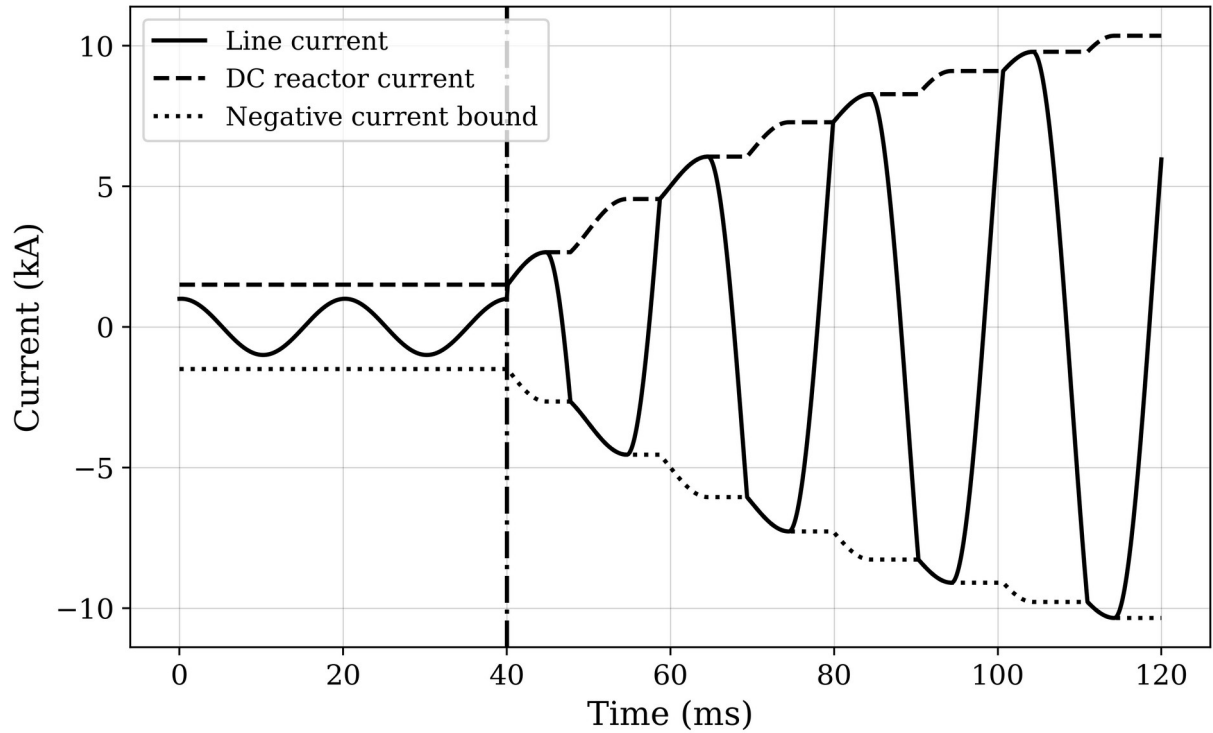


Figure 8.28. Ideal biased-bridge model specified in Section 8.5.3: initial $i_d = 1.5$ kA, prefault line peak 1 kA with an 8.75946Ω resistive load, fault at 40 ms, $R_s = 0.2 \Omega$, $L_s = 2$ mH, $L_d = 20$ mH, integration step $2 \mu s$. Bounds $\pm i_d$ follow the four-diode conduction constraint; this is an illustration without breaker clearance.

A transformer-coupled limiter places the controlled impedance on the secondary of a series transformer. A low-impedance secondary path gives a low reflected normal-state impedance; opening or commutating that path inserts the selected secondary impedance referred to the primary. If n is the primary-to-secondary turns ratio, a linear secondary impedance reflects approximately as $n^2 Z_{sec}$, with leakage and magnetizing branches included. An open secondary may rely on magnetizing impedance, but saturation and insulation stress prevent treating it as an unlimited linear reactor. A specified bridge/reactor circuit must provide energy freewheeling or absorption and actual valve commutation. Synchronizing two feeders is a transfer-switch duty, not an inherent requirement of every series FCL. Precharge, current transfer, fault limiting and recovery depend on the selected topology.

Regardless of topology, the protection sequence has five recognizable states: normal conduction, fault detection, impedance insertion, fault clearing and recovery (Figure 8.29). Detection must be fast but secure. Instantaneous current, di/dt , voltage depression, differential quantities or combinations can be used. A very low threshold risks operation during transformer energization or motor starting; a high threshold sacrifices the first-peak benefit. Coordination with downstream relays and breakers therefore belongs in the limiter design, not in a later protection review.

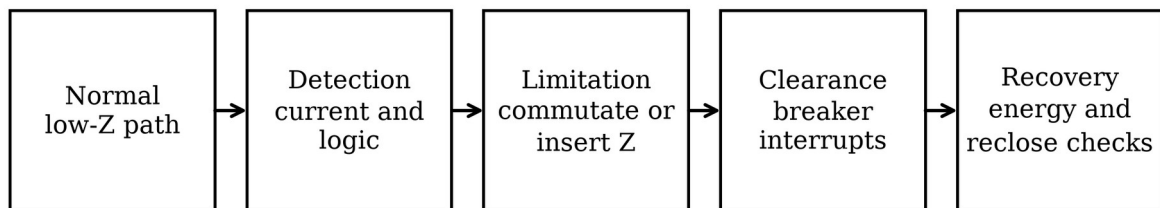


Figure 8.29. Generic operating sequence of a fast solid-state fault current limiter.

8.5.4 Solid-state breakers and transfer switches

Mechanical AC breakers normally separate contacts and extinguish the arc at a suitable current zero after their opening delay; their ability to affect an early peak depends on the breaker technology and operating time. A semiconductor breaker can act much faster with turn-off devices, or SCRs with a separate forced-commutation circuit. It must provide an energy-absorption path and withstand the transient recovery voltage. A blocked semiconductor is not necessarily a visible galvanic isolator: maintenance isolation normally requires appropriate disconnecting equipment. A breaker may combine limiting and interruption, but its interruption rating must be verified independently.

A limiter restricts current; a breaker interrupts it. A solid-state transfer switch commutates a sensitive load between feeders [23] and must avoid feeding a fault from the alternate source. Transfer time depends on failure detection, current phase, feeder phase difference and SCR recovery, so a quarter-cycle figure is not a general rating. An ordinary SCR switch needs favorable natural or source-driven commutation; a turn-off switch can interrupt more freely but carries greater voltage/energy duty. Hybrid breakers and hybrid transfer switches can use a low-loss mechanical main path plus a semiconductor transfer path, provided the sequence transfers current before mechanical interruption and provides energy absorption. Static devices do not inherently provide maintenance isolation.

8.5.5 Comparison with superconducting limiters

Superconducting limiters include resistive quench devices and inductive/shielded or biased-core arrangements; their trigger mechanism is topology dependent. A resistive device changes from superconducting to a resistive state when its critical operating conditions are exceeded. Cryogenics, thermal recovery, repeated fault duty and quench uniformity are major design constraints. Recovery ranges depend on material, cooling, absorbed energy and reclosing requirements, so no single reset time applies to all designs. Solid-state devices avoid cryogenics but add valve losses unless their main path is bypassed; neither class substitutes for a complete protection study.

8.6 The Thyristor-Controlled Voltage Limiter (TCVL)

8.6.1 Metal-oxide varistors

Zinc-oxide varistors protect equipment against overvoltages because their V-I characteristic is extremely nonlinear:

$$|I|=k|V|^\alpha \Leftrightarrow |V|=C|I|^{1/\alpha} \quad (8.23)$$

In the useful nonlinear conduction region, $I=k|V|^\alpha$ describes current magnitude, with α often of order 20–50 for ZnO. A signed implementation uses $i=k \operatorname{sgn}(v)|v|^\alpha$ so instantaneous power is nonnegative for a passive block. The equivalent magnitude law is $|V|=C|I|^{1/\alpha}$, where C is a material/geometric constant, **not capacitance**. Figure 8.30 illustrates a local power-law range; leakage, knee and very-high-current behavior need actual manufacturer curves. Continuous operating voltage, residual voltage at a specified impulse, charge/energy capability and temporary-overvoltage withstand are distinct ratings [24].

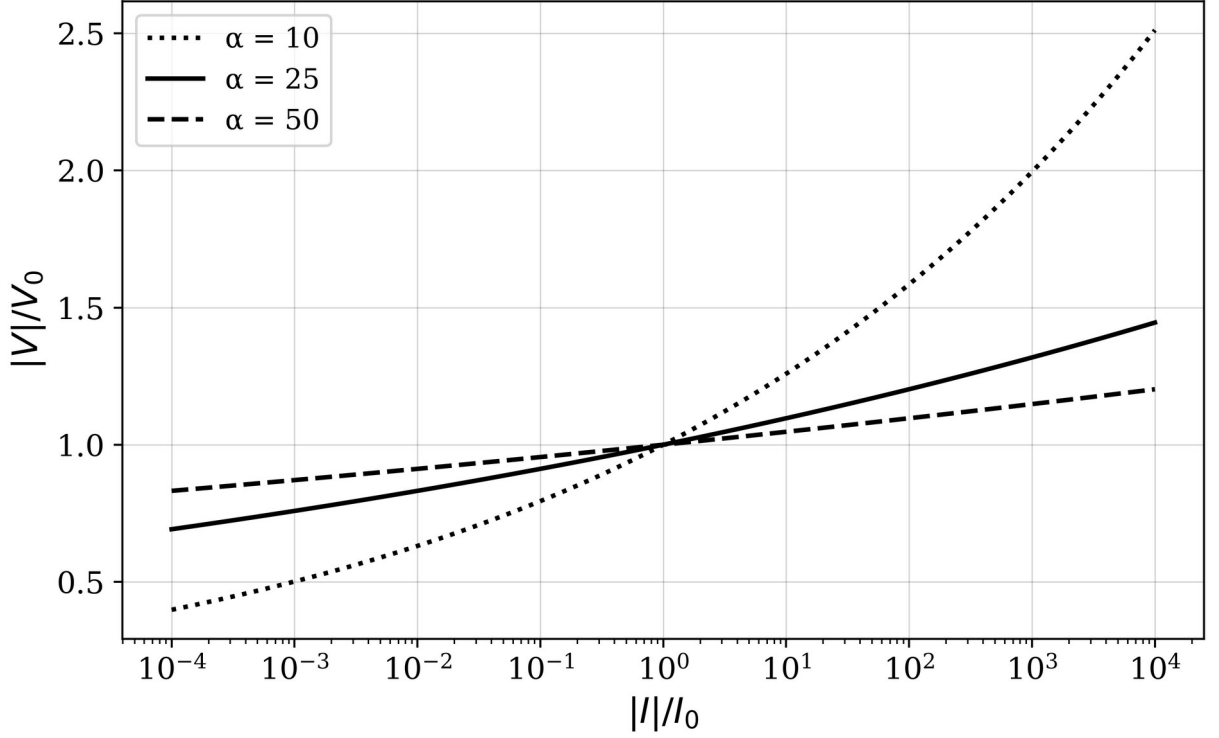


Figure 8.30. Normalized V-I characteristics of metal-oxide varistors for several nonlinearity exponents.

The power law is valuable for understanding but not sufficient for arrester design. Manufacturer data specify residual voltage at standardized current impulses, temporary-overvoltage capability and energy withstand. During a transient the absorbed energy is

$$E = \int_{t_1}^{t_2} v(t) i(t) dt \quad (8.24)$$

Lowering the clamping level reduces insulation stress but generally increases arrester current and absorbed energy, so TCVL design is an insulation-coordination and thermal problem as much as a voltage-threshold problem.

8.6.2 Principle of the TCVL

A gapless MOV stack is selected for continuous and temporary-overvoltage withstand as well as residual voltage at a specified impulse current. The protective level is therefore a design result; a ratio such as 1.7 times the normal peak is an example, not a standard value. The TCVL deliberately bypasses a fraction of an MOV stack during a controlled event (Figure 8.31), leaving fewer discs to limit voltage at a lower effective level. The proportional model below assumes identical discs, the same arrester-current reference and adequate transient voltage sharing; it is not a constant-voltage law for every waveform.

$$V_{lim} = (1 - f_b) V_{clamp} \quad (8.25)$$

where f_b is the fraction of the stack bypassed and V_{clamp} the clamping voltage of the full stack. The thyristors are fired only for the duration of the disturbance, so the full stack withstands normal voltage continuously. Once the current through the bypass reaches a natural zero after gating stops, the thyristors recover and the full stack resumes the continuous-voltage duty. The reduced protective level is used to limit dynamic overvoltages after load rejection, overvoltages during capacitor energization, and the voltage across series capacitors (Figure 8.32). In a three-phase installation each phase has its

own voltage measurement, gate isolation and energy check, so that an unsymmetrical disturbance does not force identical action in all phases.

Protected terminal

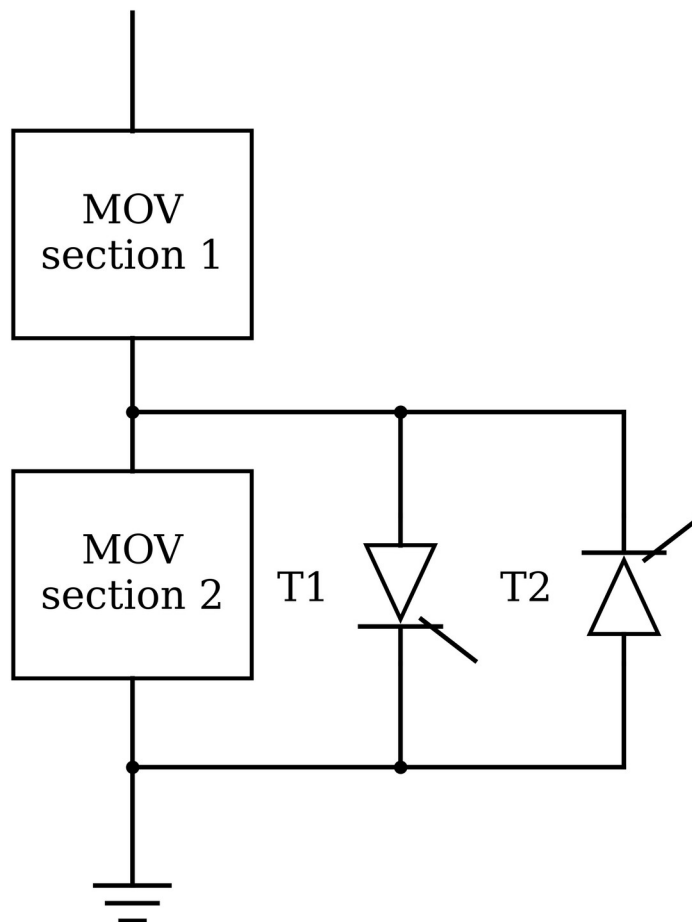


Figure 8.31. Adaptive voltage limiter equivalent: two series MOV sections, with antiparallel SCRs bypassing the lower section when fired. Conducting SCRs release at their current zero, not when the gate is withdrawn.

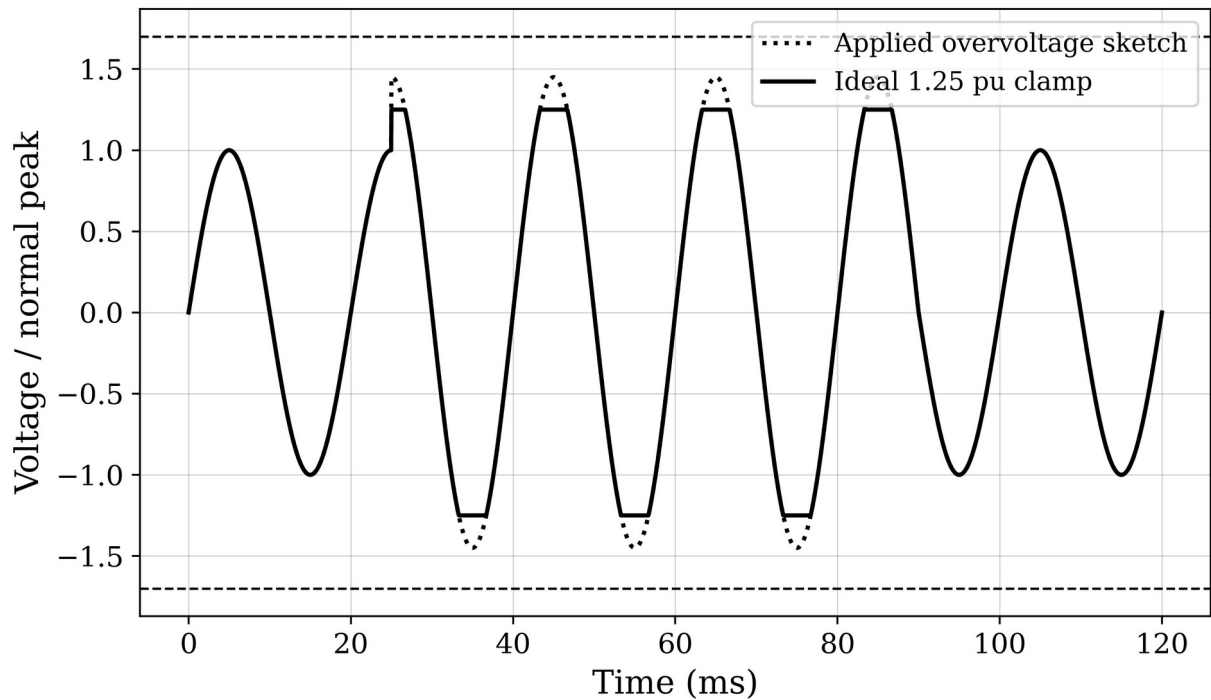


Figure 8.32. Ideal symmetric threshold illustration: applied 1.45 pu peak interval and selected clamp at ± 1.25 pu, with ± 1.7 pu comparison lines. Graphical clipping here illustrates thresholds only; it is not an energy or circuit simulation.

Trigger logic must distinguish a brief surge from an overvoltage event requiring controlled bypass, verify polarity and energy budget, and withhold further firing when the event ends. Bypass thyristors remain conducting until their current reaches zero and recovery is complete. Bypassing discs reduces available energy-bearing material and can cause a large increase in current for the same externally imposed voltage; the remaining discs and valve therefore need separate pulse, thermal and repetitive-duty checks. After the energy limit is approached, the controller must coordinate system protection rather than continue enforcing an unsafe lower threshold.



Figure 8.33. State sequence of a thyristor-controlled voltage limiter.

Example 8.13. MOV energy check

A TCVL clamps a transient at 45 kV while the average arrester current during the main conduction interval is 1.8 kA for 3 ms. Estimate the absorbed energy with a rectangular-pulse approximation.

Solution. $E \approx V_{clamp} I_{avg} \Delta t = 45\,000 \times 1\,800 \times 0.003 = 243 \text{ kJ}$. The rectangular approximation is adequate for a first sizing check; final selection should integrate the actual $v(t)i(t)$ waveform with Equation (8.24) and compare it with the manufacturer's single-event and repetitive energy capability.

Example 8.14. TCVL bypass fraction

A full MOV stack has an effective clamping level of 90 kV. A controlled event requires about 63 kV. Using the proportional stack model of Equation (8.25), estimate the fraction of the stack to bypass.

Solution. $f_b = 1 - 63/90 = 0.30$, so about 30 % of the stack is bypassed. This first-order voltage-sharing model must be confirmed with the actual residual-voltage characteristic, disc tolerances and energy distribution.

8.6.3 Capacitor-switching transients

Energizing a capacitor bank through a mechanical switch forms an LC circuit with the source inductance. The inrush oscillates at the natural frequency

$$f_0 = \frac{1}{2\pi\sqrt{LC}}, Z_0 = \sqrt{\frac{L}{C}}, \hat{I}_{inrush} \approx \frac{\Delta V}{Z_0} \quad (8.26)$$

Equation (8.26) is a step-source, short-interval approximation. For a discharged bank the ideal constant source gives a 2 pu maximum; closing from $-V_m$ against $+V_m$ gives 3 pu. If the supply is sinusoidal and $\omega_0 = 8\omega$, the exact lossless no-limiter response is $v_C/V_m = (64/63)\cos\theta - (127/63)\cos 8\theta$, where $\theta = \omega t$, $v_C(0) = -V_m$ and initial current is zero. Figure 8.34 integrates a capacitor-source model with explicit ideal bidirectional clamps at 1.7 and 1.3 pu. Clamp conduction changes capacitor current and subsequent oscillation; clipping an already computed unprotected voltage does not reproduce that energy balance. Actual MOV laws, source impedance and thyristor delays replace these ideal clamps in a design study.

For Figure 8.34 define $q = v_C/V_m$ and $u = i_L/(\omega C V_m)$. In an unclamped interval, $q' = u$ and $u' = 64(\cos\theta - q)$. At $q = \pm h$ with outward current, the ideal arrester holds q fixed, carries the inductor current and releases when u reaches zero. The energy ratio is $E_{MOV}/(C V_m^2) = \int q u_{MOV} d\theta$. Over the first full supply cycle it is 4.097 at $h=1.7$ and 6.807 at $h=1.3$. Lowering the ideal threshold reduces voltage but increases absorbed energy in this stated circuit. These energy ratios cannot be transferred to another source or real MOV stack without recalculation.

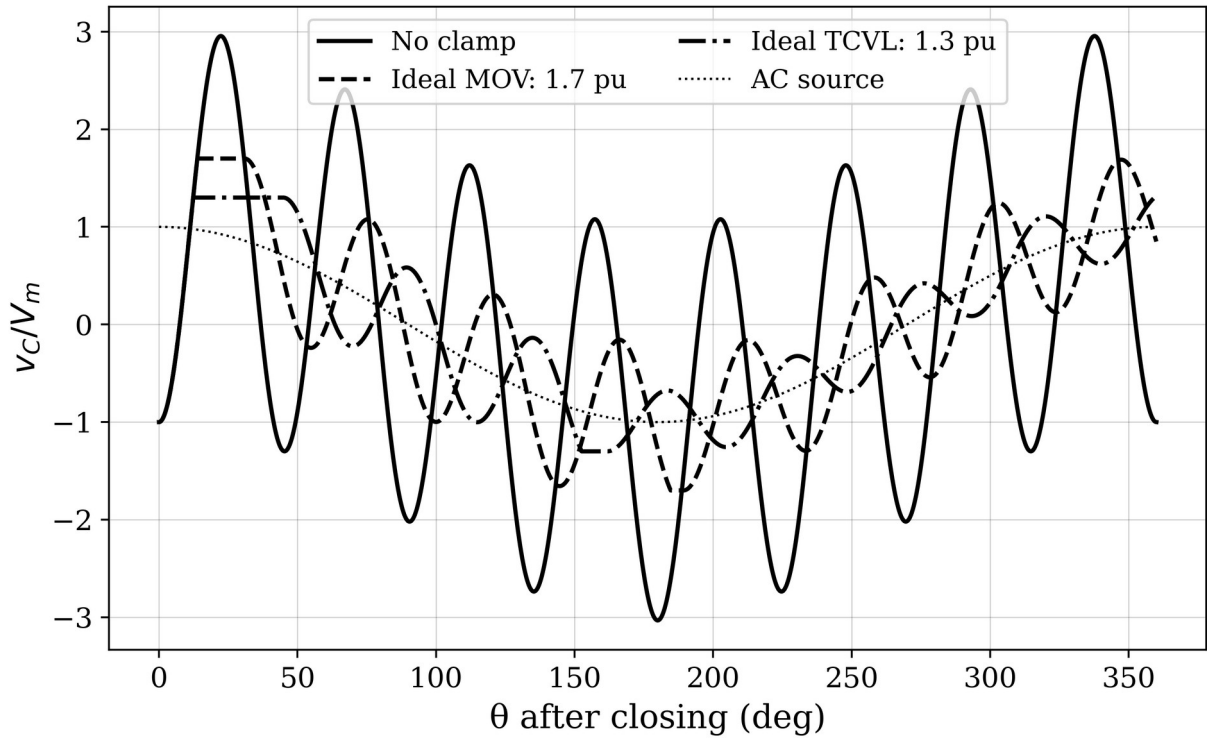


Figure 8.34. Circuit-integrated energization illustration: normalized lossless LC circuit, $\omega_0/\omega = 8$, initial $v_C/V_m = -1$ and zero current. Ideal dissipative clamp states at ± 1.7 or ± 1.3 pu obey the equations in the text; the curves are not graphical clipping or validated MOV hardware predictions.

8.6.4 Protection coordination and limitations

TCVL action must be coordinated with conventional surge arresters, breaker switching, capacitor discharge devices and insulation withstand. Triggering too early increases MOV energy and thyristor duty; triggering too late sacrifices the insulation benefit. The device is most attractive where a temporarily lower protective level is valuable but cannot be tolerated continuously because of leakage current and thermal constraints.

8.7 Comparison and Selection

The technologies in this chapter should not be ranked on a single scale. UPFC and IPFC are power-flow controllers; TCPAR and TCVR are specialized regulators; the IPC is principally a passive network-shaping device; SSCL and SSB are protection devices; the TCVL is an insulation-protection device. Table 8.1 summarizes their attributes.

Table 8.1. Comparison of the controllers of this chapter.

Attribute	UPFC	IPFC / GUPFC	TCPAR / TCVR	IPC / APST	SSCL / SSB	TCVL
Parameters controlled	V, X_{eff}, δ_{eff}	Flows in several lines	Angle or magnitude	Shape of $P-\delta$ curve	Fault current	Overvoltage level

Attribute	UPFC	IPFC / GUPFC	TCPAR / TCVR	IPC / APST	SSCL / SSB	TCVL
Independent P and Q	Locally within the combined voltage, current and energy limits	Prime-line P and Q within supporting power, current and voltage limits	No	No (passive)	Not applicable	Not applicable
Power stage	Two VSCs	n VSCs (plus shunt in GUPFC)	Thyristor tap-changer or VSC	Reactors, capacitors, transformers	Thyristors/ GTOs/IGBTs and reactors	Thyristors and MOV
External real-power port	Yes, shunt VSC	No in pure IPFC; shunt port in GUPFC	Not needed (transformer type)	None	Not a dispatch device	None
Speed	Controller and network dependent	Controller and network dependent	Transfer/commutation dependent	Passive fundamental response	Detection plus insertion/interruption time	Detection, firing and current-zero recovery
Harmonics	Modulation and transformer dependent	Modulation dependent	Phase-control harmonics; ideal steady taps sinusoidal	None in ideal linear model	Bridge, switching and load dependent	Nonlinear clamping during events
Normal-state loss	Converter and transformer loss	Converter loss	Low to moderate	Low	Topology dependent	Very low
Main limitation	Cost, losses, two ratings	DC balance, supporting margin	Range set by windings, valve duty	Fixed characteristic, branch circulation	Energy, recovery voltage	MOV energy, thyristor duty
Best-fit application	Critical corridor needing fast multi-variable control	Several lines at one substation	Fast loop-flow or voltage regulation	Tie with short-circuit constraint	Excess fault level; interruption with a rated breaker	Dynamic insulation coordination

Selection should begin with the constrained state variable, not with the device name: active power, reactive power, voltage magnitude, phase angle, fault current or transient overvoltage. The next layer is the dynamic requirement: steady-state dispatch, electromechanical damping, sub-cycle protection or insulation coordination. Only then should topology, rating, redundancy, loss and cost be compared. A UPFC is the most general controller in this chapter, but it is not automatically the best choice. If the requirement is only slow loop-flow control, a phase-shifting transformer may provide the service with much lower loss and complexity. If several lines at one substation must exchange controllable real

power, the IPFC is structurally better matched. If short-circuit level is the binding constraint, neither a UPFC nor an IPFC substitutes for a properly designed fault current limiter, and if insulation stress from a switching event is the problem, a TCVL addresses a different physical quantity altogether.

A second useful distinction is the energy path. The UPFC has an AC real-power port through its shunt converter. A pure IPFC redistributes real power among series converters but has no external real-power source. The IPC needs no semiconductor DC link. SSCL and TCVL are designed around transient energy and protection duty. Figure 8.35 places the devices on a qualitative map of response speed and steady-state controllability.

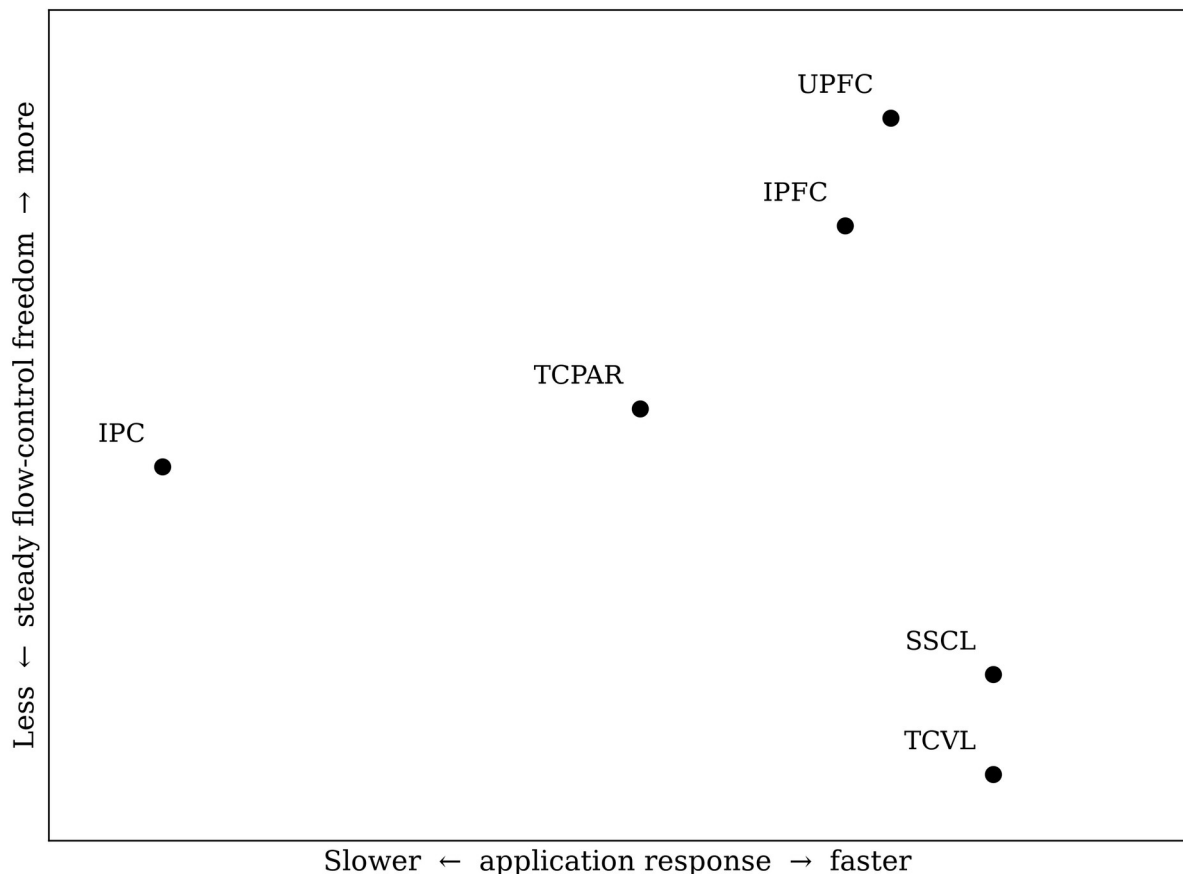


Figure 8.35. Qualitative comparison of typical application roles. Positions do not assign measured speed, cost or universal performance scores; each implementation must be assessed within its ratings and protection requirements.

Example 8.15. Selecting the controller by the binding constraint

A substation has three outgoing lines. One line is overloaded, a second has spare transfer capability, bus voltage is acceptable, and the short-circuit level is already close to the breaker rating. Which controller family is the natural first candidate if the objective is to redistribute steady line flows without increasing bus fault contribution?

Solution. The natural first candidate is a series-only arrangement: an IPFC if controllable real-power exchange between the overloaded and the underloaded line is needed, or independent quadrature SSSC modules if reactance-based redistribution is enough. Neither adds a shunt energy source at the bus, so the controller itself adds little steady fault infeed, although its transformers, control and bypass sequence still alter fault currents and relay measurements. A UPFC could also redistribute the flow, but its shunt converter contributes rating-limited current during faults, which is unwelcome when the

fault level is already close to the breaker rating. Load-flow, contingency and switching/protection studies make the final choice.

8.8 Recent Advances and Research Directions

The research question has moved from whether these controllers can regulate power flow to how they should be built, dispatched, stabilized and protected in networks dominated by power electronics [25]. This section surveys the state of the art and develops each research direction from an equation or constraint already introduced in Sections 8.1 to 8.6, so that a research scholar can see precisely which part of the classical model a new contribution must change. Each subsection follows the same pattern: what has been achieved, the governing relations, the limitations of current approaches, and the open problems worth pursuing.

8.8.1 State of the art: from multi-pulse GTO converters to modular controllers

Inez and Marcy used GTO converters with multi-pulse transformer arrangements [8], [9], [11], [12]. Later designs use modular conversion, distributed injection and alternative energy paths. Losses, redundancy and footprints depend on the actual architecture and loading; loss and footprint comparisons are meaningful only for specific designs.

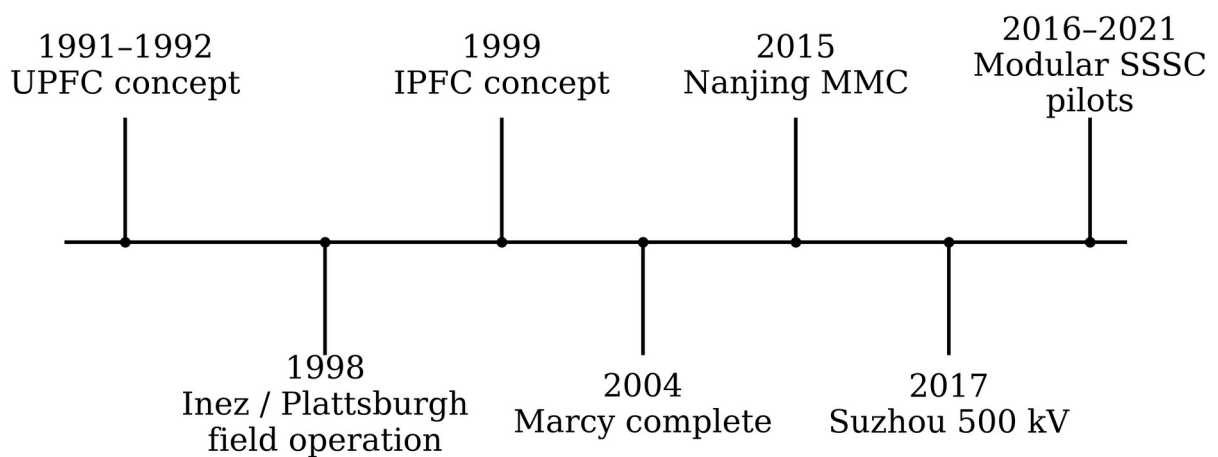


Figure 8.36. Selected concept and field milestones. Concept publications, commercial operation and modular pilots are distinct evidence categories; the timeline is not an exhaustive deployment inventory.

The Nanjing West Ring 220 kV project entered service in 2015 with a shunt MMC connected at 35 kV and two series converters in parallel 220 kV circuits [26]. In the functional terminology used here, this is a GUPFC-type arrangement. Suzhou’s southern-grid 500 kV MMC-UPFC entered operation in December 2017; the project paper describes its highest-voltage and largest-capacity status at that commissioning date [27]. MMCs reduce dependence on multi-pulse magnetic synthesis, but coupling transformers, arm reactors, cell balancing and harmonic/EMC checks remain. “Multilevel” does not automatically mean filter-free or inexpensive redundancy.

Distributed series compensation places many smaller injectors along a corridor rather than one large series converter [28]. Later modular SSSC field projects are supported by operator/project reports,

including EirGrid’s 110 kV pilot and IPTO’s FARCROSS deployment on a 150 kV line [29], [30]. They must be distinguished from conceptual papers [28], [31]. A transformerless cascaded-multilevel UPFC was experimentally demonstrated on a 4.16 kV test setup [32]; this is a demonstration, not a commissioned 4.16 kV utility fleet.

Solid-state UPFC proposals use modular conversion and compact isolated conversion stages to change the energy-transfer architecture [33]. Their study-scale results do not establish transmission-scale operating fleets. Another distinct proposal is the direct-injection controller of [34], reported as a 2025 preprint with analysis, simulation and experimental evidence for low-voltage feeders. It does not establish EHV insulation or fault performance. Recent work should therefore be read by architecture, evidence level and operating domain, rather than by “transformerless” or “solid-state” labels alone.

Important topology distinctions. The transformerless topology in [32] places its shunt converter after the series converter and uses floating cell capacitors; it does not inherit the conventional common-DC-link power balance unchanged. A matrix-converter UPFC transfers energy directly between AC ports through bidirectional switches, replacing the large common DC capacitor but requiring voltage-transfer and safe-commutation constraints [35]. The distributed power-flow controller (DPFC) of Yuan et al. [36] removes the common DC cable and uses a designed third-harmonic AC path for real-power exchange between shunt and distributed series units. It is different from a quadrature-only modular SSSC fleet. Its extra harmonic channel requires transformer/grounding-path and harmonic compatibility studies. These arrangements expand the family; none provides unlimited energy or removes component current and insulation limits.

Table 8.2. Representative field installations and demonstrators discussed in this chapter.

Year	Installation	Voltage level	Technology	Significance
1998	Inez UPFC, AEP, USA	138 kV	Two 160 MVA GTO multi-pulse VSCs	First full-scale UPFC; STATCOM, SSSC and UPFC modes [8], [9]
1998	Plattsburgh APST, NYPA, USA	115 kV	Inductors in parallel with an existing PAR	Commercial APST; June 1998 [18]
2001 to 2004	Marcy CSC, NYPA, USA	345 kV	Two 100 MVA VSCs, one shunt and two series transformers	Convertible modes; completed July 2004 [12], [13]
2015	Nanjing West Ring, China	220 kV	Three MMCs, two series and one shunt	MMC project with two series ports [26]
2017	Southern Suzhou, China	500 kV	MMC-based UPFC	Highest voltage UPFC at commissioning [27]
2016 paper	Transformerless UPFC test system, USA	4.16 kV	Two cascaded multilevel inverters	Experimental test setup, not field fleet [32]
2016 / 2021 pilots	EirGrid / IPTO modular SSSC	110 kV / 150 kV	Line- or substation-mounted single-phase modules	Documented pilots and deployment evidence [29], [30]

8.8.2 Modular multilevel and solid-state UPFC architectures

For a modular series converter with N cells whose fundamental outputs are aligned, the commanded series voltage is the phasor sum

$$V_{se} = \sum_{k=1}^N V_k \quad (8.27)$$

Equal fundamental phasor sharing $V_k = V_{se}/N$ requires comparable cell voltage, modulation and thermal margins. With bypassed cells, Equation (8.28) is a voltage-availability screen. For unequal cells the more general bound is $|V_{se}^i| \leq \sum_{k \in A} V_{k,available}$, where A is the active set. Every series cell must also carry line current, and redundancy does not increase line-current or transformer fault ratings.

$$\frac{|V_{se}^i|}{N - n_b} \leq V_{cell,max} \Rightarrow N \geq \left\lceil \frac{|V_{se}^i|}{V_{cell,max}} \right\rceil + n_b \quad (8.28)$$

Figure 8.37(a) shows how quickly the per-cell requirement rises when the cell count is small. Cell voltage is not the only constraint. Each cell of a series converter carries the full line current, and in a single-phase H-bridge cell the instantaneous power contains a component at twice the line frequency. With RMS cell voltage V , RMS current I and phase angle ϕ between them,

$$p(t) = V I \cos \phi - V I \cos(2\omega t - \phi)$$

With sine-reference RMS waveforms the second term in $p(t)$ is at twice line frequency. Its amplitude is $S_{cell} = V I$ regardless of fundamental power factor. Integration gives energy-ripple amplitude $S_{cell}/(2\omega)$ and **peak-to-peak** energy $\Delta E_{pp} = S_{cell}/\omega$. For small voltage ripple, $\Delta E_{pp} \approx C V_{dc} \Delta V_{pp}$, yielding

$$C_{cell} \geq \frac{S_{cell}}{\omega V_{dc} \Delta V_{pp}} \quad (8.29)$$

This relation explains why single-phase modular series converters are capacitor-dominated in volume and why ripple-cancelling arrangements, such as three-phase cell groups or active power decoupling, are attractive research topics.

Redundancy is usually expressed as availability. If N_r cells are required, r redundant cells are installed and each cell has availability A_c with independent failures, the converter is available whenever no more than r cells are out:

$$A_{sys} = \sum_{k=0}^r \binom{N_r + r}{k} (1 - A_c)^k A_c^{N_r + r - k} \quad (8.30)$$

Figure 8.37(b) uses the explicit independent-cell model of Equation (8.30). Redundancy greatly improves its calculated availability near high cell availability. Common-cause faults, bypass failure, controller failure and cooling/auxiliary failures violate that simple model and can dominate real system unavailability; it is a teaching estimate rather than an installation reliability guarantee.

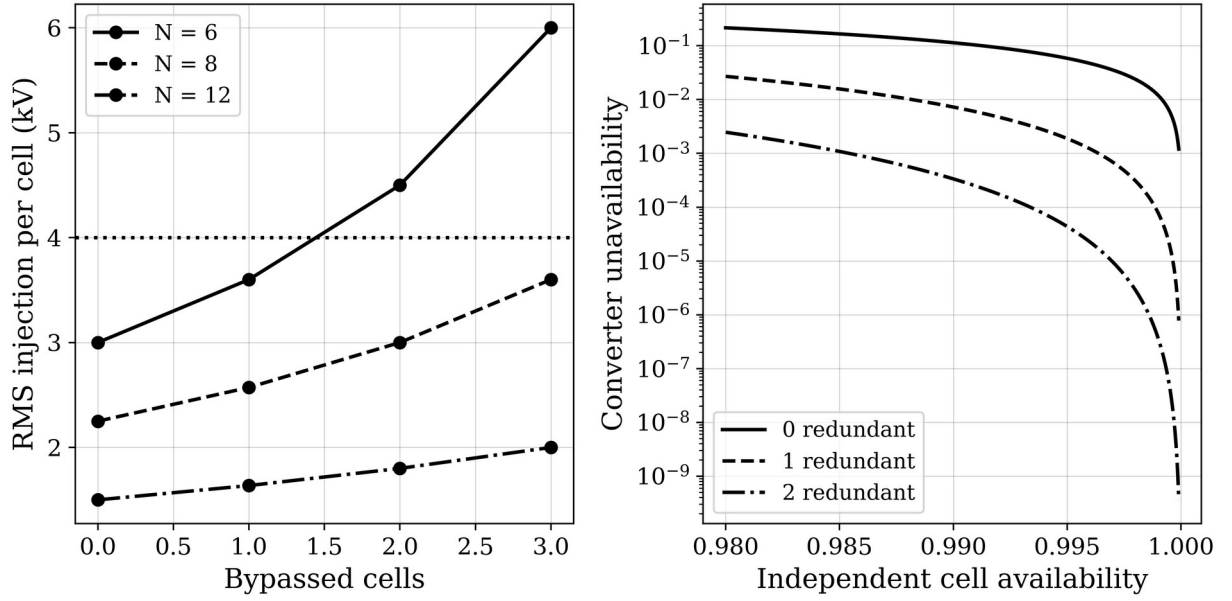


Figure 8.37. Equal-cell voltage sharing as cells are bypassed, and binomial unavailability for 12 required cells with 0, 1 or 2 redundant cells. Independence and identical cell availability are assumptions; common-cause failures are excluded.

Example 8.16. *N-1 voltage margin of a modular UPFC*

A modular series converter contains six identical cells, each rated for 4 kV fundamental output, and the operating point requires 18 kV of injection. Find the per-cell voltage before and after one cell is bypassed, and the minimum cell count for operation with two cells bypassed.

Solution. Normal sharing is $18/6=3.0$ kV per cell. After one bypass the five survivors supply $18/5=3.6$ kV each, leaving 0.4 kV of margin per cell. With two bypassed cells, Equation (8.28) gives $N \geq 18/4+2=6.5$, so seven cells are required.

Example 8.17. *Cell capacitor sizing*

A single-phase H-bridge cell of a modular SSSC injects 1.2 kV RMS into a line carrying 1.0 kA RMS at 50 Hz. Its DC voltage is 2.0 kV and the peak-to-peak ripple must not exceed 10 %. Find the minimum cell capacitance. Then find the converter availability if 12 such cells are required, one redundant cell is fitted and each cell has $A_c=0.995$.

Solution. $S_{cell}=1.2 \times 1.0=1.2$ MVA and $\Delta V_{pp}=200$ V, so Equation (8.29) gives $C \geq 1.2 \times 10^6 / (314.16 \times 2000 \times 200)=9.55$ mF. Without redundancy $A_{sys}=0.995^{12}=0.9416$. With one redundant cell, Equation (8.30) gives $A_{sys}=0.995^{13}+13 \times 0.005 \times 0.995^{12}=0.9981$, so the unavailability falls from 5.8 % to 0.19 %.

Series cells carry line current and must have a rated fast fault bypass. Floating circuits require line-potential insulation and reliable gate power. Medium-frequency isolated architectures add conversion, control and insulation duties, and their transmission-scale efficiency/fault performance cannot be inferred from a reduced prototype. Research on energy-ripple reduction, thermal-aware balancing and bypass-aware reliability should distinguish single-phase series cells, three-phase MMC arm capacitors and isolated DC-DC cells: their energy sharing and redundancy models differ.

8.8.3 Distributed and modular series power-flow control

A quadrature-only module can emulate a series voltage drop $\underline{V}_m=jX_m\mathbf{I}$; positive X_m is inductive and negative is capacitive. In this chapter's voltage-rise notation it is $\underline{V}_{se,m}=-jX_m\mathbf{I}$. Then n equal

modules give $\Delta X = n X_m$. For two lossless, purely reactive paths sharing the same fixed terminal voltages, $P_k = V_s V_r \sin \delta / X_{k,eff}$. Dividing by the sum eliminates the common sine term and gives Equation (8.31). Thus the **active-power sharing ratio is exact** under these assumptions, even when $\sin \delta \approx \delta$ is inaccurate. It does not by itself give line current, reactive power or module voltage. Those require the AC phasors, and the result changes with resistance, unequal terminal buses or additional phase shifts.

$$P_1 = P_T \frac{X_2}{X_1 + X_2 + \Delta X}, P_2 = P_T - P_1 \quad (8.31)$$

Quadrature modules exchange zero average fundamental real power ideally, although practical losses and auxiliaries require supply. They redistribute line flows by changing reactance within voltage/current bounds; they do not independently emulate resistance or arbitrary series phasor injection. Fault bypass changes compensation and apparent relay impedance. Control studies must include low-current synchronization, bypass timing and post-fault restoration rather than assuming a static impedance through the entire event.

Example 8.18. Sizing a fleet of series modules

Two parallel paths of 0.3 pu and 0.6 pu share 2.3 pu (natural sharing 1.533 pu and 0.767 pu). Each series module emulates 0.02 pu of inductive reactance. How many modules must be placed in path 1 to reduce its flow to 1.3 pu or less, and what voltage does each module inject?

Solution. $0.9 + \Delta X \geq 2.3(0.6)/1.3 = 1.06154$, so $\Delta X \geq 0.16154$ pu and nine inductive modules are required. With $\Delta X = 0.18$, $P_1 = 1.27778$ pu and $P_2 = 1.02222$ pu. If $V_s = V_r = 1$ pu and the lower-angle AC branch is selected, $\sin \delta = 2.3/(1/0.48 + 1/0.60) = 0.613333$, so $\delta = 37.8309^\circ$ and $I_1 = 2 \sin(\delta/2)/0.48 = 1.35072$ pu. Each module therefore requires $|V_m| = 0.02 I_1 = 0.0270144$ pu. The frequently used $I_1 \approx P_1$ estimate gives only 0.0255556 pu and underestimates this voltage by about 5.4%. The exact active sharing can coexist with an inaccurate current estimate; final ratings must use current and voltage together.

8.8.4 Coordinated dispatch and optimal placement

Distributed and combined controllers turn a device-design problem into a constrained network-optimization problem. For branch k between buses i and j with series impedance Z_k and a controllable series source $V_{se,k}$,

$$I_k = \frac{V_i + V_{se,k} - V_j}{Z_k} \quad (8.32)$$

and an AC optimal power flow with power-flow controllers can be written as

$$\min_{x,u} F(x,u) \text{ s.t. } g(x,u) = 0, h(x,u) \leq 0, |V_{se,k}| \leq V_{se,k}^{max}, |I_k| \leq I_k^{max} \quad (8.33)$$

Here x contains network voltage variables, u controller/generator commands, g the AC balances and h the operating limits. A pure IPFC adds $\sum_k \Re(\underline{V}_{se,k} \underline{I}_k^*) + P_{loss} = 0$ in steady state; a UPFC adds $P_{sh} + \Re(\underline{V}_{se} \underline{I}^*) + P_{loss} = 0$ and its shunt current limit. Physical per-phase models require the factor 3. Transient DC-energy equations cannot be replaced by these steady balances when studying a disturbance.

The bilinear power term $\Re(\underline{V}_{se} \underline{I}^*)$ makes the problem nonconvex even where the network part is relaxed. Second-order-cone and semidefinite relaxations of the network equations [37], McCormick

envelopes for the bilinear device terms, and mixed-integer formulations for placement are the standard tools; learning-based surrogates are increasingly used to screen many contingencies quickly. A recurring weakness of published work is that the OPF may return an injection inside the ideal control disk that violates the converter current limit in a contingency. A credible formulation enforces converter capability in every post-contingency state, not only in the base case. Security-constrained dispatch of mixed fleets (UPFC, PST and modular SSSC together), stochastic formulations for renewable variability, and exactness conditions for relaxations that include device constraints remain open.

8.8.5 Converter-grid interaction and impedance-based stability

Fast converter controls give a converter port a frequency-dependent terminal impedance. A controller that behaves well in one network can interact with another converter, an HVDC station or a weak or resonant network through the PLL, current controller, DC link and outer loops [38], [39]. Represent the converter port by an impedance $Z_c(s)$ and the network seen from it by $Z_g(s)$. For a current-controlled port the interaction is governed by the minor-loop gain

$$L(s) = \frac{Z_g(s)}{Z_c(s)} = Z_g(s) Y_c(s) \quad (8.34)$$

Equation (8.34) uses the negative-feedback interconnection characteristic $1 + L(s) = 0$ with compatible passive network and converter port signs. Count **right-half-plane poles of the chosen ratio**, even if the underlying isolated controller is described as stable. A magnitude crossing and an angular separation from -180° are local features, not a complete stability test; multiple crossings and unstable admittance poles invalidate a one-number conclusion. The Nyquist contour must provide the encirclements required by the argument principle, including imaginary-axis singularities if present. In three-phase or coupled multi-port studies apply the generalized criterion to

$$\det[I + Z_g(s) Y_c(s)] = 0 \quad (8.35)$$

The matrix determinant must be evaluated with the corresponding open-loop pole count and a well-defined impedance/admittance partition. dq or sequence matrices include off-diagonal frequency coupling. Scalar positive-sequence curves omit such coupling and cannot certify the full UPFC, IPFC or grid-forming converter.

A minimal model that already shows the mechanism is a current-controlled VSC with filter inductance L_f , proportional current gain K_p and a computation-plus-PWM delay T_d :

$$Z_c(j\omega) = j\omega L_f + K_p e^{-j\omega T_d} \quad (8.36)$$

Here $\Re Z_c = K_p \cos(2\pi f T_d)$ is negative in the bands $(4m+1)/(4T_d) < f < (4m+3)/(4T_d)$, $m=0,1,\dots$, and changes sign periodically. The first band for $150 \mu s$ is 1.667–5.000 kHz. Only frequencies inside the sampled-data/averaged model's validity should be used: no sampling rate or switching frequency is supplied here. A negative real part flags nonpassivity; it does not alone prove instability. UPFC ports also couple through finite DC energy, so their separate scalar impedances with an infinitely stiff DC bus conceal interaction modes. The numerical illustration in Figure 8.38 and Example 8.19 uses $L_f = 1 \text{ mH}$, $K_p = 15 \Omega$ and $T_d = 150 \mu s$.

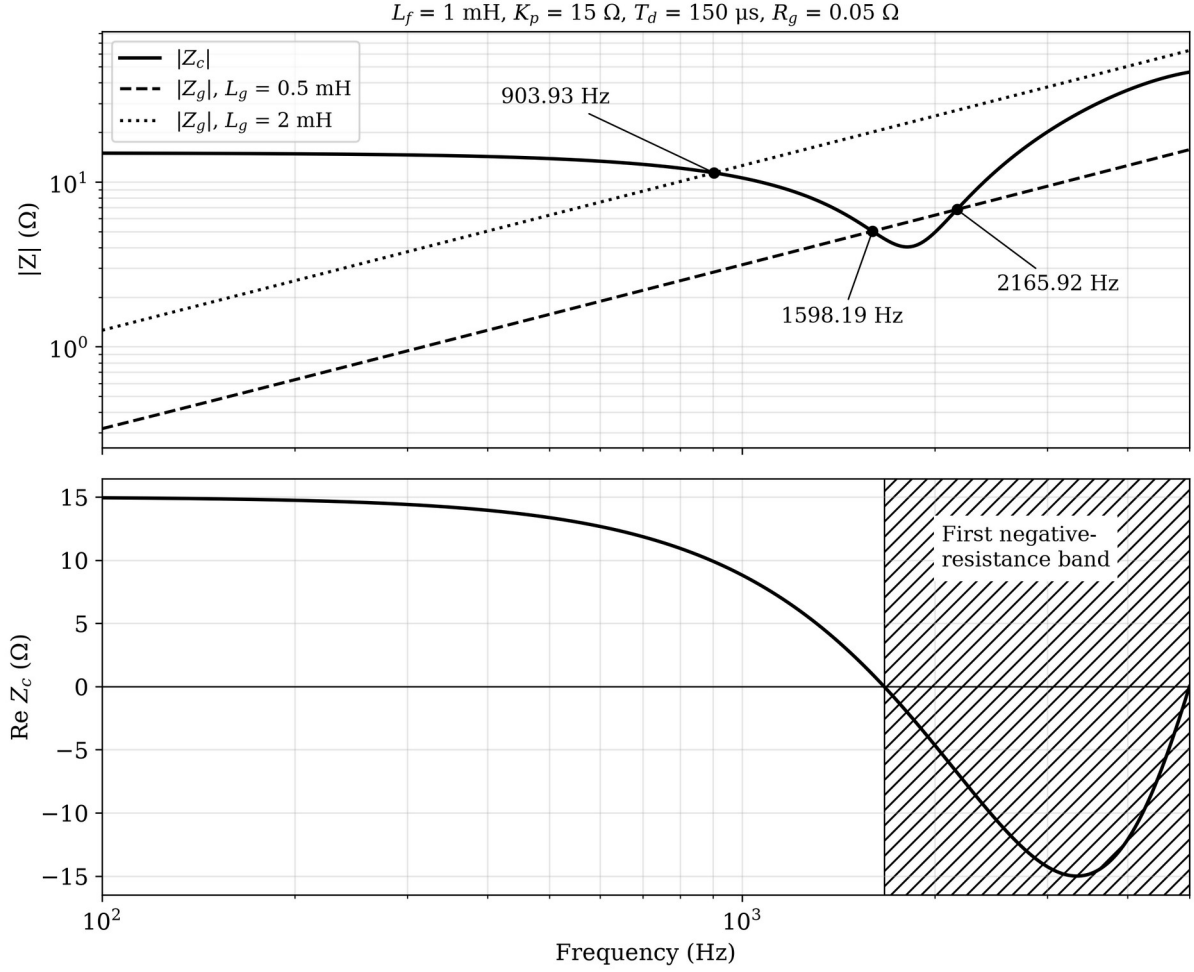


Figure 8.38. Displayed 100 Hz–5 kHz interval of the explicit delayed-impedance model. Gain crossings are 903.93 Hz ($L_g = 2$ mH) and 1598.19 and 2165.92 Hz ($L_g = 0.5$ mH). The first negative-resistance band is 1666.67–5000 Hz, periodically repeated; the 1598.19 Hz crossing is below its onset. Model parameters: $L_f = 1$ mH, $K_p = 15$ Ω, $T_d = 150$ μs, $R_g = 0.05$ Ω. Local crossing angles alone do not establish stability.

Example 8.19. Phase-margin screen

For Figure 8.38, identify the first negative-real-part band, locate **all** magnitude crossings in 0.1–5 kHz for $L_g = 0.5$ mH and 2 mH ($R_g = 0.05$ Ω), and compare the local angles. Then determine the stability of the stated continuous-delay scalar interconnection, rather than inferring it from one crossing.

Solution. The first band is 1666.67–5000 Hz. For $L_g = 2$ mH the crossing is 903.93 Hz with angular separation from -180° of 60.66° . For 0.5 mH there are **two** crossings, 1598.19 and 2165.92 Hz. The first lies below band onset and has separation 11.68° ; the second has 91.59° after using a principal phase. These are local geometrical margins. In fact $Y_c = 1/Z_c$ has a right-half-plane pole pair, so a zero-encirclement stable-open-loop rule is inapplicable. The total characteristic is $(L_f + L_g)s + R_g + K_p e^{-sT_d} = 0$. At a Hopf boundary, $\omega_h = \sqrt{K_p^2 - R_g^2} / (L_f + L_g)$ and $T_{crit} = \cos^{-1}(-R_g/K_p) / \omega_h$. For $K_p > R_g \geq 0$, the scalar retarded loop is stable for $0 \leq T_d < T_{crit}$. The thresholds are 157.41 μs (0.5 mH) and 314.83 μs (2 mH); both exceed the stated 150 μs. Hence **both specified scalar models are stable**, but the stiff case is close to its delay boundary. Its dominant pair is $-228.22 \pm j 10346.42$ s $^{-1}$ (damping ratio 0.0221), versus $-3454.90 \pm j 7658.96$ s $^{-1}$ (0.4112) for 2 mH. These conclusions belong only to this scalar continuous-delay model; a real converter still needs validated sampled-data, multi-port and EMT checks.

Reproducing the pole check. With $L_\Sigma = L_f + L_g$, the roots are $s_k = T_d^{-1} W_k \left[- (K_p T_d / L_\Sigma) e^{R_g T_d / L_\Sigma} \right] - R_g / L_\Sigma$, where W_k is a Lambert-W branch. Checking branches $k = -50, \dots, 50$ reproduces the stated dominant pairs; the delay-threshold derivation supplies the global scalar stability test rather than reliance on a finite root list. For Z_c alone, $K_p T_d / L_f = 2.25 > \pi/2$, giving the admittance's unstable pole pair. This explains why its Nyquist contour must account for two unstable open-loop poles.

8.8.6 Wide-area damping control and communication delay

Supplementary damping loops on a UPFC or IPFC are increasingly driven by remote phasor measurements. The communication and processing delay T_d adds a phase lag

$$\phi_d = -360 f T_d \text{ degrees}, e^{-s T_d} \approx \frac{1 - s T_d / 2}{1 + s T_d / 2} \quad (8.37)$$

at the mode frequency f ; the first-order Padé form on the right is used when the delay must appear in a state-space model. A delay that looks small in milliseconds can consume much of the phase lead designed into a damping controller (Figure 8.39). If the uncompensated lag rotates the damping torque far enough, a loop designed to damp a mode can excite it.

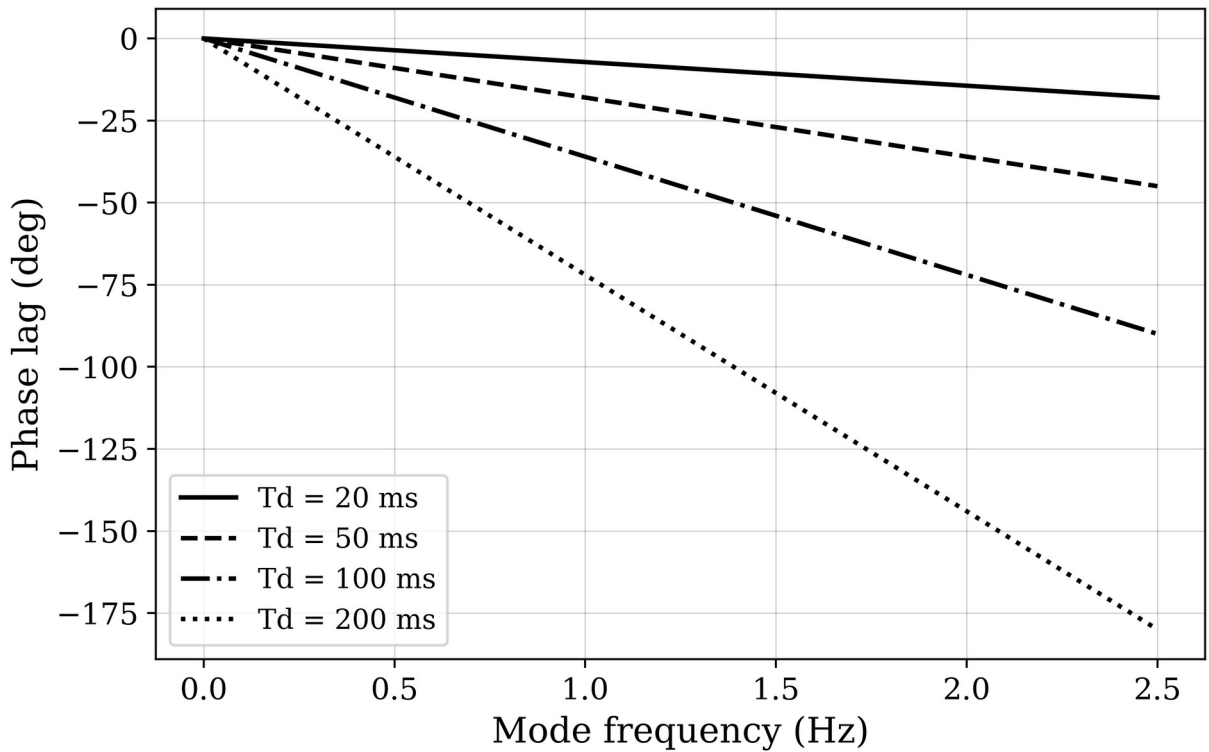


Figure 8.39. Phase lag introduced by end-to-end delay in a wide-area damping loop.

Example 8.20. Communication delay in supplementary damping

A wide-area UPFC damping loop acts on a 0.8 Hz inter-area mode. Find the phase lag caused by delays of 40 ms and 120 ms.

Solution. $\phi_d = -360 \times 0.8 \times 0.040 = -11.5^\circ$ and $-360 \times 0.8 \times 0.120 = -34.6^\circ$. The second value is comparable to the lead normally provided by one compensation stage, so delay must be included in the design and its worst-case value used in robustness checks.

Research directions here include delay-robust and adaptive damping controllers, coordination of several FACTS damping loops so that they do not compete for the same mode, and cyber-physical resilience when measurements are lost or spoofed.

8.8.7 Grid-forming concepts and current-limited operation

A conventional UPFC is synchronized to the network through a PLL and regulates power flow; it does not establish frequency. Grid-forming control is now central to converter-dominated systems, and a combined controller could in principle let its shunt port form a local voltage angle while the series port shapes corridor flow. The difficulty is energy and current coordination. Both ports draw on one DC store and on semiconductor current limits, and during a fault the shunt port's apparent-power capability shrinks with the voltage:

$$P_{sh}^2 + Q_{sh}^2 \leq (V I_{sh,max})^2 \Rightarrow |Q_{sh}| \leq \sqrt{(V I_{sh,max})^2 - P_{sh}^2} \quad (8.38)$$

Once current limiting is active, pre-fault P-Q commands cannot all remain satisfied. For a steady internal source $E e^{j\delta}$ behind physical plus virtual impedance, $|Z_{physical} + Z_v| \geq |E e^{j\delta} - V_g| / I_{max}$. The numerator is a **phasor difference**, not $E - |V_g|$ unless they are aligned. A bolted fault gives E / I_{max} ; a nonzero residual voltage needs its phase as well as magnitude. For $E=1$ pu, $V_g=0.5$ pu and $\delta=60^\circ$, the numerator is 0.866 pu, compared with only 0.5 pu under alignment. This fundamental steady-state sizing cannot guarantee peak transient current. Saturation priority, virtual-impedance dynamics, DC-energy limits and resynchronization must be checked in EMT [40].

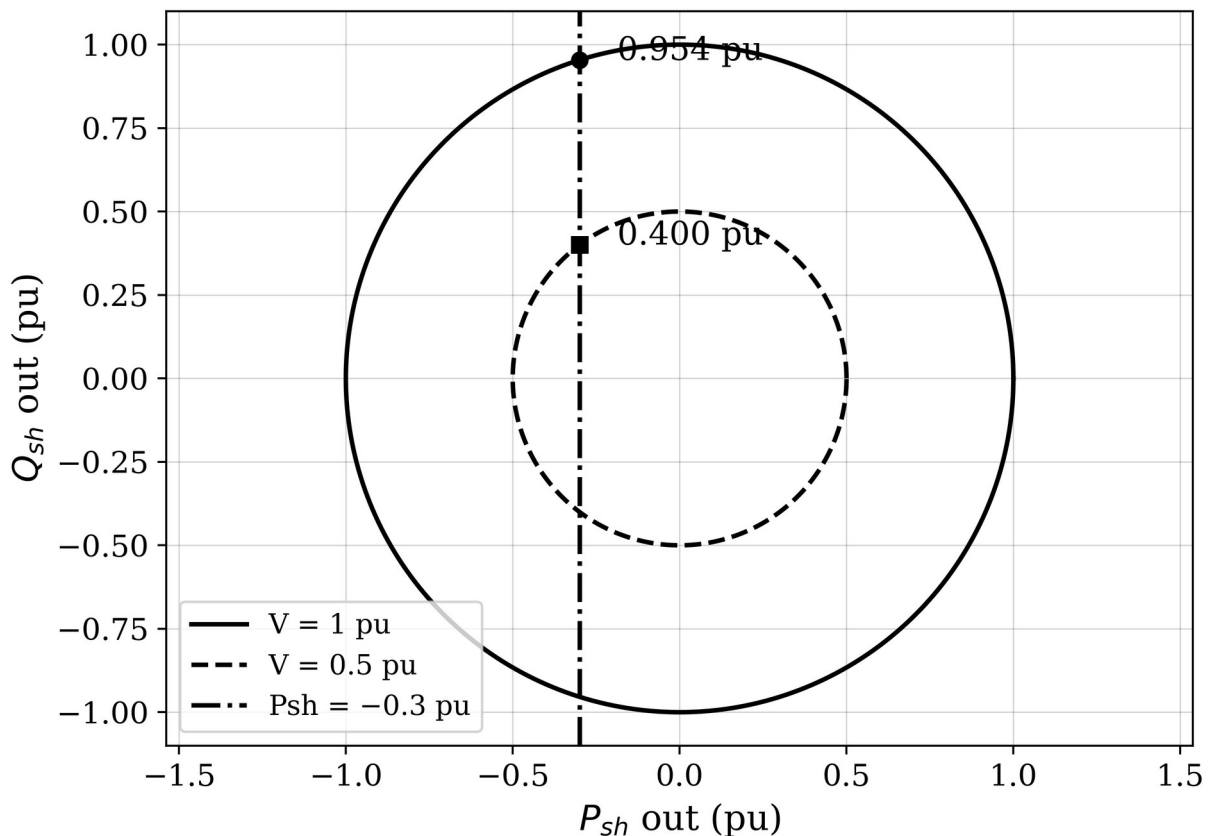


Figure 8.40. Shunt current-limit power circles for $I_{max} = 1$ pu and bus voltages 1 and 0.5 pu. With $P_{sh} = -0.3$ pu, available $|Q_{sh}|$ falls from 0.954 to 0.400 pu. Other thermal and modulation limits can reduce this capability.

Example 8.21. Reactive capability during a voltage dip

A UPFC shunt converter has $I_{sh,max} = 1.0$ pu and must keep drawing 0.3 pu of real power to support the series converter. Find its reactive capability at $V = 1.0$ pu and during a fault that depresses the bus to 0.5 pu.

Solution. At fixed drawn active power $P_{sh} = -0.3$ pu, the current circle gives $|Q_{sh}| \leq \sqrt{1^2 - 0.3^2} = 0.95394$ pu normally and $\sqrt{0.5^2 - 0.3^2} = 0.400$ pu at 0.5 pu voltage. The available magnitude falls by about 58%. If $V < 0.3$ pu, even the active reservation alone is infeasible with a 1 pu current limit: series duty, DC energy or priority must change. Bypassing series injection removes its active duty but losses and energy restoration remain. A reactive-priority scheme instead reduces active balancing and accepts a finite DC excursion.

8.8.8 Protection-aware design and solid-state interruption

Series and combined controllers change the voltages and currents measured by distance, differential and overcurrent protection. Let a series source inject V_{inj} between the relay location and the line, so that the line-side voltage is $V_{relay} + V_{inj}$. For a bolted fault at a fraction m of a line of impedance Z_L ,

$$Z_{app} = \frac{V_{relay}}{I} = m Z_L - \frac{V_{inj}}{I} \quad (8.39)$$

Capacitive emulation ($V_{inj} = +j X_c I$) shortens the apparent impedance and makes zone 1 overreach; inductive emulation makes it underreach. Because the converter saturates, blocks and bypasses during the fault, V_{inj} itself changes within the relay measurement window, so the study must reproduce those transitions rather than assume instantaneous removal of the controller.

Example 8.22. Distance-relay overreach

A line has $Z_L = j0.50$ pu and a zone-1 reach of 80 % ($j0.40$ pu). A series converter between the relay and the line emulates 0.15 pu of capacitive reactance during the fault. Up to what fraction of the line does zone 1 now operate?

Solution. $Z_{app} = j(0.50m - 0.15)$. An explicitly simplified **forward reactance element** requires $0 \leq 0.50m - 0.15 \leq 0.40$, hence $0.30 \leq m \leq 1.10$. The upper value is 110% of the protected-line impedance basis, not automatically 10% of an adjacent line's length. A real directional mho/quadrilateral relay uses polarization, resistance and transient measurement logic, so this scalar screen does not establish actual trips or blind zones. Coordination options must be validated with the converter's limiting/bypass timing and the actual relay, not assume bypass always precedes its decision.

Solid-state interruption raises a related energy problem. When a semiconductor breaker turns off current I_0 in a loop of inductance L against a driving voltage V_s , the current commutates into a surge arrester clamping at $V_c = k V_s$. The current then falls linearly, and

$$t_f = \frac{L I_0}{V_c - V_s}, E_{MOV} = \frac{1}{2} L I_0^2 \frac{k}{k-1} \quad (8.40)$$

For the constant-source, lossless-inductor model, increasing $k = V_c/V_s > 1$ shortens the fall time and lowers absorbed energy toward the original magnetic energy; as $k \rightarrow 1^{++}$ both grow without bound. This asymptote does not model an AC source reversing during a long interruption. Breaker design trades semiconductor blocking voltage, arrester energy and insulation withstand. Hybrid breakers can

reduce conduction loss by using a mechanical main path and a fast semiconductor commutation/absorption path; architecture selection is application dependent [41].

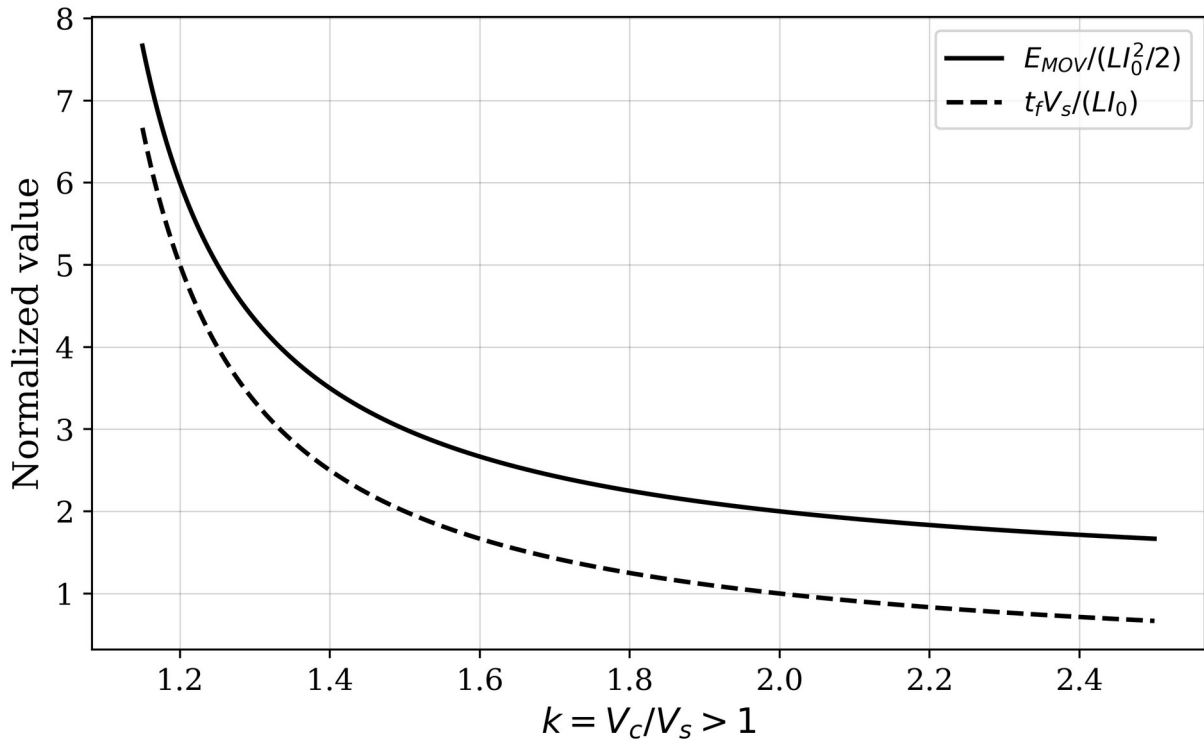


Figure 8.41. Normalized MOV energy and current fall time of a solid-state breaker versus clamping ratio.

Example 8.23. Solid-state breaker energy

A semiconductor breaker interrupts 5 kA in a 10 kV DC feeder with 1 mH of loop inductance. The arrester clamps at 15 kV. Find the current fall time and the energy the arrester absorbs.

Solution. $k=1.5$, so $t_f=10^{-3} \times 5000/(15\,000 - 10\,000)=1.0$ ms and $E_{MOV}=0.5 \times 10^{-3} \times 5000^2 \times 1.5/0.5=37.5$ kJ, three times the magnetic energy stored in the loop.

8.8.9 Data-driven methods and digital twins

Machine learning is most defensible here as a computational aid that accelerates a physics-constrained task: estimating uncertain parameters, screening many contingencies, constructing an OPF surrogate, detecting oscillatory modes or predicting converter thermal margin. The network equations and converter capability limits should remain explicit. A common way to keep a learned model physically consistent is to penalize violations of the governing equations during training:

$$L = L_{data} + \lambda_1 \|g(x, u)\|^2 + \lambda_2 \sum_i \max(0, h_i(x, u))^2 \quad (8.41)$$

where g and h are the equality and inequality constraints of Equation (8.33). Even then, a learned controller that violates $V_{se,max}$, converter current, DC-link energy or arrester energy is not physically valid, however small its training error. A digital twin of a UPFC or IPFC corridor should state what it synchronizes (line parameters, transformer data, converter loss coefficients, capacitor condition, valve temperature, protection status), which decision improves because of that synchronization, and over what operating range the model has been validated against measurements. Its contribution should be judged by measurable improvement in prediction, control or maintenance, not by the label.

8.8.10 Validation hierarchy and publishable evidence

A research claim must be matched to a model that preserves the phenomenon being claimed (Figure 8.42). Power-flow redistribution can be established with a validated steady-state model. Oscillation damping needs eigenvalue or time-domain evidence over several operating points. Harmonic, switching and impedance-interaction claims require EMT or switching models. Fault-current, bypass and relay claims require EMT with protection logic. Claims about implementation delay, sampling and gate logic are strongest when supported by controller or protection hardware-in-the-loop tests. A sound workflow is therefore to derive the reduced model, identify the expected mechanism, test sensitivity analytically, verify with EMT, and move to HIL only for claims that depend on real controller or relay timing. This avoids both extremes: a very detailed model used without understanding the mechanism, and implementation claims drawn from an oversimplified model.

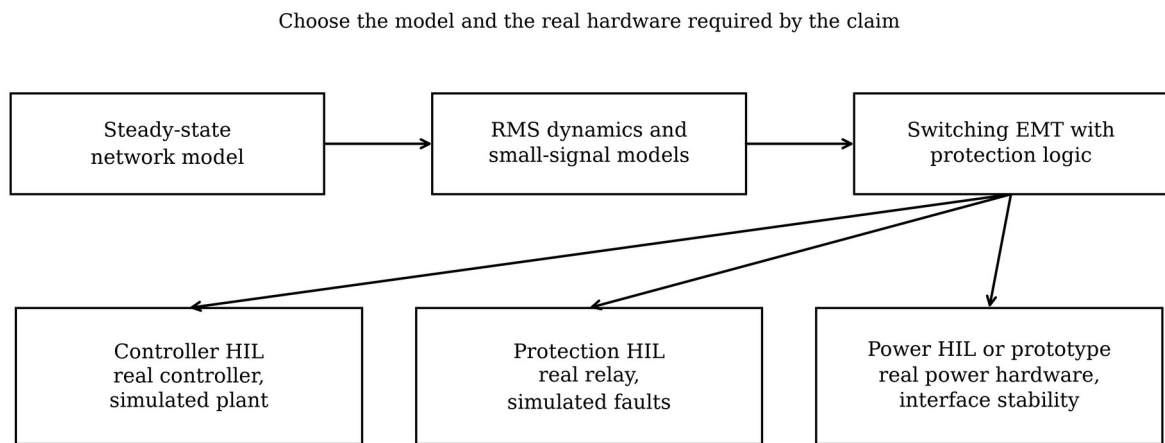


Figure 8.42. Complementary validation methods. Controller HIL, protection HIL and power HIL test different real hardware and claims; none substitutes for all equipment and field validation.

8.8.11 Open problems and potential contributions

Table 8.3 proposes research questions, candidate contributions and validation evidence. They are an outlook: novelty still has to be established through a current literature review and a reproducible comparison with established methods.

Table 8.3. Research map for combined and special-purpose FACTS controllers.

Research problem	Governing relation	Model limitation to address	Potential contribution	Minimum evidence
Fault-tolerant modular series converters	Equation (8.28), Equation (8.29), Equation (8.30)	Equal-sharing assumptions; ripple energy sizes the cells	Thermal-aware balancing, ripple cancellation, wide-bandgap cells	Switching model, loss and reliability analysis, prototype
Security-constrained dispatch of mixed fleets	Equation (8.32), Equation (8.33)	Converter limits checked only in the base case	Contingency-aware formulations with exactness guarantees	Benchmark networks, N-1 validation, solver statistics
Multi-port interaction of UPFC and IPFC	Equation (8.35) with DC-link coupling	Converters modelled separately with a stiff DC source	Three-port impedance models and design rules	Frequency scans validated against EMT
Delay-robust wide-area damping	Equation (8.37)	Fixed-delay designs; no coordination among devices	Robust or adaptive coordinated damping	Eigenvalue studies, real-time HIL with delay

Research problem	Governing relation	Model limitation to address	Potential contribution	Minimum evidence
Grid-forming combined controllers	Equation (8.38), Equation (8.7)	Priority rules unstated; DC energy shared implicitly	Current-limited dual-port control with explicit priority	EMT fault studies, synchronization-stability analysis
Protection with series injection	Equation (8.39)	Instantaneous bypass assumed in relay studies	Adaptive relay settings, converter-aware protection logic	EMT plus protection HIL
Energy-aware solid-state limiters and breakers	Equation (8.40), Equation (8.24)	Current peak or clamp level optimized alone	Co-design of clamp level, MOV energy and recovery	EMT, thermal model, high-power test
Physics-constrained learning	Equation (8.41)	Black-box models extrapolate poorly	Hybrid models with certified constraint satisfaction	Out-of-distribution tests, constraint-violation statistics

The topics in Table 8.3 address potential needs in capacity release, reliability, protection and validation. Their proposed benefits must be demonstrated against a specified baseline, operating range and cost model. Relocatability applies to suitable modular implementations; it is not an attribute of every FACTS installation.

Key Equations

Table 8.4. Key equations of Chapter 8.

Equation	Expression	Conditions
(8.2)	$P_r = \frac{V_s V_r}{X} \sin \delta + \frac{V_r V_{pq}}{X} \sin(\delta + \rho)$	UPFC at sending end, ρ measured from V_s
(8.3)	$(P_r - P_0)^2 + (Q_r - Q_0)^2 = (V V_{pq} / X)^2$	Control circle, $V_s = V_r = V$
(8.5)	$P_{sh} + P_{se} = 0$	Lossless UPFC, steady state
(8.7)	$\frac{d}{dt} \left(\frac{1}{2} C_{dc} V_{dc}^2 \right) = - (P_{sh} + P_{se} + P_{loss})$	DC-link dynamics
(8.10)	$P_{1pq} + P_{2pq} = 0$	Lossless, steady DC energy; no shunt or storage port
(8.11)	$ V_{2q} \leq \sqrt{V_{2,max}^2 - V_{2p}^2}$	Lossless steady state; $ P_{1pq} \leq V_{2,max} I_2$, $I_2 > 0$
(8.13)	$V_\sigma = 2 V_s \sin(\sigma/2) $	Ideal phase shifter
(8.14)	$P = \frac{V^2}{X} \sin(\delta \pm \sigma)$	Phase shifter on a lossless line
(8.15)	$N = 3^L, \Delta V_{max} = \frac{3^L - 1}{2} V_{step}$	Ternary windings
(8.18)	$P = P_{max} \cos \left(\delta_{SR} - \frac{\psi_1 + \psi_2}{2} \right) \sin \frac{\psi_2 - \psi_1}{2}$	Matched opposite susceptances; fundamental balanced model
(8.21)	$X_{lim} = V_{ph} / I_{f,lim} - X_s$	Limiting reactance, inductive source

Equation	Expression	Conditions
(8.22)	$i_p = \kappa \sqrt{2} I_k'', \kappa \approx 1.02 + 0.98 e^{-3R/X}$	Radial R/X approximation; fully specified fault loop
(8.23)	$ I = k V ^\alpha$	Metal-oxide varistor
(8.25)	$V_{lim} = (1 - f_b) V_{clamp}$	Stack voltage at the same current; energy duty checked
(8.26)	$f_0 = 1/(2\pi\sqrt{LC}), \hat{I} \approx \Delta V/Z_0$	Capacitor energization
(8.28)	$N \geq \lceil V_{se}^i /V_{cell,max} \rceil + n_b$	Modular converter with bypassed cells
(8.29)	$C_{cell} \geq S_{cell}/(\omega V_{dc} \Delta V_{pp})$	Single-phase cell ripple
(8.34)	$L(s) = Z_g(s)/Z_c(s)$	Compatible port signs; all crossings and RHP pole count
(8.38)	$ Q_{sh} \leq \sqrt{(V I_{sh,max})^2 - P_{sh}^2}$	Fixed P reservation feasible only if $ P_{sh} \leq V I_{max}$
(8.39)	$Z_{app} = m Z_L - V_{inj}/I$	Relay with series injection
(8.40)	$E_{MOV} = \frac{1}{2} L I_0^2 k/(k-1)$	Constant DC source, $k > 1$, ideal current decay

Common Misconceptions

- “The UPFC’s DC link carries the reactive power of the series converter.” Only real power, plus losses, passes through the link. The reactive power of each converter is generated internally at its AC terminals.
- “An IPFC can supply net real power to all its lines from the DC capacitor.” A capacitor is not a continuous energy source. Without storage or a shunt real-power port, the series-converter real powers must balance apart from losses.
- “A phase shifter raises the maximum power of a line.” An ideal phase shifter shifts the power-angle curve sideways. It raises the power at a given network angle, but the peak V^2/X is unchanged.
- “A quadrature booster is an ideal phase shifter.” A quadrature injection changes both angle and magnitude. An ideal phase shifter preserves magnitude by a different phasor geometry and needs a slightly different injection.
- “A capacitor in parallel with a PST carries power in the same direction as the PST.” A capacitive branch has negative reactance, so at a positive angle it carries power in the opposite direction. That is exactly what flattens the IPC characteristic.
- “A current limiter replaces the breaker.” A limiter restricts the current trajectory; a suitably rated breaker interrupts it. A blocked solid-state breaker does not inherently provide galvanic maintenance isolation, so appropriate disconnecting equipment is required. Hybrid devices can combine current limitation and interruption, but each duty needs its own verified rating and sequence.
- “Lowering the clamping level is free.” The energy the varistor absorbs rises steeply as the clamping level falls, which is why a TCVL bypasses part of the stack only for the duration of a disturbance.

- “A controller that can set P and Q always improves damping.” The sign and phase of the supplementary path, measurement delay, current limiting and the operating point decide whether a mode is damped or excited.

Chapter Summary

UPFC and IPFC combine controllable series injection with explicit real-power and DC-energy constraints. The ideal UPFC disk is clipped by current, voltage and both converters' duties; IPFC commands remain coupled through interline balance. Transformer phase regulation has a distinct voltage geometry, winding duty and commutation sequence. IPC matched branches cancel receiving-voltage sensitivity in the ideal fundamental model, while branch fault currents and stored energy still require protection. Current limiters restrict fault trajectories; breakers interrupt them; transfer switches move loads between feeders. TCVL operation lowers an MOV stack's effective protective level temporarily and is constrained by remaining-disc energy and valve recovery. Modular conversion, alternative energy paths and advanced control broaden the family, but stability and fault benefits must be established for stated models and validated limits.

Solved Numerical Practice Problems

Unless stated otherwise, per-unit quantities use a common three-phase base, line-to-line kilovolts and ohms give three-phase megawatts directly when both end voltages are balanced line-to-line RMS values, trigonometric angles are in degrees, and the UPFC injection angle ρ is measured from the sending-end voltage. Lossless formulas are not applied to problems that include line resistance.

Example 8.24. UPFC uncompensated point and control circle

A UPFC is installed on a lossless line with $V_s = V_r = 1$ pu, $X = 0.5$ pu, $\delta = 30^\circ$ and $V_{pq, \max} = 0.25$ pu. Find (a) P_0 ; (b) Q_0 ; (c) total reactive power absorbed by the line **with no series injection**; (d) control-circle centre and radius; (e) the ideal range of P_r .

Solution. (a) $P_0 = 2 \sin 30^\circ = 1.0$ pu. (b) $Q_0 = -2(1 - \cos 30^\circ) = -0.268$ pu. (c) $|I| = 2 \sin 15^\circ / 0.5 = 1.035$ pu, so $I^2 X = 0.536$ pu, half supplied by each end. (d) Centre $(1.0, -0.268)$, radius $0.25/0.5 = 0.5$ pu. (e) 0.5 to 1.5 pu.

Example 8.25. Maximum and minimum power and converter duty

For Example 8.24 find (a) ρ for maximum power; (b) P_r and Q_r ; (c) the line current; (d) the series-converter real and reactive power; (e) the same quantities at minimum power.

Solution. (a) $\sin(\delta + \rho) = 1$, so $\rho = 60^\circ$. (b) $P_r = 1.5$ pu and $Q_r = -0.268$ pu (unchanged). (c) From Equation (8.1), $I = 1.524 \angle 10.13^\circ$ pu. (d) $S_{se} = V_{pq} I^*$: $P_{se} = 0.067$ pu, $Q_{se} = 0.375$ pu, $|S_{se}| = 0.381$ pu. (e) $\rho = 240^\circ$: $P_r = 0.5$ pu, $Q_r = -0.268$ pu, $I = 0.567$ pu, $P_{se} = -0.067$ pu, $Q_{se} = -0.125$ pu. The real power through the link is small in both cases; the series-converter rating is governed by $V_{pq} I$, which depends on the line current as well as on the injected voltage.

Example 8.26. Two injection angles for the same active power

Use the network of Example 8.24, but for this example **increase the allowed injection magnitude to 0.45 pu** and operate at $V_{pq} = 0.45$ pu. The target is $P_r = 1.4$ pu. Find (a) both values of ρ ; (b) Q_r for each; (c) line currents; (d) series-converter apparent powers; (e) which solution has lower current and MVA duty. Other operating constraints must still be checked.

Solution. (a) $\sin(30^\circ + \rho) = 0.4/0.9$, so $\rho = -3.61^\circ$ or 123.61° . (b) $Q_r = +0.538$ or -1.074 pu. (c) Currents are 1.500 and 1.765 pu. (d) The corresponding series duties are 0.675 and 0.794 pu. (e) The first requires less current and converter MVA; its reactive delivery must also suit the receiving-bus requirement. Neither point belongs to the original 0.25 pu voltage-limit disk of Example 8.24.

Example 8.27. UPFC on a line with resistance

A 132 kV line has $X = 175 \Omega$ and $R = 70 \Omega$, with equal end voltages at $\delta = 45^\circ$. The UPFC can inject up to 45 kV (line-to-line equivalent). Find (a) P , Q_r and current without the UPFC; (b) the maximum receiving-end power and the injection angle; (c) the minimum injection that raises power by 50 % when placed at its best angle.

Solution. Use $\underline{I} = [V_s e^{j\delta} + V_{pq} e^{j(\delta+\rho)} - V_r] / (R + jX)$ and $S_r = V_r \underline{I}^*$ on the line-to-line power equivalent; physical current is $|\underline{I}|/\sqrt{3}$. (a) $P_0 = 50.637$ MW, $Q_{r0} = -49.417$ Mvar and $I_0 = 309.47$ A. (b) The active increment is maximized when the injection absolute angle is $\tan^{-1}(X/R)$, so $\rho = 68.199^\circ - 45^\circ = 23.199^\circ$. Its maximum is $V_r V_{pq} / \sqrt{R^2 + X^2} = 31.515$ MW; hence $P_{max} = 82.152$ MW and current 419.32 A. (c) $V_{pq,min} = 0.5 P_0 \sqrt{R^2 + X^2} / V_r = 36.152$ kV line-to-line. A lossless formula changes the target and optimum; it is not a valid substitute for this lossy-line calculation.

Example 8.28. Power transfer at zero angle

With $V_s = V_r = 1$ pu, $X = 0.5$ pu, $\delta = 0$ and $V_{pq} = 0.4$ pu, find (a) the uncompensated power; (b) P_r and Q_r at $\rho = 90^\circ$; (c) the current; (d) the series-converter power.

Solution. (a) Zero. (b) $P_r = 0.4 \times 1/0.5 = 0.8$ pu and $Q_r = 0$. (c) 0.8 pu. (d) $P_{se} = 0$ and $Q_{se} = 0.32$ pu. Series injection alone establishes power transfer where an uncompensated line carries none.

Example 8.29. Shunt-converter rating

In Example 8.25(a) the shunt converter must also supply 0.3 pu of reactive power to the bus. Find (a) its real power; (b) its apparent power.

Solution. (a) $P_{sh} = -P_{se} = -0.06699$ pu, drawn from the bus. (b) $|S_{sh}| = \sqrt{0.06699^2 + 0.3^2} = 0.30739$ pu. At this point the series duty is 0.38094 pu (Example 8.25); equal shunt and series ratings are not required by energy conservation. Choose the shunt rating for its full operating envelope, DC balancing, local reactive support and any independent STATCOM duty.

Example 8.30. UPFC with unequal end voltages

$V_s = 1.0$ pu, $V_r = 0.95$ pu, $X = 0.5$ pu, $\delta = 25^\circ$, $V_{pq} = 0.2$ pu and $\rho = 50^\circ$. Find (a) P_0 and Q_r without injection; (b) P_r and Q_r with it; (c) the effective sending voltage; (d) the series-converter power.

Solution. (a) $P_0 = 0.803$ pu and $Q_{r0} = -0.083$ pu. (b) From Equation (8.2), $P_r = 1.170$ pu and $Q_r = +0.015$ pu. (c) $|V_s + V_{pq}| = 1.139 \angle 32.7^\circ$ pu. (d) $P_{se} = 0.061$ pu and $Q_{se} = 0.239$ pu.

Example 8.31. A 138 kV UPFC

A 138 kV line has $X = 80 \Omega$ and operates at 35° ; the UPFC can inject 28 kV. Find (a) P_0 ; (b) the UPFC power term; (c) the power range; (d) ρ for 180 MW.

Solution. (a) $138^2 \sin 35^\circ / 80 = 136.5$ MW. (b) $138 \times 28 / 80 = 48.3$ MW. (c) 88.2 to 184.8 MW. (d) $\sin(35^\circ + \rho) = (180 - 136.5) / 48.3 = 0.900$, so $\rho = 29.1^\circ$ (the other root, 80.9° , gives a more negative Q_r). A target of 260 MW is outside the range and is physically infeasible even though a controller would accept the reference.

Example 8.32. IPFC: unity power factor at the receiving end of line 1

Two identical lines have $V = 1$ pu, $X = 0.5$ pu and $\delta = 30^\circ$; both IPFC converters can inject 0.25 pu. Converter 1 must hold $Q_{1r} = 0$ while maximizing P_{1r} . Find (a) ρ_1 ; (b) P_{1r} ; (c) P_{1pq} and Q_{1pq} ; (d) the injection angles of converter 2; (e) the resulting power in line 2.

Solution. (a) $\cos(\delta + \rho_1) = -Q_0 X / (V V_{pq}) = 0.536$, so $\rho_1 = 27.60^\circ$. (b) 1.422 pu. (c) $P_{1pq} = 0.191$ pu and $Q_{1pq} = 0.300$ pu. (d) Converter 2 must absorb 0.191 pu: $\rho_2 = -152.4^\circ$ or 122.4° . (e) $P_{2r} = 0.578$ pu ($Q_{2r} = -0.536$ pu) or 1.232 pu ($Q_{2r} = -0.711$ pu). The second choice raises the flow in both lines; the first transfers power from line 2 to line 1.

Example 8.33. IPFC DC-link dynamics

An IPFC DC capacitor has an energy-storage constant $H_{dc} = 0.01$ s, defined as the stored DC energy at $V_{dc} = 1$ pu divided by the three-phase power base. Converter 2 supplies 0.25 pu to the DC link while converter 1 absorbs 0.20 pu from it. Find (a) the net power into the link; (b) the exact DC voltage after 10 ms, neglecting losses; (c) the small-signal approximation; (d) the conclusion.

Solution. (a) $0.25 - 0.20 = 0.05$ pu. (b) Stored energy is proportional to V_{dc}^2 , so the normalized energy rises from 1.0 to $1 + 0.05 \times 0.010 / 0.010 = 1.05$ and $V_{dc} = \sqrt{1.05} = 1.0247$ pu. (c) Linearizing $W_{dc} = H_{dc} V_{dc}^2$ about 1 pu gives $dV_{dc}/dt = \Delta P / (2H_{dc}) = 2.5$ pu/s, so $\Delta V_{dc} \approx 0.025$ pu after 10 ms, in close agreement. (d) A 0.05 pu imbalance moves the DC voltage by about 2.5 % in 10 ms, so the DC-voltage controller must act quickly. Using $\Delta V_{dc} = \Delta P \Delta t / H_{dc}$ would overestimate the excursion by a factor of two.

Example 8.34. Balancing two parallel lines

Two lossless, purely reactive parallel paths with $X_1 = 0.3$ pu and $X_2 = 0.6$ pu share total active power 2.3 pu between the same unit-voltage buses. Find (a) natural active sharing; (b) inductive series reactance in line 1 that equalizes active flows; (c) the required series-voltage magnitude from AC phasors, and compare it with the unity-power-factor current approximation. Select the lower-angle solution.

Solution. (a) $P_1 = 2.3(0.6)/(0.3+0.6) = 1.53333$ pu and $P_2 = 0.76667$ pu. (b) Equal effective reactances require adding $X_{inj} = 0.3$ pu to line 1. (c) Both paths then have 0.6 pu reactance and carry 1.15 pu active power. Thus $\sin \delta = 1.15(0.6) = 0.69$, giving $\delta = 43.6301^\circ$. The actual current is $I_1 = 2 \sin(\delta/2) / 0.6 = 1.23871$ pu, so $|V_{pq}| = X_{inj} I_1 = 0.37161$ pu. The estimate $I_1 \approx P_1 = 1.15$ gives 0.345 pu, about 7.2% below the required AC magnitude. The voltage-rise phasor is $\underline{V}_{pq} = -j0.3 I_1$ for this inductive insertion.

Example 8.35. Generalized IPFC power balance

Four series converters share a DC link. Neglect conversion losses and DC-energy change. Converters 1, 2 and 3 deliver +0.15, -0.08 and -0.04 pu to their lines. Converter 4 injects 0.1 pu into a line carrying 0.9 pu. Find (a) P_{4pq} ; (b) the full-cycle injection angles relative to I_4 .

Solution. (a) $P_{4pq} = -(0.15 - 0.08 - 0.04) = -0.03$ pu. (b) $\cos \phi = -0.03 / (0.1 \times 0.9) = -1/3$, so the principal 0° – 180° solution is 109.471° . The other full-cycle solution is 250.529° . Their reactive powers are respectively $+0.08485$ and -0.08485 pu; either requires the corresponding converter and line operating limits. The real-power balance alone cannot select reactive direction.

Example 8.36. Quadrature booster on a 220 kV line

A quadrature booster injects 20 kV (phase) into a 220 kV line ($V_{ph} = 127.0$ kV) of $X = 50 \Omega$ at $\delta = 25^\circ$. Find (a) the phase shift; (b) the effective voltage; (c) the power before and after.

Solution. (a) $\tan^{-1}(20/127) = 8.95^\circ$. (b) $\sqrt{127^2 + 20^2} = 128.6$ kV. (c) $P_0 = 3V_{ph}^2 \sin 25^\circ / X = 409.0$ MW and $P = 3 \times 128.6 \times 127.0 \times \sin 33.95^\circ / 50 = 547.2$ MW. Part of the increase comes from the 1.2 % magnitude rise, which an ideal phase shifter would not produce.

Example 8.37. TCPAR on a 132 kV line

A lossless 132 kV line has $X = 40 \Omega$ and operates at 30° . Find P and the reactive power supplied at each end for $\sigma = 0, +10^\circ$ and -10° .

Solution. $P = V^2 \sin(\delta + \sigma) / X$: 217.8, 280.0 and 149.0 MW. With equal end voltages each end supplies $V^2(1 - \cos(\delta + \sigma)) / X$: 58.4, 101.9 and 26.3 Mvar, so the line absorbs twice these values in total.

Example 8.38. Ideal phase shifter

An ideal phase shifter injects 0.4 pu. Find (a) σ ; (b) the injection angle relative to V_s ; (c) the power at $\delta = 30^\circ$ on a 0.3 pu line ($V = 1$ pu) when the shift opposes the angle.

Solution. (a) Injection magnitude 0.4 pu gives $|\sigma| = 2 \sin^{-1}(0.2) = 23.0739^\circ$. Choose the positive shift for this part. (b) Its injection angle relative to V_s is $90^\circ + \sigma/2 = 101.537^\circ$. (c) To **oppose** the 30° line angle use the reflected injection angle -101.537° , or 258.463° on a full-cycle scale; then $\sigma = -23.0739^\circ$ and $P = \sin(6.9261^\circ) / 0.3 = 0.40196$ pu. The same chord magnitude supports opposite shifts, but the phasor angles are different.

Example 8.39. Ternary TCVR

A TCVR has three ternary windings with a smallest step of 5 kV. Find (a) the number of levels; (b) the maximum boost and buck; (c) the winding voltages; (d) the resolution.

Solution. (a) 27. (b) $\pm 13 \times 5 = \pm 65$ kV. (c) 5, 15 and 45 kV. (d) 5 kV per step.

Example 8.40. Thyristor tap-changer harmonics

Ideal selectable taps of 10 kV and 12 kV **peak** produce the waveform defined in Section 8.3.2, with transition at $\alpha = 60^\circ$ each half cycle. Assume an ideal selector that can realize it. Find (a) fundamental peak amplitude; (b) fundamental phase; (c) 3rd, 5th and 7th peak harmonics; (d) partial THD using only these three.

Solution. The Fourier integrals give $a_1 = -0.477465$ kV and $b_1 = 11.6090$ kV. (a) $A_1 = \sqrt{a_1^2 + b_1^2} = 11.6188$ kV peak. (b) $\phi_1 = \tan^{-1}(a_1/b_1) = -2.3552^\circ$ in $A_1 \sin(\theta + \phi_1)$. (c) Peak magnitudes of the 3rd, 5th and 7th are 0.477465, 0.275664 and 0.137832 kV. (d) Their partial THD is $\sqrt{0.477465^2 + 0.275664^2 + 0.137832^2} / 11.6188 = 4.8912\%$. As a completeness check, the exact piecewise integral gives $V_{rms} = 8.22794$ kV and total ideal THD $\sqrt{2V_{rms}^2 / A_1^2 - 1} = 5.4532\%$. The

difference is due to the omitted higher odd harmonics; neither value validates an SCR commutation circuit.

Example 8.41. Loop-flow control with a PAR

Use a **lossless DC/small-angle power-flow approximation** for two parallel paths $X_1=0.3$ pu and $X_2=0.6$ pu, with unit voltages and initial total transfer 2.3 pu. A PAR boosts the angle in path 2. Find the shift that makes the two path powers equal (a) with total transfer maintained at 2.3 pu; (b) with the corridor angle held at its original value. State the new total in (b).

Solution. Initially $\theta=2.3/(1/0.3+1/0.6)=0.46$ rad. (a) Equal powers are 1.15 pu. Path 1 gives $\theta=0.3(1.15)=0.345$ rad; path 2 requires $(0.345+\sigma)/0.6=1.15$, hence $\sigma=0.345$ rad $=19.767^\circ$. (b) With $\theta=0.46$ rad fixed, path 1 stays at 1.53333 pu. Equalization requires $(0.46+\sigma)/0.6=1.53333$, so $\sigma=0.46$ rad $=26.356^\circ$ and total transfer is 3.06667 pu. A 0.23 rad shift would give only 1.15 pu in path 2 and **would not equalize** the paths. Since these angles are not very small, the numerical values illustrate the linear network model; a full AC study must use sine power laws and actual voltage magnitudes.

Exact AC comparison. With the same unit-voltage, purely reactive paths and $0<\theta, \theta+\sigma<90^\circ$, the initial angle is $\theta_0=\sin^{-1}[2.3/(1/0.3+1/0.6)]=27.3871^\circ$. For fixed total 2.3 pu, equal flow gives $\theta=\sin^{-1}(0.345)=20.1818^\circ$ and $\sigma=\sin^{-1}(0.69)-\theta=23.4483^\circ$. At the fixed original AC angle, $P_1=1.53333$ pu and equality requires $\sigma=\sin^{-1}(0.92)-\theta_0=39.5390^\circ$; total becomes 3.06667 pu. The different shifts quantify the error of the linear angle model; do not transfer its 19.767° or 26.356° values directly to an AC design.

Example 8.42. VSC-based regulator power exchange

A series VSC injects 5 kV in phase with the bus voltage into a line carrying 500 A at 0.866 power factor lagging. Find, per phase, (a) the real power; (b) the reactive power; (c) the converter rating.

Solution. (a) $5 \times 500 \cos 30^\circ = 2.17$ MW. (b) $5 \times 500 \sin 30^\circ = 1.25$ Mvar. (c) 2.5 MVA. The real power must be supplied by a shunt converter or a store.

Example 8.43. IPC120 powers

An IPC120 with $B_1=-1$ pu and $B_2=+1$ pu connects buses with $V_S=V_R=1$ pu at $\delta_{SR}=15^\circ$. Find (a) δ_{B1} and δ_{B2} ; (b) P ; (c) Q_S and Q_R .

Solution. (a) 75° and -45° . (b) From Equation (8.16), $P=0.966+0.707=1.673$ pu, which equals $2 \times 0.866 \cos 15^\circ$ from Equation (8.18). (c) From Equation (8.17), $-Q_S=Q_R=-(1-\cos 75^\circ)+(1-\cos 45^\circ)=-0.448$ pu. Terminal powers alone do not reveal the internal branch currents.

Unsolved Practice Problems

Problem 8.1 A UPFC on a line with $V=1$ pu, $X=0.4$ pu, $\delta=45^\circ$ and $V_{pq,max}=0.2$ pu must deliver $P_r=1.8$ pu. Find P_0 , Q_0 , the circle radius, the maximum power, and the injection angle (with the corresponding Q_r) that gives 1.8 pu with the larger reactive delivery.

Problem 8.2 For $V=1$ pu, $X=0.6$ pu and $\delta=60^\circ$, find the centre and radius of the UPFC control circle for $V_{pq,max}=0.3$ pu.

Problem 8.3 A 400 kV line of $100\ \Omega$ operates at 40° with a UPFC injecting up to 30 kV. Find the power range.

Problem 8.4 Explain, with the DC-balance equation, why an IPFC cannot supply net real power to all its lines simultaneously without a shunt converter or storage.

Problem 8.5 A ternary TCPAR has four windings with a 3 kV smallest step. Find the number of levels and the maximum injection.

Problem 8.6 An injection IPC has $\psi_1 = -20^\circ$, $\psi_2 = +20^\circ$, $X = 0.5$ pu and $V_S = V_R = 1$ pu. Find P at $\delta_{SR} = 10^\circ$ and 25° .

Problem 8.7 An ideal constant-magnitude phase shifter at a 1.0 pu sending bus injects 0.25 pu. Find the positive phase shift on the 0° – 180° branch.

Problem 8.8 A quadrature booster at a 1.0 pu sending bus injects $+j0.15$ pu. Find the resulting phase shift and effective sending-voltage magnitude.

Problem 8.9 A balanced 33 kV line-to-line RMS network has a symmetrical three-phase fault current of 12 kA; the breakers are rated 8 kA. Assume a purely inductive Thevenin source, nominal-voltage factor 1, and an already inserted ideal limiter. Find the additional reactance needed to reduce the symmetrical RMS current to 8 kA.

Problem 8.10 A series-resonant SSFCL at 11 kV, 50 Hz uses $L_1 = 20$ mH with a source reactance of $0.5\ \Omega$. Find C_1 and the limited symmetrical fault current.

Problem 8.11 In a balanced 400 kV line-to-line RMS system, an illustrative arrester model clamps at 1.7 times the normal peak phase voltage. Find that voltage and the TCVL level if 25% of identical discs are bypassed at the same arrester-current reference.

Problem 8.12 A 66 kV capacitor bank of $60\ \mu\text{F}$ per phase is energized through 1.5 mH. Find f_0 , Z_0 and the inrush peak for a discharged bank closed at the voltage peak.

Problem 8.13 By what factor does the current of a ZnO block rise when its voltage rises by 10 %, for $\alpha = 25$ and $\alpha = 40$?

Problem 8.14 Compare a UPFC, an IPFC, a TCPAR and an APST for relieving a congested 400 kV corridor with a parallel underused path, a weak voltage at the receiving bus, and a tight budget.

Problem 8.15 A modular series converter must inject 24 kV with cells rated 3.2 kV and must tolerate two bypassed cells. Find the minimum number of cells and the per-cell voltage in normal operation.

Problem 8.16 A converter port has $Z_c = j\omega L_f + K_p e^{-j\omega T_d}$ with $T_d = 100\ \mu\text{s}$. Find the first positive-frequency interval in which its real part is negative, and explain why it is not negative at every higher frequency.

Problem 8.17 A series converter emulates 0.10 pu of capacitive reactance on a line with $Z_L = j0.40$ pu. Zone 1 is set at 85% of the protected-line impedance. Using a simplified forward reactance element, find the apparent-reactance operating interval and upper reach on that line's impedance basis while the converter is active.

Answer Key (Unsolved Problems)

8.1 1.768 pu; -0.732 pu; 0.5 pu; 2.268 pu; $\rho = -41.3^\circ$ with $Q_r = -0.233$ pu (the other root, 131.3° , gives $Q_r = -1.231$ pu).

8.2 (1.443, -0.833) pu; 0.5 pu.

8.3 $P_0 = 1028.5$ MW; range 908.5 to 1148.5 MW.

8.4 In lossless steady state, $\sum_k P_{k,pq} = 0$ without a shunt converter or energy store, so power delivered to some lines must be drawn from others. With converter losses, $\sum_k P_{k,pq} = -P_{loss}$.

8.5 81 levels; ± 120 kV.

8.6 1.347 pu; 1.240 pu.

8.7 14.36° .

8.8 8.53° ; 1.011 pu.

8.9 0.794Ω .

8.10 $506.6 \mu\text{F}$; 936 A.

8.11 555.2 kV; 416.4 kV.

8.12 530.5 Hz; 5.0Ω ; 10.8 kA.

8.13 10.8 and 45.3.

8.14 Open-ended. Compare voltage support, controllable path flows, real-power exchange, N-1 limits, fault behavior and lifecycle cost. A TCPAR/APST may suit loop-flow control, but weak receiving voltage may require separate shunt support; a UPFC's shunt location and network sensitivity decide whether it helps that bus. IPFC feasibility depends on line topology and supporting-converter margins. No device can be declared cheapest or sufficient without project ratings and cost data.

8.15 10 cells; 2.4 kV per cell.

8.16 First band: $2.5 < f < 7.5$ kHz. Later negative bands alternate with positive bands; use only the model-valid frequency range.

8.17 For the simplified forward reactance screen, $0 \leq 0.40m - 0.10 \leq 0.34$, so $0.25 \leq m \leq 1.10$. The upper boundary is 110% of the protected-line impedance basis. Actual relay operation depends on its characteristic, polarization and bypass transient.

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